



Blockchain-Enabled Chain of Custody for Enhanced Authentication of Electronic Evidence in the Age of AI

A proof-of-concept for tamper-evident digital evidence management

IS483 Final Report

Computing and Law Application Project Track

Team Members

Name	ID	Email	Role
Ranveer Singh Bains	01432598	ranveers.2022@scis.smu.edu.sg	Project Manager & Smart Contract/Android Developer
Nicole Michaella Daduya	01464138	nicoled.2022@scis.smu.edu.sg	Frontend, UI/UX Lead, CCTV Prototype Lead
John Shutler	01433196	jrshutler.2022@scis.smu.edu.sg	Deliverables, CM & Comms Lead
Vaibhav Mittal	01444134	vaibhavsm.2023@scis.smu.edu.sg	Backend Lead
S Bharath Raj	01422430	bharathrajs.2022@scis.smu.edu.sg	Backend Developer
Sarah Do	01457859	lifang.do.2022@scis.smu.edu.sg	Compliance Research & Frontend Developer

Sponsor

Name	ID	Email	Role
Dharma Sadasivan	Stability	dharma@stabilityprotocol.com	Chief Strategy Officer & General Counsel

Executive Summary

Artificial intelligence has undermined the trustworthiness of digital media. In January 2026, a US Senator presented a deepfaked image on the Senate floor without realising it was fabricated. Under Singapore's Evidence Act (Section 116A), electronic records are presumed to be authentic(with pre-conditions) unless someone challenges them – and challenges are now rising because of AI-generated fakes.. Singapore's judiciary has flagged AI-generated false evidence as an emerging threat, yet there exists no scalable mechanism to challenge or re-establish the authenticity of electronic evidence when challenged. The prevailing approach, reactive forensic analysis at SGD 3,000-5,000 per case over 6-8 weeks, does not scale to absorb the projected increase of cases where AI-tampered evidence is submitted.

Bedrock is a proof-of-concept web application that addresses this gap by sealing video evidence at the moment of capture. Where existing tools attempt to detect manipulation after the fact, Bedrock eliminates the need for that detection entirely by creating a fingerprint of the video as it is being recorded, making any subsequent tampering mathematically provable.

Built in partnership with STABILITY on their Global Trust Network (GTN; a tokenless, zero-fee, EVM-compatible public blockchain), Bedrock computes a unique digital fingerprint across the video stream in real time, anchoring each fingerprint to a smart contract on the GTN. No wallets, no gas fees, no crypto knowledge required.

The result is an immutable, independently verifiable chain of custody that exists on a public ledger and requires no trusted intermediary to authenticate. On recording completion, Bedrock generates a Proof of Integrity Certificate (POIC). It is a PDF containing the subcontract address, final rolling hash, timestamps, and recording metadata, ready for court submission. Any party can subsequently upload the file to the Verify module, which re-hashes it and compares the result against the on-chain record to determine authenticity and tampering(if any) of the record.

Three properties distinguish Bedrock from existing solutions. First, it captures and verifies whether any stream is legitimate, defined as a real-time, unaltered capture of actual events, recorded by a physically present person, untampered from creation to submission. Second, it architects for prevention rather than detection, sealing authenticity at creation so that forensic analysis becomes unnecessary for routine evidence cases. Third, it makes the blockchain layer invisible to the user – no cryptocurrency, no wallets, no fees, no technical knowledge required.

The team delivered a full proof of concept across three form factors in a mere 14 weeks to demonstrate how Bedrock's core engine can be repurposed across different use cases. The team built a web application (primary POC), an Android APK for mobile-first capture, and a Raspberry Pi CCTV hardware prototype with 3D-printed housing to demonstrate physical attestation capabilities. The project was executed using an agile (scrum) methodology and culminated in a final presentation to esteemed guests from industry leaders like Infocomm Media Development Authority, Singapore Academy of Law and S. Rajaratnam School of International Studies.

1. Introduction and Scope **6**
1.1 Problem Statement 6
1.2 Our Sponsor 6
1.3 Scope Definition 7
1.4 Competitor Landscape 8
1.4.1 Category 1: Capture-time Authentication (Direct Competitors) 8
1.4.2 Category 2: AI-based Deepfake Detection (Indirect Competitors) 9
1.4.3 Category 3: Industry Standards and Coalitions 10
1.4.4 Identified Gaps and Bedrock's Differentiation 10
1.5 Market Research 11
1.5.1 Total Addressable Market (TAM) 11
1.5.2 Serviceable Available Market (SAM) 11
1.5.3 Serviceable Obtainable Market (SOM) 11
1.5.4 Key market drivers 11
1.5.5 Key market challenges 12
1.6 User Personas and Segmentation 12
2. Proposed Solution: Bedrock Web Application **13**
2.1 Solution Overview & Value Proposition 13
2.1.1 How Bedrock Works 14
2.2 System Architecture and User Flow 15
2.2.1 High-level Architecture 15
2.2.2 Technology Stack 16
2.2.3 Navigation Diagram and User Flows 17
2.3 Record Module Breakdown 20
2.3.1 Liveness Verification Feature 22
2.3.2 Real-Time Hashing Algorithm 24
2.3.3 Smart Contract Architecture 26
2.4 Verify Module Breakdown 29
2.4.1 Four-Phase Verification Process 29
2.4.2 Three Terminal States 30
2.5 The Evidence Package 32
2.6 Use Cases Built Beyond the Web App 32
The Removal for Liveness check for the Phone and Raspberry Pi Prototypes 33
2.6.1 Android APK 33
2.6.2 Raspberry Pi CCTV 34
3. Project Management and Execution **34**
3.1 Project Phases and Sprint Structure 34
3.1.1 Methodology: Agile (Scrum) 34
3.1.2 Four Project Phases 35
3.2 Task Management and Progress Tracking 36
3.2.1 Jira: Kanban Board for Centralised Task Management 36
3.2.2 Jira: Gantt Chart and Sprint Timeline 37

3.2.3 Cumulative Flow Diagram 37
3.2.4 GitHub: version control and collaborative development 37
3.2.5 Bug Tracking and Documentation 38
3.3 Stakeholder Communication 38
3.3.1 Communication matrix 38
3.3.2 Fortnightly Stakeholder Updates 39
3.3.3 Single Point of Contact and Decoupled Workflows 39
3.3.4 Sponsor Handover Pack 40
4. Testing, Analysis and Findings 40
4.1 Internal Testing & Validation 40
4.1.1 Unit Testing (Sprint 1-2 and 4) 40
4.1.2 Integration Testing (Sprint 5) 42
4.2 External Testing & Validation 44
4.2.1 Usability Testing (Sprint 3, Record Module) 44
4.2.2 User Acceptance Testing (Sprints 4 & 5, Full POC) 46
4.2.3 Usability testing vs UAT comparison 48
5. Achievements and Value Creation for LegalTech Industry 48
5.1 Smart Contracts: New Evidentiary Mechanism 49
5.2 Trust and safety: Building for the Singapore Judiciary 49
5.3 Digital Forensics: Removing the Dependency 50
5.4 The Court Admissibility Pathway 50
5.4.1 Phase 1: Judicial Engagement and Dialogue with Singapore Courts 51
5.4.2 Phase 2: Pioneer Test Case to Establish Legal Precedent 51
5.4.3 Phase 3: Global legal recognition for blockchain-anchored evidence 51
6. Limitations, Mitigation Strategies, and Future Trajectory 52
6.1 Known Limitations 52
6.2 Risk Analysis and Mitigations 55
6.2.1 Risk Rating Classification Legend 55
6.2.2 Risk Register 55
6.3 Future Trajectory 60
7. Conclusion 60
8. References 62
9. Appendices 64
Appendix A: User Personas 64
A.1 Jake the Client 64
A.2 Maya the Lawyer 65
A.3 Ron the Data Verification Specialist 66
Appendix B: User Stories 67
B.1 Easy Flow (Persona: Jake the Client) 67
B.2 Hard Flow (Persona: Maya the Lawyer) 68
Appendix C: Competitive Analysis Matrix and Gartner Magic Quadrant 69
Appendix D: Consent and Data Disclosure Form 70

Appendix E: Brand Guide 71
Appendix F: Website Copy Brief 73
Appendix G: User Flow Diagrams 74
G.1 Record Module User Flow Diagram 74
G2: Verify Module User Flow Diagram 75
Appendix H: Project Charter 76
Appendix I: Stakeholder Register 76
Appendix J: Risk Register 77
Appendix K: Work Breakdown Structure 77
Appendix L: RACI Matrix 78
Appendix M: Lessons Learned Register 78
Appendix N: Usability Testing Raw Data 79
Appendix O: User Acceptance Testing Raw Data 81
Appendix P: Proof of Integrity Certificate 83
Appendix Q: Meeting Notes Summary 84
Appendix R: Sponsor Handover Pack 86
Appendix S: Bedrock Privacy Policy 87

1. Introduction and Scope

1.1 Problem Statement

AI-generated media is now indistinguishable from real footage. Generative AI has broken the baseline assumption that digital evidence is trustworthy, and legal systems worldwide lack the capacity to handle the resulting spike in synthetic evidence.

The scale of the problem is difficult to overstate. 75% of individuals cannot identify a high-quality deepfake (Deepstrike, 2025). Deepfake video content has grown 550% since 2019 (Security Hero, 2023), and deepfake volume is projected to increase 900% annually (Keepnet Labs, 2026). This trend is fuelled by the low cost of fabricating false evidence. Any individual with access to consumer-grade AI tools can produce synthetic media at effectively zero cost. This was evidently shown in January 2026 when a fatal shooting in Minneapolis generated a wave of AI-manipulated images on social media. Many of these altered images were based on verified photographs and mirrored reality closely enough to mislead the public. This included US Senator Dick Durbin, who displayed one such deepfake during a speech on the Senate floor without realising it was fabricated (Perlo, Scott & Horvath, 2026).

Singapore has recognised this as an emerging threat to its legal system. Justice Aidan Xu, head of Innovation and Transformation at the Singapore Judiciary, flagged false evidence as a major concern during a speech at the IT Law Series 2025 (Xu, 2025). His warning pointed to a specific structural vulnerability in Singapore's evidentiary framework.

Under s 116A of the Evidence Act (Cap. 97), electronic records are presumed authentic unless the opposing party adduces sufficient evidence to raise doubt about that presumption. Currently, the opposing party challenging the authenticity of evidence (scenario 1) or the proponent of the evidence itself (scenario 2) currently has no affordable, scalable and deterministic way to question or re-establish its integrity respectively. The prevailing approach forces litigants to rely on three reactive methods: chain-of-custody records maintained by the submitting party, testimony from IT custodians regarding the handling and storage of the evidence (both 1 and 2 do not actually analyse the legitimacy of the video evidence itself), and mostly importantly, forensic expert reports analysing the file for signs of manipulation. Each of these methods suffers from fundamental limitations. Chain-of-custody records depend on the integrity of the custodian, and they only prove process, not content. IT custodian testimony is only as reliable as the systems and personnel involved. Last but not least, forensic analysis is prohibitively expensive (typically SGD 3,000-5,000 for basic services), time-consuming (6-8 weeks for a standard report), and fundamentally non-scalable as every contested file requires its own forensic expert (Asia Top Investigation LLP, 2023).

Therefore, if one can certify the authenticity of digital evidence before it enters a courtroom, and before the authenticity question arises, countless hours and dollars can be saved in the near future.

1.2 Our Sponsor

Bedrock was developed in partnership with STABILITY (stabilityprotocol.com), a blockchain infrastructure company building public blockchain technology for enterprise and government use. STABILITY's flagship product is the Global Trust Network (GTN); a tokenless, zero-fee, EVM-compatible public blockchain.

Unlike Bitcoin, Ethereum, and other widely known blockchains, GTN is entirely cryptoless and public. Blockchain interactions are gasless and not associated with volatile crypto tokens. Network access is managed through standard API keys. This eliminates the regulatory, compliance, and reputational hurdles that typically block law firms, governments, and corporations from adopting blockchain technology, while maintaining absolute data immutability. This ultimately means that no wallets, no gas fees and no crypto knowledge are required to operate on STABILITY.

GTN is not a speculative or untested technology either. STABILITY has partnered with Singapore's Infocomm Media Development Authority (IMDA) to integrate GTN as infrastructure for TradeTrust, a government-backed framework for digitising international trade documents (STABILITY, 2024). As such, our sponsor also invited senior IMDA directors and members of the judiciary to our final presentation to evaluate Bedrock's applicability in legal and public sector contexts.

Our primary sponsor contact was Dharma Sadasivan, Chief Strategy Officer and General Counsel at STABILITY. His dual role made him an ideal sponsor for a project sitting at the intersection of computing and law. Throughout the project, Dharma provided fortnightly guidance on the technical architecture and legal positioning.

1.3 Scope Definition

The original project brief from STABILITY called for a web application proof-of-concept (POC) that could capture video evidence in-app and anchor it to the GTN blockchain in real time. Over the course of five sprints, the scope evolved substantially in response to the midterm panel feedback, user testing insights, and sponsor input. The table below summarises the key scope changes between the original brief and the final product.

Component	Original Brief (Pre-Midterm)	Final Delivery (Post-Midterm)	Driver
Liveness Verification	Luxand.cloud passive liveness API	AWS Rekognition with active colour challenges + Cognito session management	Midterm panel concerns about the artificial injection of pre-recorded videos and deepfakes
Android APK	Not in original scope	Functional prototype: mobile-first capture, on-device blockchain signing	Sponsor stretch goal (Sprint 5 delighter)
Raspberry Pi CCTV	Not in original scope	Functional prototype: hardware attestation, standalone hashing and anchoring	Sponsor stretch goal (Sprint 5 delighter)

Table 1: Changes made to the scope over the course of the project

1.4 Competitor Landscape

The market response to the deepfake threat falls into three categories: AI-based detection tools that analyse media after creation, authentication platforms that seal media at the point of recording, and industry standards and coalitions that define provenance frameworks. Bedrock sits in the second category.

1.4.1 Category 1: Capture-time Authentication (Direct Competitors)

Bedrock's closest comparators, using cryptographic methods to establish authenticity at capture.

Truepic (truepic.com)

Basis of Comparison	Description
Key Features and Functionality	San Diego–based company founded in 2015. Originally used Bitcoin-based hash notarisation; has since shifted to a PKI-driven "Controlled Capture 2.0" system using Keyfactor's EJBCA. Co-founder of the C2PA standard alongside Adobe, Microsoft, and the BBC. Secures content at capture for insurance, lending, and identity verification use cases.
Pricing models	Enterprise tiers.
Target Audience	Fintech, enterprise operations, media and journalism, device manufacturers.
Strengths	Most mature company in this space. Strong leadership, hardware partnerships, and enterprise traction.
Weaknesses	Has moved away from blockchain. Primarily focused on photos, not video. Enterprise pricing excludes individual users.

Table 2: Truepic Breakdown (Truepic, 2025; Keyfactor, 2025).

ETH Zurich Cryptographic Sensor Chip (ethz.ch)

Basis of Comparison	Description
Key Features and Functionality	Published March 2026 by ETH Zurich researchers. A prototype sensor chip that cryptographically signs images, video, and audio directly at capture. The on-chip signature makes post-capture manipulation nearly impossible without physically attacking the hardware. Signatures can be stored on a public blockchain for independent verification.
Pricing models	N/A (research prototype).
Target Audience	Journalists, social media/news companies, public authorities.
Strengths	Hardware-level attestation is the strongest possible trust anchor. Signing occurs inside the sensor itself, eliminating software-layer attack vectors.
Weaknesses	Requires hardware manufacturer adoption at scale. Not commercially available. No application layer, unusable by a layperson. Requires an entire ecosystem to form before it delivers value to end users.

Table 3: ETH Zurich Chip (Gross, 2026).

1.4.2 Category 2: AI-based Deepfake Detection (Indirect Competitors)

These tools analyse media after creation to determine whether it has been manipulated. They address a different stage of the evidence lifecycle from Bedrock.

Sensity AI (sensity.ai)

Basis of Comparison	Description
Key Features and Functionality	Amsterdam-based, founded as Deeptrace in 2018. Enterprise-grade deepfake detection across video, images, and audio using deep neural networks. Claims 95-98% accuracy on benchmarks. Offers API access, real-time monitoring of 9,000+ sources, forensic reports with confidence scores, and KYC/identity verification integration.
Pricing models	Enterprise tiers.
Target Audience	Law enforcement, media organisations, cybersecurity firms.
Strengths	Multi-modal detection. Traceability of fake media spread.
Weaknesses	Outputs probabilistic confidence scores, not binary certainty. Reactive by nature. Requires constant retraining as generative AI improves.

Table 4: Sensity AI Breakdown (CloudSEK, 2026; SOCRadar, 2025).

Reality Defender (realitydefender.com)

Basis of Comparison	Description
Key Features and Functionality	New York–based. Secured USD 15M Series A. Named top finalist at RSAC 2024 Innovation Sandbox. Multi-model platform detecting deepfakes across video, images, audio, and text. Uses probabilistic detection rather than watermarks or prior authentication. Adopted by public broadcasters in Asia and multinational banks.
Pricing models	Enterprise tiers.
Target Audience	Government, media, financial institutions.
Strengths	Multi-format coverage. Strong institutional traction.
Weaknesses	Probabilistic, not deterministic. Does not establish provenance at creation.

Table 5: Reality Defender Breakdown (CloudSEK, 2026; SOCRadar, 2025).

1.4.3 Category 3: Industry Standards and Coalitions

C2PA/Content Authenticity Initiative (c2pa.org/contentauthenticity.org)

Basis of Comparison	Description
Key Features and Functionality	Open-source technical standard for embedding tamper-evident "Content Credentials" into media files. Uses cryptographic hashing and digital signatures to allow verification of origin, edit history, and AI involvement. Backed by Adobe, Microsoft, and other major technology and media organisations. Functions as a provenance metadata layer where unauthorised alterations break the cryptographic seal.
Pricing models	Open standard (free).
Target Audience	Content creators (varies).
Strengths	Globally standardised provenance framework with hardware-to-software coverage. Backed by the largest technology and media organisations in the world.
Weaknesses	Does not verify truth or fact-check content. Tracks provenance and editing history only, not a capture-and-verify system. Relies on PKI trust models (certificate authorities), anchoring trust in institutional infrastructure. Metadata can be trivially removed by malicious actors, leaving the image appearing "unverified" rather than "tampered with"

Table 6: C2PA and Content Authenticity Initiative Breakdown (C2PA, 2026; TrueScreen, 2026).

1.4.4 Identified Gaps and Bedrock's Differentiation

The analysis that follows is based on the Competitive Analysis Matrix Based and the Gartner Magic Quadrant drawn up in Appendix C. It is clear existing solutions optimise for either technical integrity or practical deployability, but not both. C2PA and Truepic deliver deterministic, provenance-based verification with strong ecosystem adoption, but both sign or verify content at a single moment of time and not across its evidence lifecycle. In fact, STABILITY recently joined the C2PA, lending it provenance-based expertise. Reactive detection tools like Reality Defender and Sensity AI are easier to deploy but produce probabilistic outputs. Hardware approaches like the ETH Zurich chip represent the strongest technical vision but face significant barriers to real-world adoption. This creates four structural gaps in the market:

1. **No deployed solution combines liveness, hashing, and blockchain in a single pipeline.**

ETH Zurich's chip achieves hardware-level signing but has no application layer. Truepic has an application layer but has abandoned blockchain. Bedrock is the only deployed system that fuses liveness verification, continuous real-time hashing, and public blockchain anchoring into a single unbroken pipeline.

2. **No solution targets the legal sector in Singapore or ASEAN.**

Every competitor targets enterprise security, media verification, or insurance workflows. None is built for the courts or designed to address a specific statutory framework. Every competitor identified is headquartered in North America or Europe.

3. **The hardware approach requires ecosystem-wide adoption before it delivers value.**

ETH Zurich's chip needs camera manufacturers to embed it, social media platforms to build verification layers, and an entirely new supply chain to form. Bedrock is a software application. It runs on existing hardware and is live today.

4. **Continuous rolling hash is unique to Bedrock.**

ETH Zurich's chip signs at the moment of capture. Truepic signs at capture (one-time). C2PA embeds provenance at signing. Bedrock's rolling hash chain hashes every video segment continuously during recording. Every next video segment will be hashed, and combined with the previous hash before it.

1.5 Market Research

1.5.1 Total Addressable Market (TAM)

The Total Addressable Market for digital forensics is valued at USD 8.46 billion in 2026 and projected to reach USD 15.37 billion by 2031, reflecting a CAGR of 12.68% (Mordor Intelligence, 2026). This growth is driven by cloud-native forensic investigations, deepfake countermeasures, and the integration of digital forensics within Extended Detection and Response (XDR) platforms, alongside legislated mobile device extraction mandates and sustained public-sector investment.

1.5.2 Serviceable Available Market (SAM)

The Serviceable Available Market focuses on the Asia-Pacific (APAC) region, which is the fastest-growing corridor for digital forensics globally, expanding at a 13.66% CAGR through 2031 (Mordor Intelligence, 2026). Within this region, demand for deterministic integrity tools is rising as agencies contend with the proliferation of synthetic media and the legal necessity for unalterable digital signatures. Bedrock's target SAM within APAC, specifically the litigation support and institutional compliance sectors (considering the LegalTech lens required for this Final Year Project), is estimated at approximately USD 1.2 billion by 2031. This is derived from the global digital forensics market's government and law enforcement segment, which accounts for 36.83% of total market revenue (Mordor Intelligence, 2026), applied to the APAC region's projected share of the global market. If Bedrock can move into the hardware space then the SAM would definitely increase substantially to cover use cases such as CCTV, drone and bodycam footage.

1.5.3 Serviceable Obtainable Market (SOM)

Bedrock's immediate SOM is Singapore itself. The beachhead market is estimated at around USD 25-35 million over the next three years. The Ministry of Law's LIFT Initiative (June 2025) acts as the national catalyst that anchors this estimate. It has introduced dedicated legal-tech consultancy and co-funding to bridge the "change management" gap for small and medium law firms, creating a direct government-backed pipeline for Bedrock's adoption (Ministry of Law, 2025).

1.5.4 Key market drivers

The market is being propelled by five forces: (1) Exponential growth of AI-generated synthetic media, with deepfake volume projected to increase 900% annually (Keepnet Labs, 2026). (2) Increasing regulatory attention globally, including the EU AI Act, Singapore's AI governance framework, and US executive orders on AI safety. (3) Rising cost and frequency of identity fraud; the US identity fraud rate increased from 1.04% in 2021 to 1.68% in 2023 (The Insight Partners, 2026). (4) Growing judicial concern about evidence

integrity, of which Justice Aidan Xu's remarks are one example of a broader global trend. (5) Maturation of feeless blockchain infrastructure, which removes the historical barriers to institutional adoption.

1.5.5 Key market challenges

Adoption faces three headwinds: (1) Courts are unlikely to accept blockchain-anchored evidence without precedent, and precedent requires a first case. Bedrock resolves this by targeting litigants first: when a litigant submits Bedrock-verified evidence, the court is compelled to evaluate it, creating the first opportunity for precedent to form. (2) Well-funded detection companies such as Sensity AI and Reality Defender have established enterprise sales channels and brand recognition in adjacent markets. (3) Blockchain technology remains closely associated with cryptocurrency speculation in institutional perception. GTN's feeless, tokenless architecture eliminates the technical basis for this concern, but the association persists. Partnerships with government organisations such as IMDA and the Judiciary are critical to establishing credibility and creating early adoption momentum.

1.6 User Personas and Segmentation

The design of Bedrock was informed by two primary user personas and one secondary persona (see Appendix A). Each persona represents a distinct use case and set of requirements. These personas were developed during the Planning & Scope phase (Weeks 1-2) and validated through both usability testing (Sprint 3, 7 participants) and UAT (Sprints 4, 8 participants). A summary of their characteristics and behaviours can be found in the table below:

Feature	Jake the Client	Maya the Lawyer	Ron the Data Verification Specialist
User Archetype	Primary User: Litigant	Primary User: Lawyer	Secondary User: Compliance
Tech Literacy	Low (Smartphone-literate)	Intermediate (SaaS-comfortable)	High (Hashing, Metadata, Logs)
Core Motivation	Peace of mind. Evidence admissibility without technical friction.	Winning cases through clear, court-explainable integrity proofs.	Eliminating manual forensic workflows with immutable truths.
Primary Pain Point	Confusing legal/technical jargon and fear of "losing" evidence.	Reliance on slow, expensive experts to prove authenticity.	Incomplete chain-of-custody and late-stage discovery issues.
Solution Usage	Simple "Record & Seal" interface with immediate confirmation.	Uses dashboard for rebuttals and generates "Proof -of-Integrity" Certificates.	Compares hashes against anchored instances and reviews audit trails.
Key Preference	Extremely simple UI. Plain-language explanations.	Human-readable verification reports for court submission.	Exportable technical reports and batch verification capabilities.

Potential Use Cases	Record virtual or in-person meetings. Recording high-stakes in-person events where speed and discretion are needed.	Record resolutions or conflicts. Conferences, depositions, and hearings. Virtual or in-person meetings.	Any video in need of verification. Media organisation verification workflows handling drone, body cam, CCTV footage etc.

Table 7: User persona & segmentation

These three personas, along with sponsor feedback, inspired Bedrock's three form factors (web application, Android APK, and Raspberry Pi CCTV). They collectively represent the spectrum of use cases in which proactive evidence authentication is needed and inform the subsequent user testing. Additional elaboration of how these use cases may arise in each persona, as well as how they go about deploying our solution, can be seen in the user stories found in Appendix B.

2. Proposed Solution: Bedrock Web Application

2.1 Solution Overview & Value Proposition

Bedrock seals audio-visual evidence at the moment of capture. Existing tools attempt to detect manipulation after the video is recorded, whereas Bedrock instead creates a fingerprint of the recording as it is being captured, making any subsequent tampering mathematically provable and pinpointing the exact moment it began.

Bedrock computes a unique digital fingerprint across the media stream in real time, anchoring each fingerprint to a smart contract on Stability Protocol's Global Trust Network (GTN), a cryptoless, public blockchain. Any party can create an authentic, immutable record that can be independently verified at any time. The result is an immutable, independently verifiable chain of custody that exists on a public ledger and requires no trusted intermediary to authenticate. On recording completion, Bedrock generates a Proof of Integrity Certificate (POIC) – a PDF containing the smart contract address, final rolling hash, timestamps, and recording metadata, ready for court submission.

While the individual technologies underlying Bedrock are not new in isolation, no existing solution combines them into a pipeline designed specifically for legal evidence. With that we are able to list out our specific value proposition as follows:

#	Value Proposition	Description
1	**We capture and verify stream legitimacy**	Bedrock guarantees three properties for every recording: (1) it was captured live, not replayed from a file; (2) a physical person operated the camera, not a synthetic source; and (3) no byte has been altered from capture to court submission. We call this a **legitimate stream**.
2	**We architect for prevention, not detection.**	Deepfake detection is reactive and locked in a permanent arms race with generative AI. Bedrock seals authenticity at the point of creation, so its guarantees do not degrade as AI improves.

3	**We make blockchain invisible & scalable**	Bedrock eliminates the most persistent barriers to adopting blockchain technology. By building on a cryptoless, public blockchain and embedding smart contract interactions on the backend layer, we provide a seamless experience for the user. No crypto, no gas fees and no blockchain expertise is needed. The blockchain is invisible to the user but auditable by anyone. This makes our solution easy to use and easily scalable.

Table 8: Bedrock Value Propositions/X-Factors

What Bedrock does not prove

However, a critical distinction must be drawn between what Bedrock proves and what it does not. Bedrock confirms that a recording has not been altered since capture and that the person who initiated it was live. It guarantees the file's authenticity but does not judge the credibility of its content. A witness may record a genuine, untampered video of themselves making a false statement.

2.1.1 How Bedrock Works

The pipeline operates in four sequential steps, followed by an independent verification pathway:

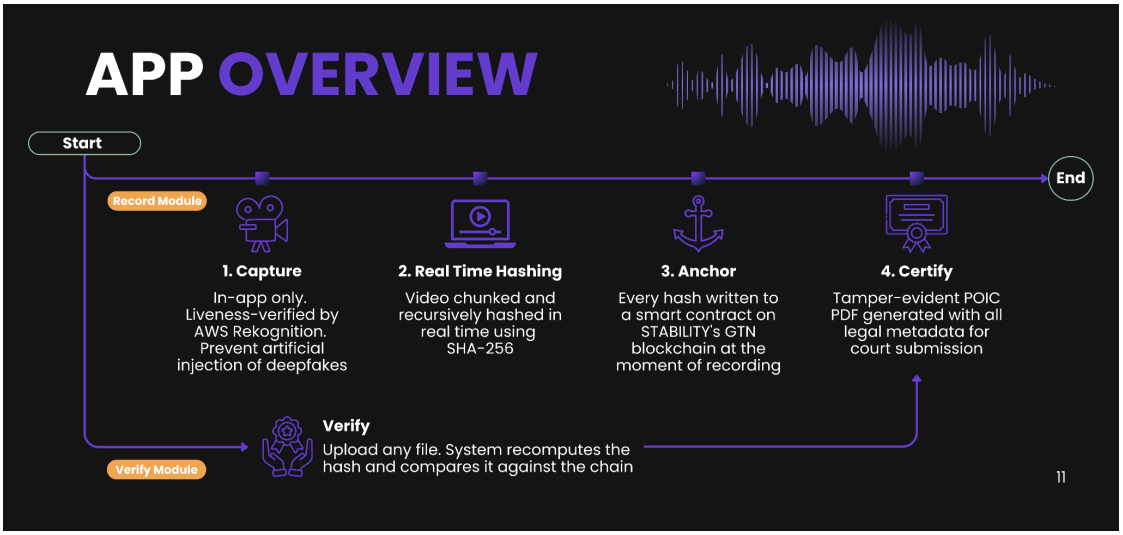


Figure 1: Bedrock Web Application Overview

<u>The Record Module</u>

The Record Module handles the end-to-end evidence capture pipeline in the following steps:

1. **Capture:** Evidence begins recording here. A liveness verification in the Web-Application via AWS Rekognition confirms real human presence through a series of active challenges with randomised colour prompts, preventing spoofing/injections, ensuring that the recording is live. *(Injection prevention is handled via OS/Hardware attestation for CCTV/Phone App use cases)*
2. **Real-Time Hashing (fingerprint):** As the evidence is being recorded, there is real-time cumulative SHA-256 hashing of video stream segments into segments of 1,310,720 bytes (~1.3MB in size). Each segment hash is computed from the current segment combined with the cumulative hash of all previous segments, producing a comprehensive hash-trail. A single byte alteration causes every subsequent hash in the chain to change, making partial tampering impossible to conceal.

3. **Anchor:** Every hash is sent to a queue system and written in real-time to a dedicated smart contract on STABILITY's GTN. This on-chain record is immutable and publicly verifiable. Note: Only the SHA-256 hash of the evidence is written onto the network and not the evidence itself, this is due to PDPA concerns which are highlighted in Section 6.1. SHA-256 is a one-way hash with no feasible way to revert it at the current state of technology.
4. **Certify:** Upon completion, the user is prompted to download the evidence recorded and a POIC. The POIC includes all legal metadata required for court submissions, including the capture timestamp, recording duration, smart contract address and the final cumulative hash.

The Verify Module

The Verify Module allows any party to upload any evidence. By comparing the hash values of the uploaded evidence what is written on-chain, the user can receive a deterministic answer on whether the file is

1. A verified, legitimate video recorded through our platform
2. Tampered video, which was originally recorded through our platform
3. Never recorded through our platform

Together, these two modules form a closed loop: evidence is sealed at the point of creation and verifiable at any point thereafter.

2.2 System Architecture and User Flow

This section describes the high-level architecture, the technology stack across all three form factors, the navigation and user flows for the Record and Verify modules, and the REST API surface that connects the frontend to the backend and blockchain layers.

2.2.1 High-level Architecture

Layer 1: Frontend (Client Device)

The user-facing interface handles video capture, liveness verification, real-time hashing, and evidence package download. For the web application, this is a React single-page application (SPA) served via Google Cloud Run. The website and web app share a single deployment split by hash routing: /#/ serves the marketing website (landing page, instructions, FAQ, contact) while /#/app serves the web application (Record and Verify modules). For the Android APK, the frontend is a native Kotlin application using Jetpack Compose. For the Raspberry Pi CCTV, the "frontend" is a physical interface consisting of a 128×160 LCD status display and a single-button input with hold-duration gestures.

Layer 2: Backend (API & Workers)

The Node.js/Express server handles session management, liveness verification via AWS Rekognition, hash submission queueing via BullMQ/Redis, smart contract deployment, and interaction via ethers.js. The backend enforces a critical security boundary: it owns the liveness session (via AWS Cognito), preventing the frontend from bypassing or forging liveness results. For the Android APK, there is no backend; the app signs transactions on-device and communicates directly with the blockchain. For the Raspberry Pi, a local Python pipeline handles capture and hashing, with a BullMQ worker draining the hash queue to the blockchain.

Layer 3: Blockchain (STABILITY GTN)

Each recording session deploys a dedicated Subcontract through a Factory Contract. Cumulative hashes are written to the Subcontract in real time during recording. The Factory Contract maintains reverse lookups (firstHash/finalHash → subcontractAddress), enabling the Verify module to locate any recording's blockchain record from the file alone without requiring a contract address.

2.2.2 Technology Stack

The technology stack spans four product surfaces (marketing website, web application, Android APK, and Raspberry Pi CCTV) with a shared core hashing algorithm and blockchain integration layer. The table and image below condense the stack by functional layer.

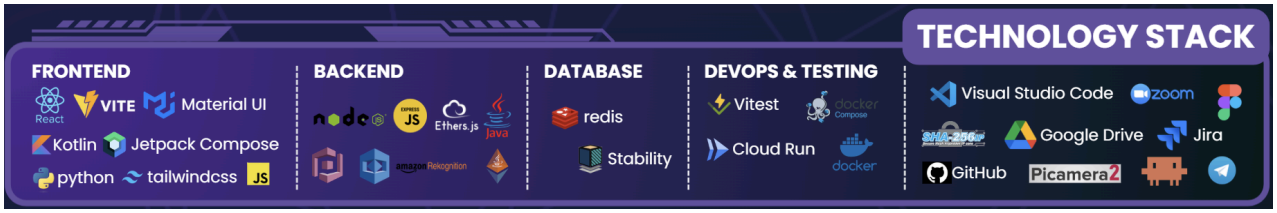


Figure 2: Bedrock Technology Stack

Layer	Platform	Technology
Frontend	Web App	*React 18, Vite 6 (build tool), Tailwind CSS 4, Material-UI 7, Radix UI, Three.js, Recharts, Motion, Lottie React (3D/animations/charts)*
	Mobile App	*Kotlin 2.0.21, Jetpack Compose, Material3, Camera2*
	CCTV	*Pillow, adafruit-circuitpython-rgb-display, adafruit-circuitpython-busdevice, RPi.GPIO*
Backend	Web App	*Node.js 20, Express 4*
	Mobile App	*web3j 4.12.2, Java MessageDigest (native)*
	CCTV	*picamera2, requests, BullMQ (Python)*
Blockchain	Web App	*Ethers.js 6 (interacts with Stability Protocol (ZGT) network), Custom smart contracts (Factory + Subcontract pattern), SHA-256 rolling hash chain per ~1.3 MB video segment*
	Mobile App	
	CCTV	
Cloud & APIs	Web App	*AWS Rekognition FaceLiveness (biometric liveness detection), AWS Amplify + Cognito (guest auth for Rekognition access)*
	Mobile App	*Stability Protocol RPC*
	CCTV	*Stability Protocol RPC, Backend REST API*

Infrastructure & Deployment	Web App	*Redis (Docker) + BullMQ (job queue for sequential blockchain writes), Docker + Docker Compose, Google Cloud Build / Cloud Run / Artifact Registry : CI/CD & hosting*
	Mobile App	*Gradle, Manual APK distribution*
	CCTV	*Redis, systemd, Node.js, dotenv, USB export*
DevOps & Testing	Web App	*Vitest 3*
	Mobile App	*JUnit, Kotlin Coroutines Test*
	CCTV	*Manual Testing with Physical Device and .sh Startup Script*

Table 9: Bedrock Full Technology Stack Breakdown

2.2.3 Navigation Diagram and User Flows

The Record Module

The Record module is the primary user-facing workflow. It guides the user from the landing page through consent, liveness verification, recording, and evidence package download. The flow is designed to be linear and unambiguous. The user cannot skip steps, and the system enforces each prerequisite before proceeding. A navigation diagram of this module can be found below:

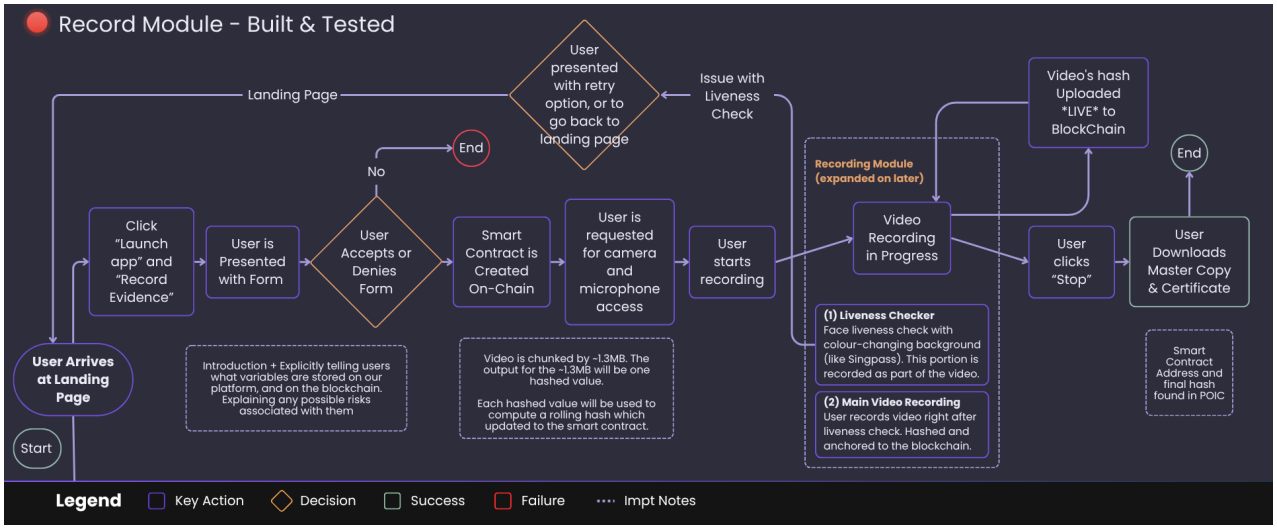


Figure 3: Record Module Navigation Diagram

The flow proceeds as follows:

1. **Landing page.** The user arrives at the Bedrock landing page and navigates to the web application.
2. **Consent and data disclosure.** The user is presented with a consent form (see Appendix D) explicitly stating what data will be collected, what will be stored on the platform, and what will be written to the blockchain. The form explains that blockchain records are permanent and cannot be deleted. The user must accept to proceed; declining returns them to the landing page. A more detailed Privacy Policy (see Appendix S) can be found on the website for the users viewing as well.

3. **Smart contract deployment.** Upon consent, the backend deploys a new Subcontract on GTN via the Factory Contract. A loading screen shows deployment progress. This step creates the on-chain "vault" that will store the recording's hash chain.
4. **Camera and microphone access.** The browser requests camera and microphone permissions. Both are required for recording to proceed.
5. **Liveness verification.** The AWS Rekognition FaceLiveness challenge is then triggered. The user is prompted with a randomised colour-changing background (similar to Singpass's liveness check) that they must follow with their face. This portion is recorded as a part of the video evidence. If liveness passes, recording continues. If it fails, the user is presented with a retry option or the choice to return to the landing page. On failure, all buffered hashes are discarded, and no data reaches the blockchain.
6. **Recording in progress.** The main video recording begins. As the user records, the video stream is processed in real time: encoded video segments are captured via the browser's MediaRecorder API and segmented into ~1.3 MB segments, and each segment is hashed with SHA-256. The rolling cumulative hash is computed (each new hash incorporates all previous hashes) and sent to the backend via HTTP Post. The backend sequentially enqueues each hash to BullMQ to ensure idempotency, and a blockchain worker writes it to the Subcontract on GTN.
7. **Stop and finalise.** The user clicks "Stop." The final hash is submitted and the Subcontract is finalised on-chain . The Factory Contract registry is updated with the first and final hashes, enabling reverse lookups.
8. **Evidence package download.** The success screen displays the video playback, subcontract address, and final hash. The user downloads the video recording and the Proof-of-Integrity Certificate (POIC) as a PDF. The recording uses a descriptive filename, whereby the capture date and time in hours and minutes are automatically appended to the file prefix (e.g. bedrock_evidence_YYYY_MM_DD_HHMM.mp4).

Two-phase recording architecture. A critical design detail is the two-phase buffering during liveness. While liveness verification is in progress (state: LIVENESS_PENDING), video segments are captured and hashed on the frontend but held in a pendingHashRecords buffer rather than being submitted to the backend. If liveness passes, the buffer is flushed. All held hashes are submitted and the phase transitions to LIVE_HASHING. If liveness fails, the buffer is discarded entirely, the recorder stops, and no data reaches the backend or blockchain. This ensures that no evidence is anchored until a real person's presence has been confirmed.

This entire process can be visualised through the user flow diagram found in Appendix G1.

The Verify Module

The Verify module enables any party to upload a video file and determine whether it has been tampered with since recording. A navigation diagram of this module can be found below:

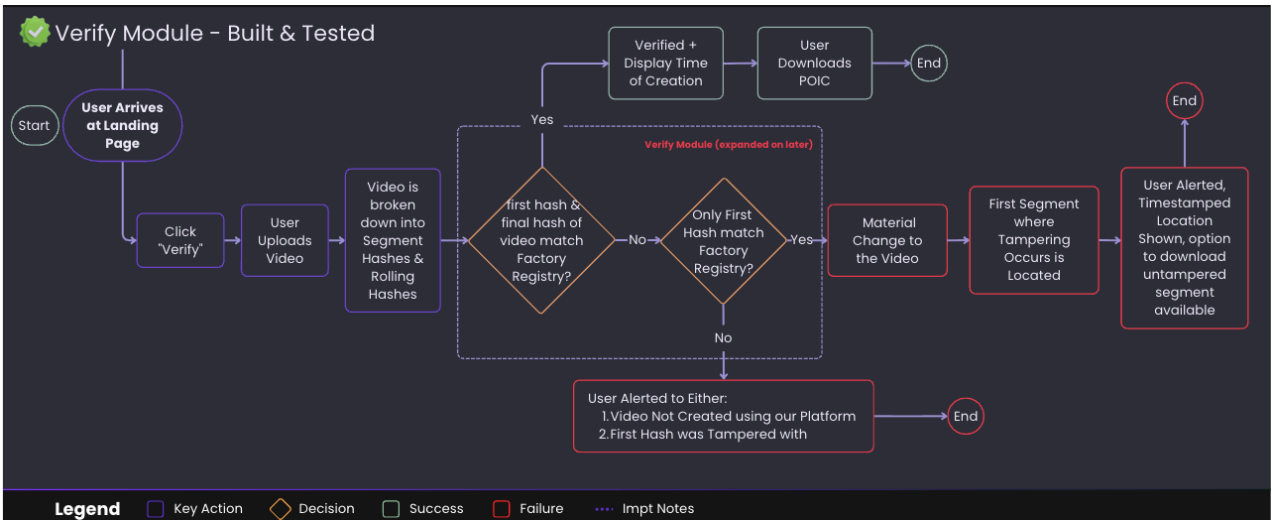


Figure 4: Verify Module Navigation Diagram

The flow proceeds as follows:

1. **Landing page.** The user clicks "Verify" from the landing page, navigating to the Verify module within the web application.
2. **File upload.** The user uploads a video file via drag-and-drop or file selection. No other input is required.
3. **Re-hashing.** The system breaks the uploaded video into the same ~1.3 MB segments used during recording and recomputes the SHA-256 rolling hash chain. This produces a complete set of segment hashes and cumulative rolling hashes.
4. **Blockchain comparison.** The computed hashes are compared against the on-chain record. The system checks for a match between the computed first and final hashes with the Factory Contract's Registry. If both matches, Authentic. Else (if the final hash does not match), will query the chain for the hashtrail and compare segment hashes individually to identify the first tampered segment.
5. **Result determination.** Three outcomes are possible, each with distinct visual treatment:
    a. Authentic: Hashes match. POIC generated. Video unmodified.
    b. Tampered: Hashes diverge. Video tampered. The file was altered starting from [Timestamp].
    c. No Record: Video not recorded through Bedrock, or the first segment was tampered with.

This entire process can be visualised through the user flow diagram found in Appendix G2.

2.3 Record Module Breakdown

The Record Module forms the core of the solution. It was architected with consideration of industry standards - the client device handles the liveness check, recording and hashing while the backend handles the throttling of the hashes to the blockchain. This is to ensure idempotency - that no hash is submitted twice corrupting the chain of hashes. Figure 5 illustrates this architecture as a three-column process flow.

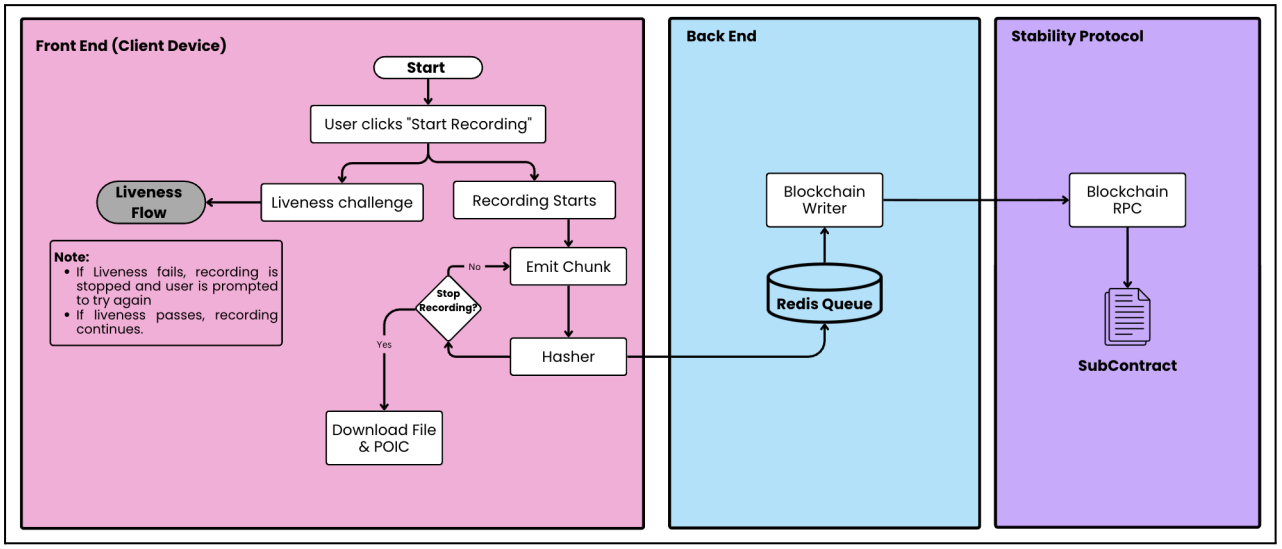


Figure 5: Process Flow Diagram of the Record Module

Frontend Layer: Capture, Hash, and Buffer

Prior to recording, a Subcontract is deployed on chain to store the hash trail. When the user starts recording, two things happen simultaneously – the liveness check and the recording of the video (and the hashing thereof). This was an intentional design choice to allow detection of fraud: if someone were to add a deepfake overlay after the liveness check passes, it would appear in the evidence recording itself.

The purpose of the liveness check is to ensure that the video recorded through the platform is live, meaning no pre-recorded or deepfake spoofing videos. The liveness check is outsourced to AWS Rekognition.

For the recording, the browser's MediaRecorder API captures camera and microphone input in small segments, which accumulate into ~1.3 MB segments. The 1,310,720-byte (≈1.3 MB) segment size is not arbitrary but was tuned to satisfy two constraints simultaneously. Smaller segments produce more per-segment transactions per minute of recording, which can build up the Redis/BullMQ submission queue faster than it de-queues. Additionally, the 1.3MB segment is sufficient to ensure that the recording duration per segment is long enough for AWS Rekognition's liveness verdict to return before the first segment anchors on-chain, so a failed liveness can abort before any identity hash is written to the Factory registry.

Subsequently, these segments are SHA-256 hashed in real-time, building a rolling hash chain that gets submitted to the blockchain. When recording stops, the final partial segment is flushed, and all segments are assembled into a downloadable MP4/WebM file.

This rolling hash chain structure is fundamental to Bedrock's tamper detection because each cumulative hash incorporates every hash before it. Any byte altered anywhere in the file cascades all the way to the

final rolling hash $roll_{N-1}$. Practically, this means verification collapses to a **single** on-chain equality check as comparing our recomputed $roll_{N-1}$ against the finalHash in the Factory registry is O(1), yet cryptographically certifies the entire recording is byte-identical to the original. This is further elaborated in Section 2.4.

Backend Layer: Queue, Sequence, and Write

The backend acts as the secure bridge between the frontend and the blockchain. It runs on Node.js with Express, and is built around two subsystems: the API Gateway and the Blockchain Worker.

The API Gateway receives hash records from the frontend and feeds them into a transaction queue (BullMQ). The queue is set to process one hash at a time, in strict order. This is necessary as the blockchain expects hashes to arrive in sequence, and concurrent writes would corrupt the chain.

The queue is also configured to attempt each write only once. This is a deliberate design decision: if a write fails, it is abandoned rather than retried. Since the blockchain has no deduplication guard, retrying a failed write risks writing the same hash twice, silently corrupting the chain. A failed write is the safer outcome.

A single Blockchain Worker handles all on-chain interactions, with different functions called at different points in the flow:

- **Deploy Contract**: fires at session start, deploying a unique contract for the recording. This behaves as a "database" where the hashes are stored.
- **Update Hash Value**: drains the queue by processing each hash through a four-step pipeline:
    1. **Preflight check**: simulates the transaction without executing it. If it fails, the hash is rejected before any transaction is made.
    2. **Submit**: send the transaction to the blockchain.
    3. **Await receipt**: waits for confirmation.
    4. **Validate**: checks the result against expectations and updates the verification cache.

When the final hash is submitted, the contract is marked as finalised, enabling the Verify module to look it up later.

Blockchain Layer: Immutable Record on GTN

All on-chain interactions are submitted through the STABILITY Protocol ZGT RPC endpoint, the gateway that routes requests to the GTN nodes, using ethers.js. The blockchain layer consists of two smart contract types, both written in Solidity.

Think of Bedrock's smart contracts as a filing system. The Factory Contract is the filing cabinet; each Subcontract is an individual case file. The cabinet's index lets any party find a case file by its contents (the hash) rather than needing to know the drawer. What follows is how that system is implemented. This will be further explained in-depth at Section 2.3.3.

2.3.1 Liveness Verification Feature

Figure 6 illustrates the liveness verification flow across the Frontend, Backend, and AWS Rekognition layers and will be referred to extensively in the upcoming sections.

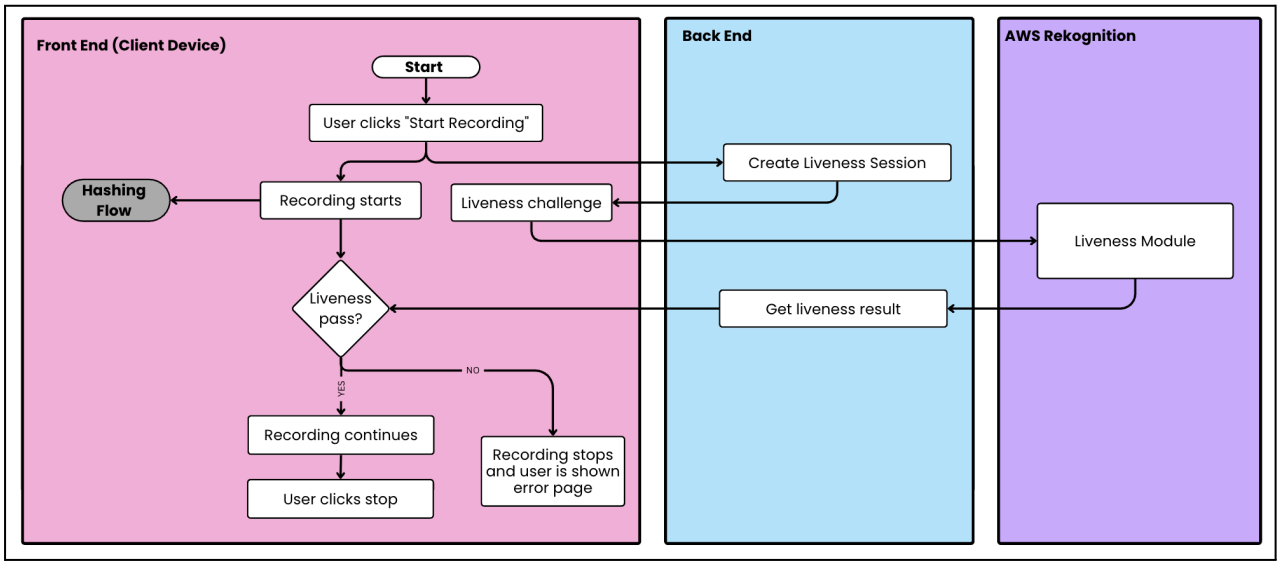


Figure 6: Liveness Verification Process Flow Diagram

Why Liveness Matters

Liveness verification addresses a fundamental question in any evidence capture system: is the person in front of the camera physically present and real? Without it, three categories of attack undermine the system's integrity guarantee:

- **Presentation attacks**: an attacker holds a photo, screen, or mask in front of the camera. The system records what appears to be a real person, but no real person is present.
- **Digital injection attacks**: an attacker bypasses the physical camera entirely, feeding pre-recorded or AI-generated video directly into the capture pipeline via a virtual camera driver or hardware injector. This is the more sophisticated threat as no physical anti-spoofing measure applies.
- **Mid-session substitution**: an attacker passes the liveness check with their real face, then substitutes the feed with synthetic content for the remainder of the recording.

Without addressing all three, a blockchain-anchored hash chain proves only that a file passed through the system, but not that what was recorded was genuine. Bedrock's liveness implementation addresses the first two. Mid-session substitution is a known limitation, but is mitigated by recording the liveness segment allowing for manual verification.

The Migration from Luxand.cloud to AWS Rekognition

At midterm, Bedrock used Luxand.cloud's passive liveness detection, the most frictionless approach, requiring no user action. Frames were captured and sent to Luxand's endpoint, returning per-frame scores and a binary result over a 3-second window. This raised a valid concern: passive liveness analyses static image characteristics without requiring any user response, meaning a sufficiently high-quality pre-recorded video of a real person can pass the check. This is a documented limitation of all passive liveness systems, and is particularly unsuitable for a system intended for legal evidence.

The team determined passive liveness was insufficient and migrated entirely to Amazon Rekognition Face Liveness. The migration touched three layers:

- **Frontend** – Luxand API calls were replaced with the AWS Amplify FaceLivenessDetector component, which handles the full challenge UI in the browser.
- **Backend** – the Luxand proxy endpoint was deprecated and replaced with two new endpoints: one to create a liveness session, and one to retrieve the result.
- **Security model** – session-based ownership was introduced, detailed in the section below.

Active Liveness with Randomised Colour Challenges

AWS Rekognition Face Liveness uses the FaceMovementAndLightChallenge mode, combining facial movement with a randomised colour sequence for the highest accuracy.

The challenge runs in two stages. First, the user moves their face into an on-screen oval and holds still – defeating static presentation attacks, as a photograph cannot respond to a positional instruction. Second, the device screen flashes a sequence of randomised coloured lights, and the user's face must reflect these colours in the expected pattern. Because the colour sequence is generated randomly at the moment of the challenge, a pre-recorded video cannot produce the correct reflections, and a virtual camera injecting synthetic video has no prior knowledge of the sequence.

The check produces a confidence score between 0 and 100. Bedrock requires a score above 90 to pass – a threshold higher than a typical consumer login flow, reflecting the elevated stakes of legal evidence.

Session-Based Architecture with AWS Cognito

A core design principle is that the backend owns the liveness session. The frontend cannot create, manage, or forge one independently.

When the user clicks "Start Recording," the frontend requests a session from the backend, which calls AWS Rekognition to generate a unique session ID. This ID is passed to the frontend's Amplify component, which handles the challenge and streams the biometric video directly to AWS – bypassing the backend entirely. This direct streaming means the backend never handles raw biometric data, reducing both the attack surface and privacy exposure.

Once the challenge completes, the frontend requests the result from the backend, which retrieves it directly from AWS using the session ID. The backend makes the authoritative pass/fail determination – not the frontend.

This architecture prevents two specific attack vectors: a malicious frontend cannot forge a session ID and claim liveness passed, and session creation being gated by the backend prevents clients from probing AWS Rekognition directly.

Guest credentials for the Amplify component are provisioned through AWS Cognito Identity Pools, allowing the frontend to stream to AWS without requiring the user to create an account. This preserves Bedrock's anonymous-by-default privacy model.

2.3.2 Real-Time Hashing Algorithm

Figure 7 illustrates the two-track hashing architecture and will be referred to throughout this section.

Overview

SHA-256 hashing produces a unique digital fingerprint for any given data. If there is even a slight change, the output changes entirely. Hashing a completed video can confirm whether it was altered, but this traditional approach is reactive: it cannot protect footage while recording is still in progress, and it cannot pinpoint where a modification occurred.

Bedrock addresses this by hashing the video in real-time across ~1.3 MB segments, building a rolling chain of cryptographic dependencies as the recording happens. Any tampering breaks the chain at the exact point of modification.

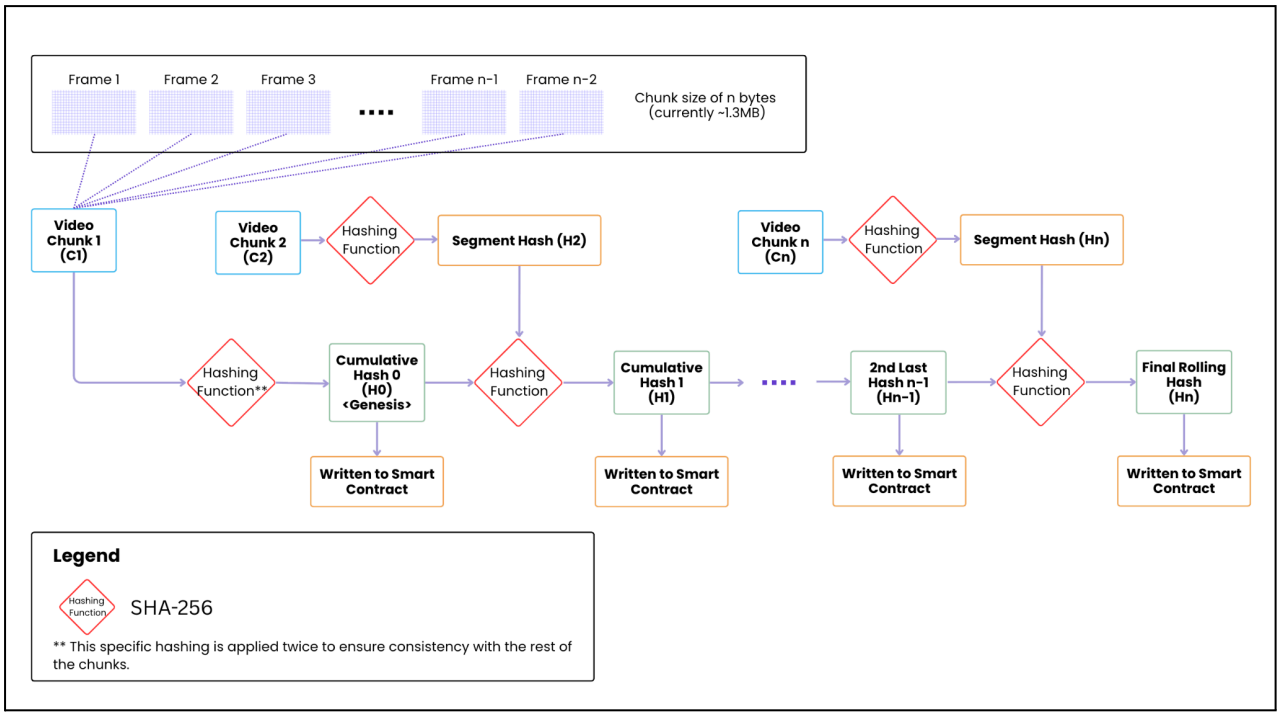


Figure 7: Hashing Algorithm Process Flow Diagram

Data Segmentation

The browser's MediaRecorder API generates a continuous stream of encoded video data. Bedrock partitions this stream into fixed ~1.3 MB segments. Once a segment reaches this threshold, it is passed to the hashing pipeline. The final segment is zero-padded to exactly ~1.3 MB to ensure consistent input regardless of where the recording ends.

The ~1.3 MB segment size was chosen to balance three factors:

- **Tamper detection** – at standard bitrates, ~1.3 MB represents roughly 4 seconds of video, enabling precise localisation of edits.
- **Blockchain efficiency** – it regulates the frequency of write operations to the ledger.
- **System performance** – it minimises computational overhead during real-time hashing.

The Two-Track Hashing Architecture

As shown in Figure 7, the algorithm runs two parallel tracks for every segment:

- **Segment hash (segHash)** – a SHA-256 fingerprint of the raw segment bytes alone.
    - *note: only the last segment will be padded with zero bytes to fill it up to ~1.3MB*
- **Cumulative rolling hash** – a SHA-256 fingerprint computed from the previous rolling hash combined with the current segment hash.

For each segment hash calculation:

$$segHash_i = SHA\text{–}256(segment_i)$$

For last segment hash calculation:

$$segHash_{N-1} = SHA\text{–}256(paddedSegment_{N-1})$$

For the genesis segment (segment 0):

$$roll_0 = SHA\text{–}256(segHash_0)$$

For every subsequent segment:

$$roll_i = SHA\text{–}256(roll_{i-1} + segHash_i)$$

This creates a continuous chain of dependency from the first second of the recording to the last. Each cumulative hash encodes the entire history of everything recorded before it – altering any single segment causes every subsequent cumulative hash to change, cascading forward through the entire chain.

Strengths of Rolling Chain and Blockchain Anchoring of each Cumulative Hash

The rolling chain defeats three specific attack scenarios:

- **Single-frame edit**: altering any content within a segment changes its segment hash, which cascades through all subsequent cumulative hashes. The system detects and localises the exact segment.
- **Segment deletion or insertion**: removing or adding a segment changes the total entry count. Even if an attacker recomputes all subsequent hashes, the total will not match the Subcontract's record, and the first and final hashes will not match the Factory Contract's registry.
- **Segment reordering**: swapping segments changes the cumulative hashes at both positions and all positions downstream. The chain is order-dependent by construction.

A critical design decision is that every cumulative hash is written to the blockchain individually, not just the final hash. This has two consequences:

- **Real-time protection**: each segment is anchored within seconds of being recorded, leaving no window of vulnerability between recording and protection.
- **Segment-level tamper localisation**: during verification, the system compares the recomputed chain against the on-chain record position by position, identifying the exact segment where divergence first occurs.

The trade-off is transaction volume. A 10-minute recording produces approximately 150 segments, requiring 150 blockchain transactions. On conventional blockchains, transaction fees would make this economically unviable. On STABILITY's GTN, all transactions are gasless, eliminating this cost entirely. Per-hash anchoring is therefore only practical because of GTN – a fundamental technology enabler that allows Bedrock to scale in a way competitors cannot.

Cross-Platform Hash Consistency

Bedrock is deployed across three products, each using a different SHA-256 implementation:

Platform	Implementation	Language	Library
Web App (Record)	WebCrypto API	JavaScript	Native browser API (crypto.subtle.digest)
Web App (Verify)	Node.js crypto	JavaScript	Native Node.js module (crypto.createHash)
Android APK	Java MessageDigest	Kotlin	java.security.MessageDigest
Raspberry Pi CCTV	Python hashlib	Python	hashlib.sha256()

Table 10: SHA-256 Implementations Across Platforms

Discrepancies can arise from differences in how platforms handle byte encoding, padding, and endianness. Two measures ensure consistency across all implementations:

- All platforms operate on **raw byte arrays** – the hash is computed on the raw binary content of each segment, not on any string representation.
- The **segment size is defined as a shared constant** across all platforms. The final segment is zero-padded to exactly ~1.3 MB on every platform, ensuring padding logic does not introduce variation.

2.3.3 Smart Contract Architecture

Overview

Bedrock uses a Factory–Subcontract pattern on STABILITY's GTN (Solidity ^0.8.24). The Factory deploys one Subcontract per recording session and acts as a registry; each Subcontract stores that recording's cumulative hash trail and syncs the registry only on its first and final hashes.

The Factory Contract

The Factory Contract does three things.

1. Deploys a dedicated Subcontract per recording session on demand.
2. Maintains forward and reverse-lookup mappings on a registry *(firstHash / finalHash ↔ subcontract address)* so any subcontract can be located from hashes recomputed at verification time.
3. Marks each deployed subcontract *isAuthorizedChild = true*, gating registry writes so only Factory-deployed Subcontracts can update it.

The Subcontract

A per-evidence contract owned by our deployer EOA (*owner = msg.sender*), tracking four state variables:

Variable	Set When
currentHash	updated on every *addCumulativeHash* call
totalEntries	incremented on every call
isInitialized	Genesis hash is written *(isFirst = true)*
isFinalized	Final hash is written *(isFinal = true)*

Table 11: Subcontract Variables

Storing only first/final hashes in the registry gives O(1) lookup from either anchor hash to the subcontract address, so users never need to know their subcontract address to verify authentic evidence.

For further verification, the full hash trail is then retrieved via paginated HashRecorded event queries scoped to that subcontract and bounded by its deploymentBlock – O(N) in trail length, but with a tight block range and address filter that keep it to a few RPC round-trips even for long recordings.

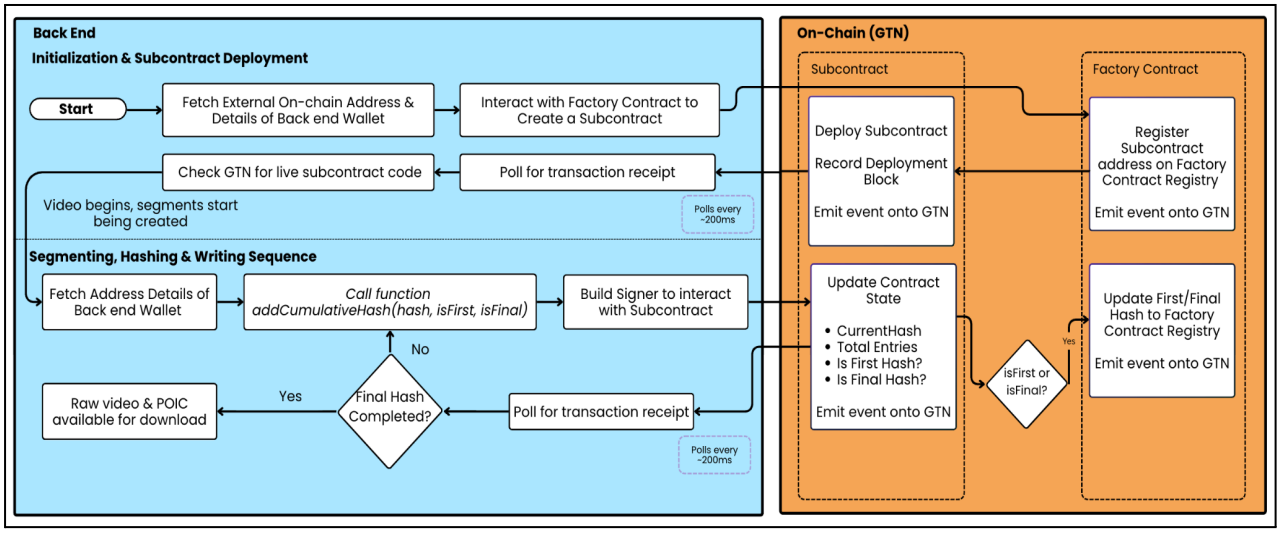


Figure 8: Smart Contract Interaction Diagram

Initialization & Subcontract Deployment

On "Start Recording", the frontend calls *POST /api/subcontract/deploy.* The backend builds an ethers v6.9 signer from PRIVATE_KEY in the environment variables against GTN's RPC, then calls *Factory.createSubcontract()*.

In a single transaction the Factory;

1. sets owner = msg.sender
2. registers the new subcontract address
3. marks it *isAuthorizedChild = true* (enables subcontract to edit the factory contract registry)
4. stores the deployment block and emits *SubcontractCreated* event

The backend polls for the receipt at 200 ms intervals, parses *SubcontractCreated* to extract the subcontract address, then calls *provider.getCode(addr)* to confirm bytecode is synced (guards against a known GTN propagation lag flagged to STABILITY).

Segmenting, Hashing & Writing Sequence

As recording proceeds, the hashing pipeline emits cumulative segment hashes onto BullMQ. The worker then drains the queue sequentially (concurrency=1) and calls *Subcontract.addCumulativeHash(hash, isFinal, isFirst)*.

The Provider and signer are built once at module load; a per-address contract cache reuses ethers. Contract instances for each subsequent *Subcontract.addCumulativeHash()* call.

On-chain effects per call:

- *currentHash* ← new value; totalEntries++
- If *isFirst = true* → set *isInitialized,* emit *Initialized(hash)*
- If *isFinal = true* → set *isFinalized*, emit Finalized(hash); subsequent calls will revert
- Always emit *HashRecorded(previousHash, newHash, sequenceNumber)* event

The BullMQ queue is also configured to *attempts=1* per job. This is deliberate The Subcontract has no dedup guard for middle entries, so retrying a write that already landed risks double-writing and corrupting the chain. A failed-and-abandoned recording is a safer failure mode than a silently corrupted chain producing a false "Tampered" verdict on authentic evidence.

Registry Anchor (first & final only)

When *isFirst || isFinal*, the Subcontract calls *Factory.updateRegistry()* directly on-chain (gated by *require(isAuthorizedChild[msg.sender])*). The Factory updates its forward map (*registry[subcontract] = {firstHash, finalHash})* and reverse maps *(firstHashToSubcontract, finalHashToSubcontract)*, then emits *RegistryUpdated*. Middle entries skip the Factory entirely; the registry is anchored only at recording start and end.

Receipt Validation & Completion

After each write, the worker calls *awaitReceipt(txHash)*, parses the *HashRecorded* event, and verifies the emitted newHash matches what was submitted. A mismatch fails the job.

On the final hash, the worker writes *verify:<subcontractAddress> = { status: "complete" }* to Redis. The frontend polls *GET /api/subcontract/:address/status* and transitions to the success screen when it flips to complete.

Verification

Verification is user-initiated, not automatic. On *POST /api/verify*, the backend queries everything from the GTN. It starts with an O(1) anchor lookup: the recomputed firstHash and finalHash from the uploaded file resolve directly to the subcontract address through *Factory.registry(subcontractAddress)* and its reverse maps. If both anchors match, the file is authentic. No further GTN reads needed.

If the finalHash doesn't match, tampering happened somewhere in the segment stream, so the backend pulls the full hash-trail to find where. It queries *HashRecorded* events on the Subcontract, scoped to its

address and bounded by deploymentBlock → latest. This query is O(N) in trail length, but the address filter and tight block range keep it to a handful of eth_getLogs round-trips in practice.

2.4 Verify Module Breakdown

The Verify module answers a single question: **is this video file identical to what was originally recorded through Bedrock?** The answer is binary - yes or no. There are no confidence scores, no probabilistic judgments, and no human interpretation needed.

Any party can run a verification provided they have the video they want to verify. By default, the system locates the corresponding blockchain record automatically from the file's content. No account, login, or prior knowledge of the recording is needed.

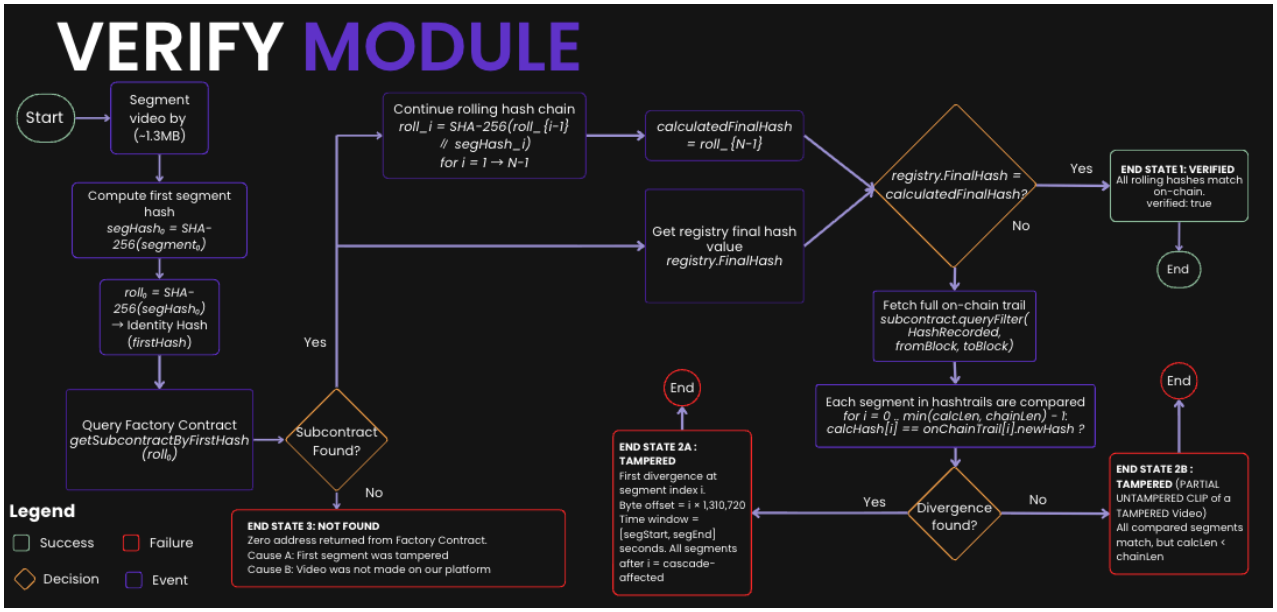


Figure 9: Detailed Process Flow Diagram of the Verify Module

2.4.1 Four-Phase Verification Process

The Verify module executes a four-phase pipeline when a user uploads a video file via POST */api/verify*.

Phase 1: Ingest and Compute the Genesis Hash

The uploaded video is read as a raw byte buffer. The system takes the first 1,310,720 bytes (~1.3 MB), zero-pads if the file is shorter than one full segment, and computes the genesis hash. This genesis hash is computed by the same algorithm as above and serves as a lookup key that identifies which subcontract it belongs to. We do this by $getSubcontractByFirstHash(genesisHash)$. This mapping was stored during the live recording, i.e., when the first hash was written to the Subcontract, the Subcontract called $Factory.updateRegistry()$, which stored the $firstHash \rightarrow subcontractAddress$ association.

If the Factory Contract returns a valid address, verification proceeds against that Subcontract. If not, the pipeline terminates at End State 3: **Not Found** .

Phase 2: Re-Hash the Entire File

The system computes the full rolling hash chain from the uploaded file, following the same algorithm used during recording (Section 2.3.2). This output is an ordered list of rolling hashes $[roll_{0,}\ roll_{1,}\ldots roll_{N-1}]$, with the last value being the $calculatedFinalHash$.

Phase 3: Compare calculated final hash against the Factory registry

Using the subcontract address from Phase 1, the system first confirms the Subcontract is an authorised child of the Factory contract via $Factory.\ isAuthorizedChild(SubcontractAddress)$. If not, verification aborts before any trail is read.

The system then reads the Subcontract's state (totalEntries, isInitialized, isFinalized, deploymentBlock, owner) and the Factory's registry entry for this subcontract, which stores $\{firstHash,\ finalHash\}$.

Once completed, the system compares the *calculatedFinalHash* from Phase 2 against the Factory Register's stored *finalHash*. Because rolling hashes chain every prior segment, equality here cryptographically implies the entire trail matches. If they match, the pipeline terminates at **End State 1: Verified.**

Phase 4: Position-By-Position Trail Comparison

If the values do not match, (if final hashes differ), the system queries the full *HashRecorded* event trail: $subcontract.\ queryFilter(HashRecorded,\ fromBlock,\ toBlock)$ from *deploymentBlock* to the latest block. The events are sorted by sequence to reconstruct the complete on-chain trail.

The system aligns the two trails and compares them element by element:

$For\ each\ position\ i\ from\ 0\ to\ min(calcLen,\ chainLen)\ -\ 1$:

does $calculatedHash[i]$ equal $onChainTrail[i].newHash$?

If any position diverges, the pipeline terminates at **End State 2A: Tampered**, with the first divergence point identified.

If all compared positions match but the calculated trail is shorter than the on-chain trail ($calcLen\ <\ chainLen$), the pipeline terminates at **End State 2B: Partial Clip from a Verified Recording** (which has been tampered due to the video being cut).

2.4.2 Three Terminal States

The Verify module produces exactly one of three deterministic outcomes. End State 2 has two sub-states, 2A and 2B, both originating from tampered videos. Each outcome has a distinct meaning, distinct causes, and a distinct user-facing presentation.

End State 1: Verified (Authentic)

The final hash values match and thus the video is untampered. The file is confirmed as practically identical to what was originally recorded through Bedrock. Not a single byte has been altered since the moment of capture.

When this state is reached, the system displays the verification result with the recording's time of creation (extracted from the *Initialised* event timestamp) and offers the user a downloadable POIC.

End State 2A: Tampered

The calculated trail and the on-chain trail diverge at one or more positions after the first genesis. The system identifies the first divergence point (i.e., the lowest index i where $calculatedHash[i] \neq onChainTrail[i].newHash$) and reports it to the user:

- Segment index: The exact segment number where tampering was first detected
- Byte offset: $i \times$ ~1.3MB the exact byte position in the file where the alteration begins
- Time window: The estimated time range in the video corresponding to the affected segment, calculated from the segment index and the recording's bitrate.
- Cascade notation: All segments after index i are flagged as "cascade-affected". Their hashes also diverge due to the rolling chain structure (each subsequent cumulative hash depends on the corrupted hash at position i), at this moment, we cannot tell if there was more tampering in subsequent segments but we can pinpoint when the first-instance of tampering occurred at position i.

Future Improvement

$segHash_i$ is independent per-segment - it only depends on that segment's bytes. At a divergence at position i, we can substitute the $onChainRolling_i$ into the corrupted $segHash_i$ value and recompute the hash trail for subsequent segments to produce a per-segment tampering map instead of a single divergence point followed by cascade.

The Tampered state confirms that: (1) the file was originally recorded through Bedrock (the first hash was found in the Factory registry), (2) the file has been modified since recording.

When tampering is detected, the user can play back the segment where the first divergence occurred. For all untampered segments before tampering begins, the user can download the untampered segments of the evidence.

End State 2B: Partial Clip from a Verified Recording

If a user downloads untampered segments (from End State 2A) or already has a partial clip from a verified recording, they can feed it back into the Verify module. The system re-hashes it, compares each of the uploaded file's X segments against the first X entries of the on-chain trail, and confirms all X segments match. The partial clip was authentic-but-still tampered due to the original legitimate video being cut.

This state is reached when the file is a byte-prefix of the original recording: the firstHash lookup succeeds, the $calculatedFinalHash$ differs from the Factory's stored «, but **every** segment of the calculated trail matches the corresponding segment of the on-chain trail up to position $calcLen - 1$. Because $calcLen < chainLen$, the file is an untampered segment of the video with authentic bytes, but incomplete.

End State 3: Not Found

The Factory Contract returned a zero address for the computed $firstHash$ which means one of two things.

Cause A: No Subcontract was ever registered with this identity hash and thus the video was not recorded through Bedrock.

Cause B: The first segment was tampered with. Because Bedrock identifies a recording by the hash of its first segment, modifying that segment changes the lookup key. The Factory Contract can no longer find the record, and the result is identical to Cause A. This means Bedrock cannot distinguish a forged video from a tampered-first-segment legitimate video. The "Future Improvement" below addresses this gap.

2.5 The Evidence Package

When a recording completes successfully, Bedrock produces the video recording as an mp4 and a POIC as a PDF (see Appendix P). It is a one-page PDF document that serves as a self-contained attestation of the recording's information and metadata. The certificate contains the following fields: certificate header, description, timestamp of when the recording was captured, length of the recording, on-chain smart contract address where the hash chain is stored, final rolling SHA-256 hash of the entire recording and the generation timestamp. The POIC is designed to be comprehensible to a non-technical reader while containing the precise technical identifiers needed for independent verification on a blockchain explorer or verification on the Bedrock application.

2.6 Use Cases Built Beyond the Web App

The web application is Bedrock's primary POC but the core engine is platform-agnostic by design. To demonstrate this, the team deployed Bedrock across two additional form factors: a native Android APK and a Raspberry Pi CCTV hardware prototype. They are proof that the same fundamental pipeline works on fundamentally different platforms as seen below.

Platform	Description	Use Cases
Web Application (primary POC and future MVP)	Full-featured POC with liveness, recording, verification, POIC generation, and chain-of-custody audit logging.	Record resolutions or conflicts. Record conferences, depositions, and hearings. Record virtual or in-person meetings. Media organisation verification workflows.
Android APK (mobile prototype)	Mobile-first evidence capture for outward-facing scenarios. Liveness removed by design.	Recording of high-stakes in-person events where speed and discretion are needed. e.g. Investigative journalism, war & conflict, recording crimes, etc.
Raspberry Pi CCTV (hardware prototype)	Standalone embedded device with Camera Module v2.1. Hardware-level attestation. Records, hashes, and	Drone and security footage. Law enforcement bodycam footage.

	anchors to GTN in real time. Outputs master file to USB. Verification done on the web application.	
And many other possibilities…		

Table 12: Cross Functional Applications of the same Core Bedrock Engine

The Removal for Liveness check for the Phone and Raspberry Pi Prototypes

As mentioned in Section 2.3.1, on a browser, the navigator.mediaDevices.getUserMedia() API has no way to distinguish a real camera feed from a software-injected one. Any user can install a virtual-camera driver (OBS, ManyCam, XSplit VCam) that presents a pre-recorded MP4 or a real-time deepfake as a "camera device" to the browser. The browser's security model treats the OS's camera list as authoritative - whatever the OS says is a camera, is a camera. This is the reason the web flow outsources a liveness challenge to AWS Rekognition.

The Android APK and Raspberry Pi CCTV prototypes operate in more constrained channels where virtual-camera injection is materially harder than in the browser. On Android, the Camera2 API binds to the OS's camera HAL and does not enumerate arbitrary user-space virtual-camera drivers in the way desktop browsers do; an attacker seeking to inject a synthetic feed typically needs root access, a custom ROM, or a rebuilt HAL. On the Raspberry Pi, the CSI-attached camera module is physically wired to the board and the device is provisioned with Bedrock as its only user-facing application, leaving no general-purpose OS surface on which an end-user could install virtual-camera software. Hence, in both cases, the trust assumption shifts from "the camera stream is not software-tampered" (browser model) to "the physical device has not been compromised at the OS or hardware level" (hardware-rooted model).

These are narrower attack surfaces than the browser, but they are not closed. A rooted Android device, a compromised supply chain, or physical tampering with a Raspberry Pi deployment all remain viable paths for a determined adversary, and these residual exposures are captured in Risk #3 of the Risk Register Section 6.2.2. Future iterations could layer on context-aware attestation - device integrity checks, GPS and telemetry correlation, and signed attestation tokens from the OS.

2.6.1 Android APK

The Android APK was built for scenarios where evidence must be captured in the field quickly and discreetly. The camera faces outward at the event being documented rather than at the user.
The APK is a native Kotlin application using Jetpack Compose for the UI, Camera2 for camera capture, and Material3 for the design system. Critically, the APK has no backend server and it signs blockchain transactions directly on-device using web3j 4.12.2, communicating with the GTN RPC endpoint without any intermediary. The private key for transaction signing is injected at compile time via local.properties/BuildConfig. Hashing uses Java's native MessageDigest class for SHA-256, producing identical outputs to the web application and operating in the same way.

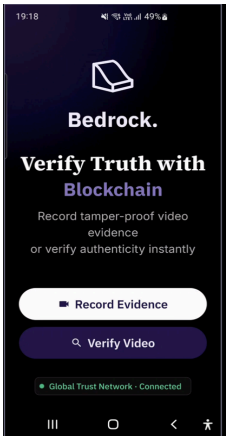


Figure 10: Bedrock Mobile Application

2.6.2 Raspberry Pi CCTV

The Raspberry Pi CCTV prototype represents the most ambitious form factor, a standalone embedded device that records, hashes, and anchors evidence to the GTN.



Figure 11: Raspberry Pi CCTV Prototype being Modelled and Built using CAD Software and 3D Printers at SMU Makerspace

The Raspberry Pi CCTV prototype was developed iteratively across two sprints, built on a Raspberry Pi 4B. Unlike the web application, which asserts user integrity through liveness detection, the CCTV prototype serves as a hardware attestation of the technology. Verification is performed externally via the web application's Verify module. This is appropriate for fixed-installation surveillance scenarios where the camera's physical security can be assured and is an ambitious step made to fulfil one of the most sought after delighter of the sponsor. With this, STABILITY now has a working POC to push for Bedrock applications across different hardware applications like drone and body cam footage.

3. Project Management and Execution

3.1 Project Phases and Sprint Structure

3.1.1 Methodology: Agile (Scrum)

The project adopted an Agile/Scrum methodology to enable rapid, iterative development suited to a zero-to-one product. As the solution space and sponsor requirements evolved throughout the project, a plan-driven approach like Waterfall would have locked the team into early architectural decisions requiring costly rework. Scrum's sprint-based cadence allowed new requirements to be absorbed at natural breakpoints (fortnightly meetings with the sponsor and supervisor) rather than treated as scope creep. Its emphasis on shippable increments ensured a demonstrable build was always available, which was critical for gathering feedback at both the midterm and final presentations.

3.1.2 Four Project Phases

The project was structured into four sequential phases spanning the full academic term.

Phase 1: Initiation (Pre-term, Weeks 0-1)

Before the term began, the team established foundational project governance. This phase produced the Project Charter (defining the project's purpose, scope boundaries, key stakeholders, and success criteria - see Appendix H), the Business Case , the Stakeholder Register (see Appendix I), and the initial Risk Register (see Appendix J). The team also conducted preliminary research into the deepfake landscape, Singapore's Evidence Act framework, and STABILITY's GTN blockchain infrastructure.

Phase 2: Planning & Scope (Weeks 1-2)

The team translated the Project Charter into deliverable components through a Work Breakdown Structure (WBS - see Appendix K), across frontend, backend, blockchain, testing, and deliverables workstreams. The RACI Matrix (see Appendix L) was established, assigning Responsible, Accountable, Consulted, and Informed roles for each major deliverable. User personas were developed as well (see Section 1.6 for more details). Research outputs included competitor analysis, market research, technology research (evaluating blockchain options, liveness verification providers, and hashing strategies), and a PDPA (Personal Data Protection Act) compliance assessment (very critical) was done to inform the application's design. High-fidelity UI builds were produced on Figma before any frontend development began, ensuring that the UX was designed and validated before code was written.

Phase 3: Execution & Development (Sprints 1 to 5, Weeks 3-12)

The core development phase, structured as five two-week sprints. Each sprint followed the standard Scrum cadence of sprint planning at the start, daily standups throughout (conducted via Telegram) and weekly huddles through zoom.

Sprint reviews were conducted at the end of every two weeks and documented as formal Sprint Review documents that captured what was delivered, what was deferred, and what was learned. Sprint reviews were done fortnightly (internally) and the outcomes shared with the sponsor and our supervisor during our progress meetings.

Sprint	Weeks	Key Milestones	Key Deliverables
Sprint 1	3–4	Architecture finalised, Figma prototypes approved	Frontend scaffold (React + Vite) and high fidelity prototypes. Backend scaffold (Express + BullMQ), Smart contract deployment, Liveness integration (Luxand.cloud) and Hashing algorithm process flow diagrams.
Sprint 2	5-6	Record Module functional end-to-end	Video capture + real-time hashing + blockchain anchoring working, POIC generation (initial).
Sprint 3	7-8	Usability testing (7 participants), Midterm presentation	Usability testing sessions and report, Midterm deck and demo.

Sprint 4	9-10	Liveness migration, Verify Module built, UAT (8 participants), Plan Delighters	AWS Rekognition replaces Luxand.cloud, Verify Module fully integrated, Preparation and planning for Android APK and Raspberry Pi CCTV complete.
Sprint 5	11-12	Delighters developed and deployed	UAT sessions and report completed, Android APK prototype, Raspberry Pi CCTV prototype, Final UAT and bug fixes implemented, Full frontend integration

Table 13: Sprint Timeline with Key Milestones

Phase 4: Closing (Weeks 13-14)

The final phase focused on deliverable production which was the final academic report, the final presentation deck and script, the one-minute elevator pitch/trailer, the booth poster and run through, the technical brief for guests attending the final presentation, and the sponsor handover pack.

Lessons learned were captured in a Lessons Learned Register (see Appendix M) documenting what worked well, what could be improved, and recommendations for future teams working on Bedrock or blockchain-based LegalTech as a whole.

3.2 Task Management and Progress Tracking

3.2.1 Jira: Kanban Board for Centralised Task Management

All development tasks were tracked in Jira using a Kanban board with three columns: To Do, In Progress, and Done. Before a task could move to "In Progress", it required three properties: acceptance criteria (a clear definition of "done"), an assignee, and a sprint tag.

The Kanban board was used actively during weekly standups to surface blockers in real time. When a task remained in "In Progress" for more than three consecutive standups, it was flagged as a potential blocker and escalated to the sprint review for resolution. This was done either through pair programming, scope adjustment, or task reassignment.

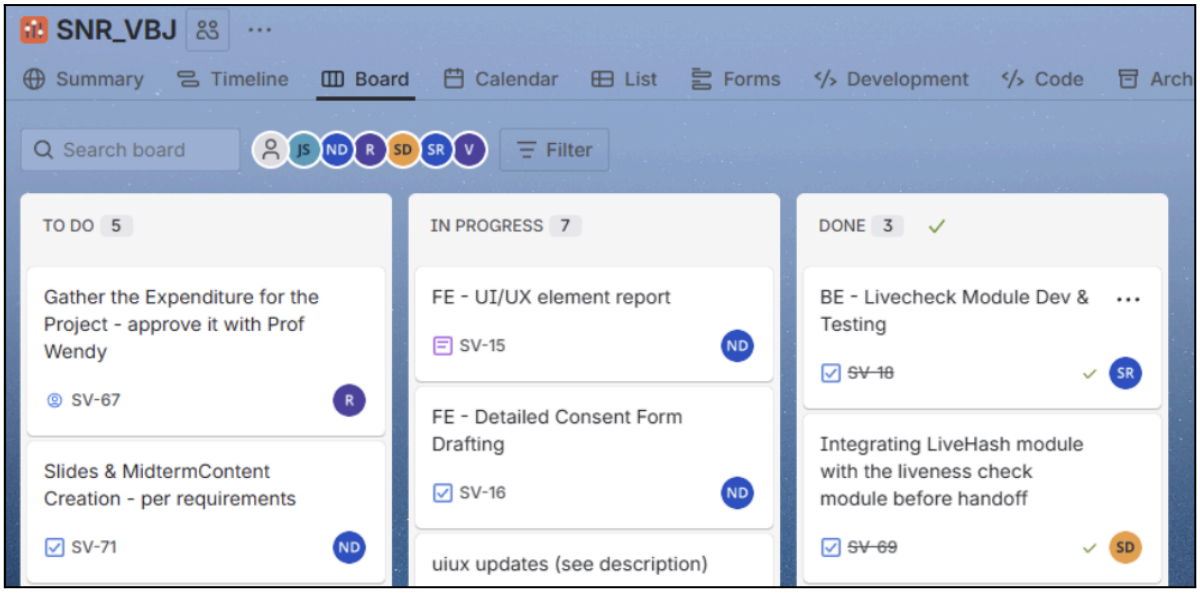


Figure 12: Jira Kanban Board Screenshot

3.2.2 Jira: Gantt Chart and Sprint Timeline

The Jira Gantt chart (also referred to as the Sprint Timeline) provided a helicopter view of progress across all five sprints. Milestone markers were placed at key delivery points to monitor deliverable progress. For example, midterm presentation (end of Sprint 3), Verify Module completion (first half of Sprint 4), UAT completion (second half of Sprint 4), and final deliverable submission (Phase 4).

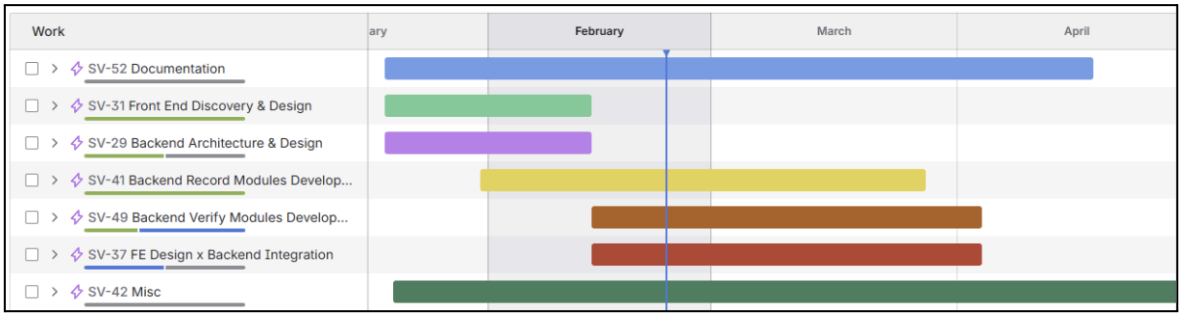


Figure 13: Jira Gantt/Sprint Timeline

3.2.3 Cumulative Flow Diagram

The Jira Cumulative Flow Diagram (CFD) provided a visual representation of work-in-progress over time across the three Kanban states. The CFD was monitored for the width of the "In Progress" band (a widening band indicates growing work-in-progress, suggesting the team is starting tasks faster than it is finishing them), and the gap between the "In Progress" and "Done" bands (a widening gap suggests bottlenecks or blockers preventing tasks from being completed).

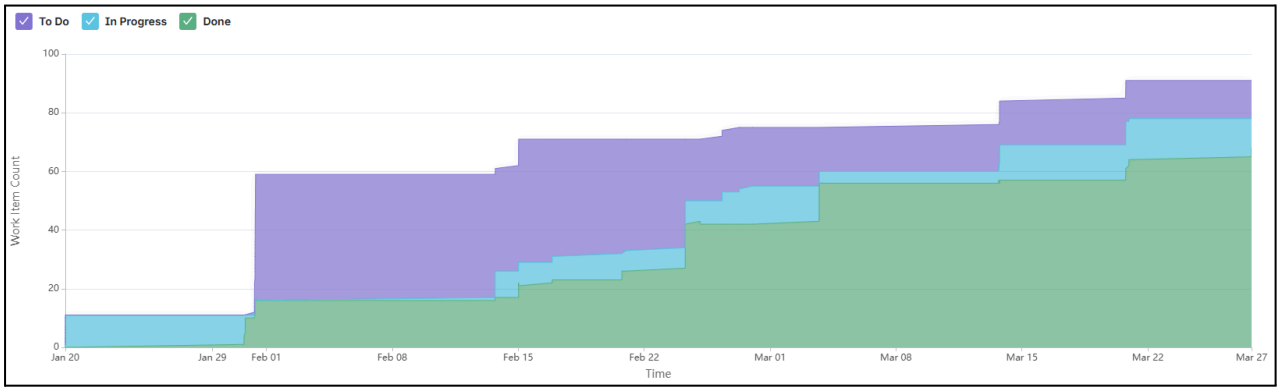


Figure 14: Jira Cumulative Flow Diagram

The CFD shows a flat "In Progress" band through Sprint 1 as the team built foundational infrastructure, followed by a large influx of "To Do" items around Feb 1 reflecting backlog grooming and finalisation of scope, features, and deliverables.

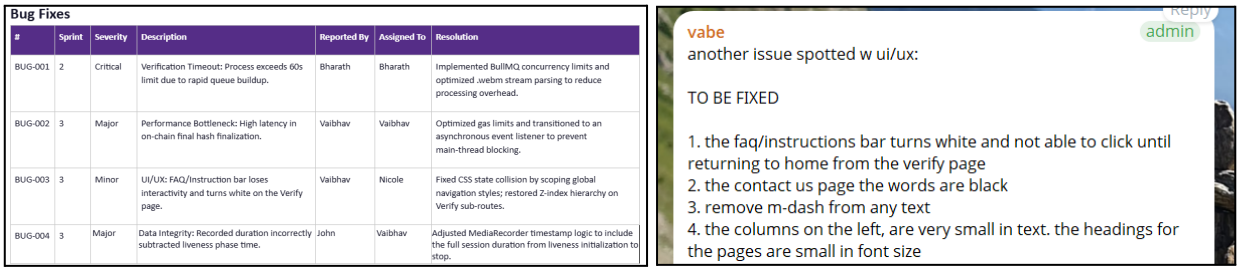
Through Sprints 2–3, the "In Progress" band (cyan) stays thin, a positive signal that items were flowing through the pipeline rather than stalling, before the "Done" area (green) climbs steeply in Sprints 4–5, with a spike in Sprint 4 as the liveness migration, Verify Module, and UAT preparation finished in parallel, and convergence in Sprint 5 as the final delighters wrapped up and "Done" caught up to the total scope, indicating a strong finish into the capstone deadline.

3.2.4 GitHub: version control and collaborative development

GitHub served as the team's version control and collaborative development platform throughout the project. The repository was structured as an npm monorepo with frontend and backend as separate workspace packages. Pull requests were required for all merges to main, with at least one peer review per PR. GitHub's issue tracking complemented Jira's sprint-level management by providing code-level visibility into bugs and fixes.

3.2.5 Bug Tracking and Documentation

The team tracked bugs through two channels: Telegram for immediate discussion and a formal 'Bug Fix' Document for logging severity, assignee, resolution, and date. Hence, balancing speed with traceability. Bugs were categorised by severity: Critical (blocks sprint demo or end-to-end flow), Major (affects a core feature but has a workaround), and Minor (cosmetic or UX issues that don't block functionality).



Bug Fixes

#	Sprint	Severity	Description	Reported By	Assigned To	Resolution
BUG-001	2	Critical	Verification Timeout: Process exceeds 60s limit due to rapid queue buildup.	Bharath	Bharath	Implemented BullMQ concurrency limits and optimized .webm stream parsing to reduce processing overhead.
BUG-002	3	Major	Performance Bottleneck: High latency in on-chain final hash finalization.	Vaibhav	Vaibhav	Optimized gas limits and transitioned to an asynchronous event listener to prevent main-thread blocking.
BUG-003	3	Minor	UI/UX: FAQ/Instruction bar loses interactivity and turns white on the Verify page.	Vaibhav	Nicole	Fixed CSS state collision by scoping global navigation styles; restored Z-index hierarchy on Verify sub-routes.
BUG-004	3	Major	Data Integrity: Recorded duration incorrectly subtracted liveness phase time.	John	Vaibhav	Adjusted MediaRecorder timestamp logic to include the full session duration from liveness initialization to stop.

Figure 15: Screenshot of Bug Fix Documentation and Telegram Reporting

3.3 Stakeholder Communication

3.3.1 Communication matrix

Bedrock involved three distinct stakeholder groups, each with different cadences and channels.

Stakeholder	Wendy	Dharma	Internal Team
Role	Supervisor	Sponsor (STABILITY)	-
Channel	Zoom + Telegram (ad hoc)	Zoom + Telegram (ad hoc)	Telegram (sub-channels) + standups
Cadence	Fortnightly progress updates	Fortnightly progress updates	Daily (async) + weekly (sync) + Fortnightly sprint reviews
Content	Sprint progress, rubric alignment, deliverable review, feedback	Technical architecture decisions, legal positioning, GTN integration guidance, scope evolution, commercial direction	Task updates, blockers, code reviews, design decisions
Documented Output	Meeting minutes (scribed, summarised, and recorded)	Meeting minutes (scribed, summarised, and recorded)	Standup notes via telegram, Bug fixe docs, Jira task updating

Table 14: Stakeholder Communication Matrix

3.3.2 Fortnightly Stakeholder Updates

External stakeholder meetings (Wendy and Dharma) and internal sprint reviews both ran fortnightly, covering progress updates, demos, open issues, and next steps. Minutes were scribed live, summarised afterwards, and stored with the recording, ensuring no decisions were lost and absent members could reconstruct full context. A detailed breakdown is in Appendix Q.

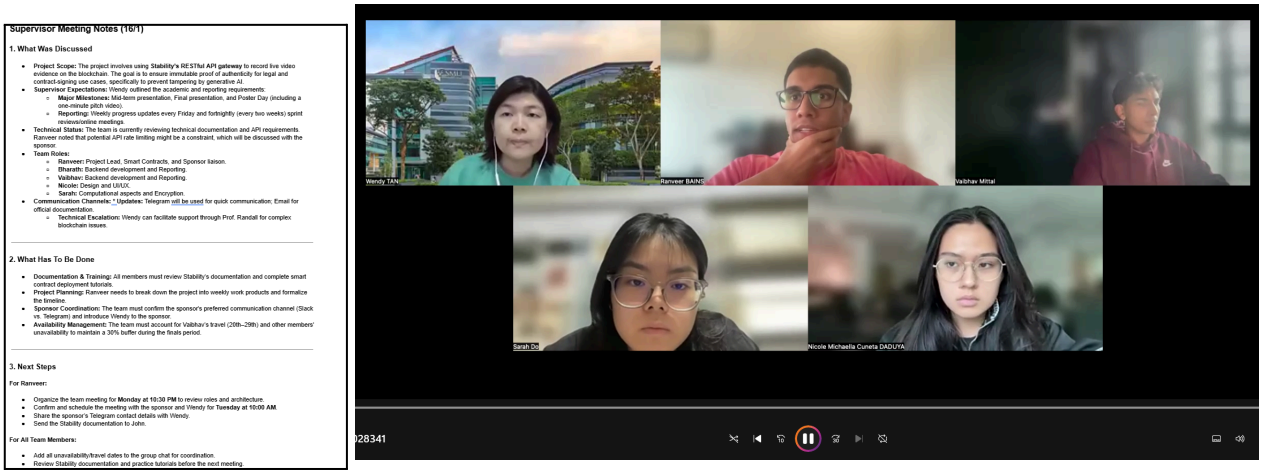


Figure 16: Snapshot of Meeting Minutes Documentation

3.3.3 Single Point of Contact and Decoupled Workflows

A deliberate communication design was the single point of contact principle. Ranveer (Project Manager) served as the primary contact for both stakeholders and maintained relationships throughout the project. This decoupled the two communication streams, preventing crossed wires between academic requirements and sponsor expectations.

Internally, the Telegram group was structured with sub-channels to prevent information overload. General (announcements and cross-team items), Admin (logistics, scheduling, deadlines), Frontend (React/UI discussions), Backend (Node.js/blockchain discussions), and Sponsor. A shared Google Drive folder served as the document repository for all stakeholder-facing materials and documentation.

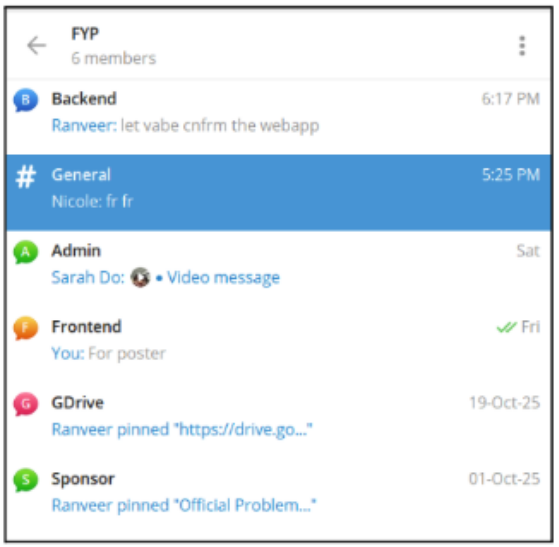


Figure 17: Snapshot of Bedrock's Telegram Group

3.3.4 Sponsor Handover Pack

The final deliverable to STABILITY is a comprehensive handover pack designed to enable the sponsor to continue development, deploy the product, or onboard a new development team without dependency on the original capstone team. The handover pack contains the follow (see Appendix R):

- Source code
- Final report and presentation slides
- Technical documentation, Environment configuration guide, CCTV hardware guide
- Website copy brief (see Appendix F)
- Brand guide (see Appendix E)

While we acknowledge our rigorous approach to documentation very much exceeds the lean benchmarks of traditional Agile methodologies, its role as a strategic lever for team synchronisation and project success was foundational considering our packed schedules as students. By maintaining these documents and processes we institutionalised a culture of transparency and asynchronous collaboration.

Beyond internal efficiencies, this comprehensive knowledge base serves as a critical asset for the handover to the STABILITY technical team. It provides a longitudinal record of the decision logic and architectural evolution of the Bedrock platform.

4. Testing, Analysis and Findings

4.1 Internal Testing & Validation

The backend testing strategy was built on three objectives:

1. **Unit Testing:** Verify each module works correctly in isolation (Sprint 1-2 and 4)
2. **System Integration Testing:** Verify modules work correctly together and with external dependencies (Sprint 5)
3. **Edge Case Testing:** Verify system behaves predictably under abnormal conditions (Sprint 5)

All backend tests were implemented using Vitest 3.2.4, a Vite-native testing framework that integrates with the project's existing Vite build toolchain. Vitest was selected over alternatives (Jest, Mocha) because the Bedrock backend uses ES modules (import/export syntax) natively. Vitest supports ES modules without transpilation, whereas Jest requires additional configuration. This alignment with the project's module system eliminated an entire category of test configuration issues and allowed tests to import backend modules directly.

4.1.1 Unit Testing (Sprint 1-2 and 4)

Unit tests verify that individual functions and modules produce correct outputs for known inputs, in isolation from external dependencies (the blockchain, AWS services, Redis). External dependencies were mocked using Vitest's built-in mocking capabilities (vi.mock, vi.fn).

Test Module	Description
Hashing (fingerprinting) Module (verification/hashing.js)	• SHA-256 segment hash of a known input matches the expected output (determinism)

	• Rolling hash chain computation follows the documented formula *(roll_0 = SHA-256(segHash_0) for genesis, roll_i = SHA-256(roll_{i-1} || segHash_i) for subsequent hashes)* • Final segment zero-padding produces correct results for files that are not exact multiples of the ~1.3 MB segment size • Test cases included: ○ Single-segment file ○ Multi-segment file ○ File requiring zero-padding (2 full segments + 1 padded segment) ○ Empty file (this edge case should produce a deterministic hash of a fully zero-padded segment)
BullMQ Job Logic	• Hash records submitted via */api/hashes/submit* are correctly serialised into BullMQ job payload • isFinal and isFirst flags are set correctly based on the submission context • Queue configuration (concurrency=1, attempts=1) is enforced The Redis dependency was mocked using an in-memory stub to isolate the queue logic from the actual Redis instance.
Liveness Session Management	• */api/liveness/session* endpoint correctly calls CreateFaceLivenessSessionCommand with the expected parameters • */api/liveness/session/:id/result* endpoint correctly parses the GetFaceLivenessSessionResultsCommand response • Pass/fail determination is applied at the correct confidence threshold. The AWS SDK was mocked to return predetermined responses (pass with score 95, fail with score 30, session not found).
Verification Pipeline (*/api/verify*)	Three scenarios tested: • Authentic file (all hashes match, expected result: Verified) • Tampered file (hash divergence at segment 3, expected result: Tampered with first divergence at index 3) • File with no on-chain record (Factory Contract returns zero address, expected result: Not Found) The blockchain reads were mocked to return predetermined hash trails, allowing the verification logic to be tested independently of the actual blockchain state.
Subcontract Deployment and deployInfo Management.	• */api/subcontract/deploy* endpoint correctly calls *Factory.createSubcontract()*

	• SubcontractCreated event is parsed correctly from the transaction receipt, and that *provider.getCode(addr)* is called to verify deployment. • Deployment metadata (deployment block, signer EOA) is stored correctly in the in-memory deployInfo Map. • Anti-spoofing check when a hash submission references a Subcontract address not present in deployInfo, the endpoint rejects the request

Table 15: Unit Testing Checklist

4.1.2 Integration Testing (Sprint 5)

System integration tests (SIT) verified that the backend modules work correctly together and with their actual external dependencies (i.e. the blockchain, AWS Rekognition, and Redis). Unlike unit tests, SIT tests were run against STABILITY's live GTN, a real Redis instance, and real AWS endpoints.

The SIT suite covered two end-to-end flows:

- **Record flow (end-to-end)**
  - A full session was simulated. *POST /api/liveness/session* created a real AWS Rekognition session;
  - GET */api/liveness/session/:id/result* confirmed it entered the Created state. Liveness challenge completion was not automated (it needs a real face on a real camera), but both API paths were validated against the live AWS endpoint.
  - Recording then proceeded: a Subcontract was deployed via the Factory, hashes were streamed through */api/hashes/submit*, and the Subcontract's on-chain state was read after each write to confirm the hash landed.
  - The final hash was submitted with *isFinal=true*, after which the test verified the Subcontract's isFinalized flag was set and the Factory's registry held the recording's first and final hashes. The pipeline under test: API → BullMQ → blockchain writer → on-chain.
- **Verify flow (end-to-end).**
  - Building on the Record flow, a known video file was pushed through the recording pipeline and then re-submitted to *POST /api/verify* by dragging and dropping it into the verify field of the web application. The endpoint returned Verified with the correct Subcontract address and hash summary.
  - The same file was then modified (a section excised from the middle) and re-submitted. The endpoint returned Tampered with *firstDivergenceIndex = 2*, confirming that both the success and the tamper-detection paths round-trip correctly against live on-chain state.

The web application backend exposes six active endpoints that mediate between the frontend client and the blockchain and AWS services, all served via Express on a single Node.js process. Each endpoint's expected response and error handling is summarised in Table 16.

Endpoint	Success Response	Error Response
POST /api/liveness/session	Returns session ID	403 if AWS credentials invalid
GET /api/liveness/session/:id/result	Returns confidence score and pass/fail	404 if session ID not found
POST /api/hashes/submit	Accepts hash array, returns enqueued count	400 if address format invalid
POST /api/subcontract/deploy	Returns Subcontract address and block number	500 if RPC unreachable
GET /api/subcontract/:address/status	Returns "complete" with hash summary, or "pending"	404 if address unknown
POST /api/verify	Returns Verified/Tampered/Not Found	413 if file exceeds 500 MB

Table 16: SIT API Endpoint Validation Checklist

Bedrock's SHA-256 rolling hash must produce identical output on every platform that implements it, otherwise a file recorded on one device would appear tampered when verified on another. Four implementations exist:

- WebCrypto: web frontend, live hashing during recording
- Node.js crypto: web backend, rehashing uploaded evidence at verification time
- java.security.MessageDigest: Android APK (Kotlin)
- Python hashlib: Raspberry Pi CCTV

Across all four implementations, *tools/rehash.py* is the canonical reference. Parity was checked by hashing the same test files with every implementation and requiring the segment hashes, rolling hashes, and final hash to match *rehash.py* byte-for-byte. This had to hold even though each platform (WebApp,Phone App, CCTV) submits hashes differently.

One divergence surfaced in Sprint 4 (BUG-008). The WebCrypto path was hashing a UTF-8 string version of each segment instead of the raw ArrayBuffer, so its output disagreed with every other implementation. Passing the Uint8Array straight into crypto.subtle.digest() fixed it, and all four platforms have matched *rehash.py* exactly since.

4.2 External Testing & Validation

Usability Testing	User Acceptance Testing (UAT)
Evaluates how easily users interact with the product during development to improve experience and behaviour. Testing done internally and externally. **Scope: Record Module only | Sprint 3, Week 7 | 7 participants**	Final phase before release. Users verify the product meets business requirements and is ready for production. Testing done externally. **Scope: Record + Verify Modules (Full POC) | Sprint 4, Weeks 9–12 | 8 participants**

4.2.1 Usability Testing (Sprint 3, Record Module)

Session Details

Conducted to validate the Record Module with real users before finalising the design. Seven participants completed moderated task-based sessions in a focus group setting without guidance.

Participants	7 (mix of lawyers and non-technical users)
Format	Moderated task-based sessions (remote Zoom and in-person)
Timing	Sprint 3
Team Leads	Nicole, John, Sarah (UX Leads)
Post-session	SUS survey (10-question Likert scale) administered after each session

Table 17: Session Details (Usability Testing)

Tasks & Success Metrics

Task	Success Metric
"Navigate through the web app and give a brief description of Bedrock's UVP"	Accurate description provided
"Capture a video of yourself stating your name using the app"	Completed without assistance
"Download your Proof-of-Integrity Cert and the video evidence you just captured.”	Both files downloaded within 30s without assistance

Table 18: Task Success Metrics (Usability Testing)

Key Insights & Fixes Applied

All raw data in which these insights were based can be found in Appendix N.

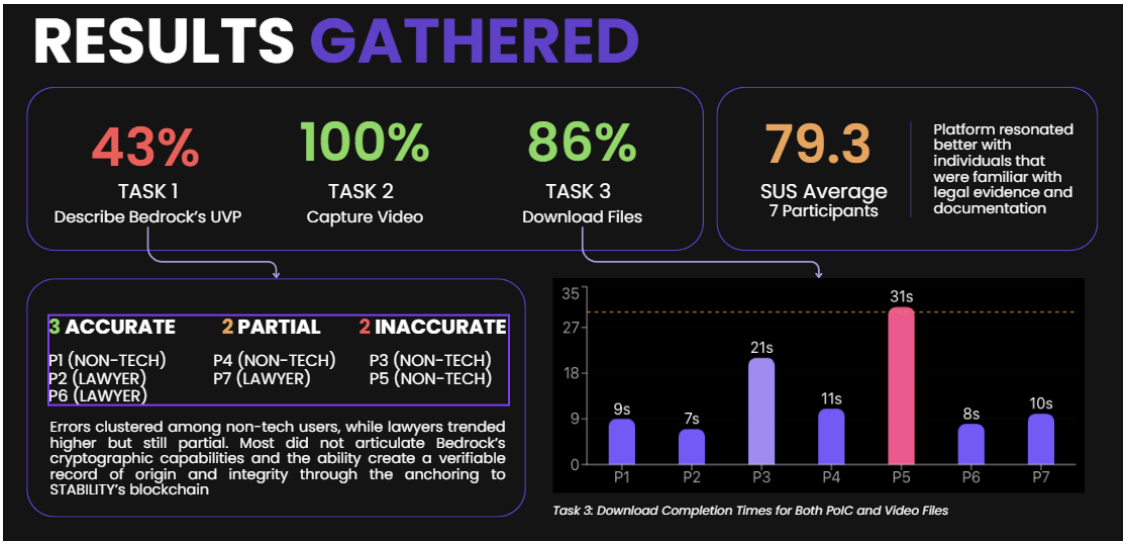


Figure 18: Usability Testing Insight Snapshot

SUS scores were interpreted using the adjective rating scale established by Bangor, Kortum, and Miller (2009), in which scores below 52 are rated Poor, 52-70 as OK with 68 being the industry average, 71-85 as Good, and above 85 as Excellent (Bangor et al., 2009; Sauro & Lewis, 2016). Taking into account the 79.3 SUS score and pass rates in this usability test, the web application performed well but needs further refinement as elaborated below.

Insight 1: Landing Page Did Not Communicate Blockchain UVP

- 4 out of 7 participants (57%) failed or only partially described the UVP accurately in Task 1. Struggled to articulate what differentiated our platform from standard recording applications.
- Recommendation: Refine landing page to include more keywords (e.g. 'blockchain-anchored', 'cryptographic hashing', 'STABILITY') so the UVP is immediately visible on arrival.

Insight 2: Improve Upon File System (Naming Convention)

- Data: Task 3 had a 86% pass rate. P5 >30 seconds due to unfamiliar and generic labelling. P2 and P6 raised concerns that the file names were too vague. P6 requested a date tag at minimum to be added to the file name. P2 mentioned that there may be an issue with identifiability where multiple recordings are concerned. Not suited for high-volume workflows.
- Recommendation: Adopt a descriptive filename convention at the point of download (e.g. bedrock_evidence_2026_03_01_2328.mp4). Possible addition of UID in the future.

Insight 3: Improve Upon File System (Bundle Video and Certificate Together)

- P5 had required moderator intervention during T3 to download the POIC as he misunderstood that the files were already bundled together, hence only downloading the video file. P7 suggested bundling the certificate and video files together for convenience.

- Recommendation: Future iterations can zip the POIC and video file together, allowing the user to download both files in a single click (e.g. bedrock_evidence_2026_03_01_2328.zip). However, this recommendation was marked as low priority to make way for other tasks.

4.2.2 User Acceptance Testing (Sprints 4 & 5, Full POC)

Session Details

UAT is the final validation phase before the POC is published. 8 participants complete end-to-end workflows in a focus group setting spanning both modules, confirming the system meets its business requirements and is ready for a production demonstration. All sessions were moderated and task-based, conducted both remotely (via Zoom screen-share) and in-person. Each session followed the same protocol where the participant was given a task prompt, asked to complete it without guidance, then debriefed with follow-up questions. At the end of all tasks, participants completed a standard 10-question System Usability Scale (SUS) survey.

Participants	8 (mix of lawyers, law students, and non-technical users)
Format	Moderated task-based sessions (remote Zoom and in-person)
Timing	Sprint 4
Team Leads	Nicole, Sarah, John (UX Leads)
Post-session	SUS survey (10-question Likert scale) + verbal debrief
Scope	Record Module + Verify Module (ie. full end-to-end POC flow)

Table 19: Session Details (UAT)

UAT Tasks & Business Requirements

#	Task	Module	Business Requirement	Pass Criteria
1	"Navigate through the web app and give a brief description of Bedrock's UVP"	Record	System communicates UVP clearly on first visit	Accurate description provided without prompting
2	"Capture a video of yourself stating your name and download the evidence package"	Record	End-to-end record flow completes; evidence bundle delivered, both files downloaded	Both files downloaded within 30s without assistance
3	"Upload your downloaded video to the Verify module and confirm its authenticity"	Verify	Verify module correctly identifies an unmodified Bedrock file as authentic	User interprets authentic result without assistance
4	"We have modified a copy of your video. Upload it and tell us what the result means"	Verify	Verify module detects tampering and communicates result in plain English	User correctly identifies file was modified

Table 20: Tasks and Business Requirements (UAT)

Key Insights & Fixes Applied

All raw data in which these insights were based can be found in Appendix O.

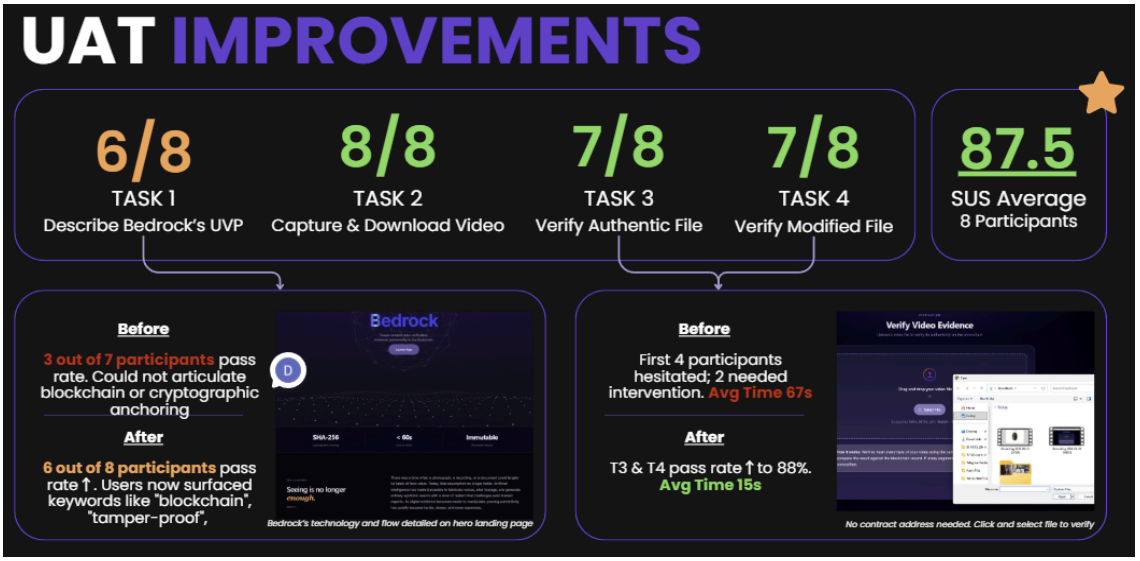


Figure 19: UAT Insight Summary

SUS scores were interpreted using the adjective rating scale established by Bangor, Kortum, and Miller (2009), in which scores below 52 are rated Poor, 52-70 as OK with 68 being the industry average, 71-85 as Good, and above 85 as Excellent (Bangor et al., 2009; Sauro & Lewis, 2016). Taking into account the 87.5 SUS score and pass rates in this UAT, the web application performed exceptionally and only needs minimal improvements before launching the POC proper.

Insight 1: Landing Page Did Not Communicate Blockchain UVP

- Web page copywriting was tested for communicating Bedrock's UVP. Previously, 43% pass rate. Users could not articulate blockchain or cryptographic anchoring.
- Now, 75% pass rate (32% increase). Users now surfaced keywords like "blockchain", "tamper-proof" when describing the platform.

Insight 2: File-based verification replaces contract address input

- The first 4 participants (observed during T3 and T4) had to enter a blockchain contract address to initiate verification manually. The first 4 participants hesitated, 2 required moderator intervention, and the average task completion time was 67 seconds. Users found the contract address unintuitive and were unsure where to locate it. In subsequent testing and when redoing the testing for this portion, verification changed to become initiated by file upload only, no contract address required.
- Contract address input field removed entirely. Verification is now initiated by uploading a video file. The system automatically looks up the corresponding blockchain record. Average verification time dropped from 67 seconds to under 15 seconds across all 8 participants for both T3 and T4.

4.2.3 Usability testing vs UAT comparison

Usability Testing vs Final UAT Testing

The table below compares results from the Sprint 3 usability testing round (Record Module only, 7 participants) with the Sprint 4 UAT round (full POC, 8 participants).

Metric	Usability Testing (Sprint 3)	UAT (Sprints 4–5)	Change
Participants	7	8	+1 participant
Participant Mix	External	External only	More rigorous
Tasks Tested	3 (Record Module only)	4 (Record + Verify)	Full POC coverage
T1 Pass Rate (UVP)	43%	75%	+32%
T2 Pass Rate (Capture and Record)	100%	100%	Maintained
Usibility Testing T3 (Download) vs. UAT T3 (Verify Authentic)	86% (Download Files)	88% (Verify Authentic)	UAT T3 is an extension of Usibility Testing T3 done prior
T4 Pass Rate (Verify Modified)	NA	88%	New task
SUS Average	**79.3 (Good)**	**87.5 (Excellent)**	**+8.2 points**

Table 21: Usability Testing and UAT Comparison

UAT confirms that Bedrock's full POC (covering capture, real-time hashing, blockchain anchoring, certification, and verification) is functional, usable, and ready for final demonstration. The platform achieved a SUS average of 87.5 (Excellent) among 8 external participants. All four tasks met their pass rate targets. No critical usability issues remain outstanding. The platform resonates most strongly with participants who have legal or institutional backgrounds, confirming alignment with Bedrock's target user base. Non-technical users benefit from the UX improvements made since usability testing, but may require additional onboarding support in a production deployment.

5. Achievements and Value Creation for LegalTech Industry

This section addresses the value Bedrock's technology creates in the LegalTech sector, per the final year project's course requirements. As elaborated later in Section 6.3, the sponsor and team view the stronger commercial potential of Bedrock lying in adjacent sectors like media verification and hardware attestation (CCTVs), where demand is more pronounced. Nonetheless, the LegalTech analysis that follows fulfills the computing and law assessment requirement of the original project scope.

5.1 Smart Contracts: New Evidentiary Mechanism

The current legal procedure for establishing the authenticity of electronic evidence is fundamentally reactive. Under Section 116A of the Evidence Act, authenticity is presumed and typically only interrogated after a dispute arises. This leaves a window of vulnerability between the moment of capture and the moment a challenge to authenticity is raised. During this period, the custodian acts as the sole holder of truth, exercising centralised control over the authoritative copy. This creates a private custody vulnerability where the integrity of the evidence is merely assumed but never independently proven.

Bedrock eliminates this vulnerability by replacing the human-centred chain of custody with one that is on-chain. No single entity, including Bedrock and STABILITY, retains centralised control over the authoritative record. This collapses the traditional [Capture → Store → Dispute → Forensic Expert] workflow into a streamlined [Capture → Anchor → Verify].

The existing model permits only a probabilistic opinion. Forensic experts can verify that no artefacts of tampering were detected, but cannot confirm whether tampering occurred before the evidence was submitted. Bedrock replaces this investigative process with a deterministic lookup. The smart contract records the cryptographically unique fingerprint of the original file, so any independent party can upload and prove the integrity of the recorded media with a degree of certainty that is computationally infeasible to falsify. Hence, removing the need for specialist forensic knowledge.

Furthermore, this technological shift provides a robust mechanism for easing the evidentiary burden under s 116A of the Evidence Act. By allowing the court or the opposing party to independently verify the file through Bedrock, the platform provides a reproducible verification method. This does not supplant judicial discretion, but instead changes the nature of the court's task when a piece of evidence is questioned. Rather than weighing competing expert testimonies (which costs money and time), the court can rely on Bedrock to prove evidence authenticity.

5.2 Trust and safety: Building for the Singapore Judiciary

Currently, no existing tools within the courts, in Singapore or globally, verify the authenticity of an electronic record at the moment of submission. Bedrock secures evidence at the moment of creation rather than analysing it after storage.

Bedrock implements two complementary controls. The first control gate is the liveness verification module, as documented in Section 2.3.1. Before recording commences, AWS Rekognition performs an active liveness challenge using randomised colour sequences to confirm that a live human being is physically operating the camera. This prevents the injection of pre-recorded video, deepfake footage, or other synthetic media at the point of capture. The second is blockchain anchoring (Section 2.3.2). The system slices the video stream into segments, computes cumulative SHA-256 hashes, and anchors them onto STABILITY's GTN in real time. Once written, the on-chain record cannot be altered or deleted by any party.

These controls are complementary, not redundant. Liveness verification without blockchain anchoring would confirm a real person recorded the video but not that the file was unaltered afterwards. Blockchain anchoring without liveness verification would confirm the file was unaltered after capture but not that the source was live and authentic rathan than synthetic. Together, they establish the "legitimate stream" standard defined in Section 2: a real-time, unaltered capture of actual events, recorded by a physically present person, untampered from creation to submission.

This architecture operationalises the core principles of IMDA's Model AI Governance Framework. STABILITY's open-source, independently auditable codebase provides the required transparency. Furthermore, the deterministic nature of SHA-256 verification, which produces an identical result regardless of who performs it, ensures that the system's outputs remain explainable to any non-technical user. Lastly, the two-gate design addresses the Framework's safety requirement because neither synthetic media at capture nor post-capture alteration can compromise the evidentiary record without detection.

However, a critical distinction must be drawn between what Bedrock proves and what it does not. Bedrock confirms that a recording has not been altered since capture and that the person who initiated it was live. It guarantees the file's authenticity but does not judge the credibility of its content. A witness may record a genuine, untampered video of themselves making a false statement. In addition, as bad actors develop new means to bypass security measures, Bedrock must evolve beyond this two-control architecture to remain foolproof. Future recommendations are detailed in the risk register in Section 6.2.2.

Ultimately, Bedrock functions as a foundational layer for digital trust, not a replacement for legal judgment. It solves the objective problem of file authenticity with cryptographic precision, leaving the subjective 'honesty' of the content as a matter for judicial determination.

5.3 Digital Forensics: Removing the Dependency

When a dispute concerning the authenticity of evidence is raised, the current procedure relies on forensic expert analysis. This process is costly and takes 6 to 8 weeks to generate a court-ready report. Since each piece of evidence demands manual expert intervention, the process cannot scale. As AI-tampered evidence grows in litigation, these delays compound into a systemic bottleneck.

Bedrock eliminates the need for forensic re-verification by proving authenticity at capture. This reduces the forensic cost and delay burden to near zero for evidence captured through Bedrock. The verification process is algorithmic, reproducible and instantaneous. It requires no expert testimony to explain and no chain of custody documentation to maintain beyond the POIC itself. Bedrock architects for prevention, sealing integrity at the moment of creation so that the detection question does not arise for evidence captured through the system.

It is important to note that Bedrock does not replace forensic analysis for contested or complex scenarios. Where evidence was not captured through Bedrock, forensic analysis remains the only method for assessing integrity. In the long run, where novel forms of manipulation emerge that operate below the segment-level granularity of Bedrock's hashing algorithm, expert testimony may be required to assess the significance of more advanced anomalies. Bedrock's objective is to reduce the caseload reaching forensic experts, allowing them to focus on genuinely contested matters that require specialist skills.

5.4 The Court Admissibility Pathway

The current judicial landscape reveals a critical structural gap because no native infrastructure exists for courts to verify the authenticity of electronic evidence at the point of submission. Bedrock occupies a unique position where real-time cryptographic hashing meets blockchain immutability for evidentiary purposes. As a pioneer, Bedrock's transition from a POC to judicially recognised evidentiary tool requires a structured engagement programme with the Singapore courts. The following details a three-phase pathway that defines the sequence, objectives and dependencies for achieving court admissibility and hence, international legal recognition.

5.4.1 Phase 1: Judicial Engagement and Dialogue with Singapore Courts

The first phase consists of establishing technical credibility with the Singapore Judiciary and securing formal channels for dialogue on blockchain-anchored evidence.

Bedrock must present its technical architecture to judicial stakeholders such as the Supreme Court's IT Committee and the State Court's Centre for Dispute Resolution. The system's alignment with the evidentiary requirements of the Evidence Act must be made clear and address the concerns as addressed by Justice Xu regarding synthetic media. A formal technical submission to the court will be made, alongside an independent security audit of the smart contract architecture detailing the factory contract and subcontract, and a white paper mapping Bedrock's verification outputs to the burden-of-proof requirements under s 116A. This phase would also be an opportunity for the courts to share industry knowledge about existing bad actors, informing additional controls Bedrock should develop.

However, there is a possibility that the courts will not be receptive as they may require regulatory or legislative clarification before accepting blockchain as a method of authentication. Bedrock notes that neither institutional engagement timelines nor technical feasibility are not within the project's control. It is noted however, that the judiciary would use Bedrock in the capacity of the secondary persona of (data verification agencies or personnel, see Section 1.6). It is therefore more impactful to first reach litigants, the primary personas, to adopt the platform so that legal precedent can be set. Phase 2 elaborates on this.

5.4.2 Phase 2: Pioneer Test Case to Establish Legal Precedent

The second phase establishes a legal precedent for the admissibility of Bedrock-authenticated evidence through a controlled test case in the Singapore State Courts.

This requires the identification of a suitable matter in which Bedrock-captured evidence can be submitted as part of the evidentiary record, and must be selected to present the system's verification capabilities under conditions where the authenticity of electronic audio-visual evidence is in dispute. Bedrock's team must prepare expert testimony capable of explaining the architecture behind Bedrock, the various algorithms used (e.g. cumulative hashing, liveness and verification) to a non-technical judicial audience. The POIC should also be presented as a way to certify that Bedrock has authenticated the evidence.

Bedrock notes that an adverse ruling in the test case may set a negative precedent and delay adoption. The selection process is of importance, where selecting an inappropriate case would fail to establish the intended precedent. The availability of a willing litigant prepared to adopt the risk of submitting Bedrock-verified evidence is a dependency outside the project's control.

5.4.3 Phase 3: Global legal recognition for blockchain-anchored evidence

The third phase extends legal recognition of blockchain-anchored evidence beyond Singapore. Following successful validation in Singapore, the admissibility framework must be adapted for jurisdictions with analogous evidentiary standards: the United Kingdom, the United Arab Emirates, and common law regions in the Asia-Pacific.This requires localised legal mapping to align Bedrock's verification outputs with jurisdiction-specific evidence rules. The utilisation of STABILITY's GTN, being a public, cryptoless blockchain remains a strategic advantage by removing common regulatory objections associated with cryptocurrency.

Bedrock recognises that evidentiary standards vary materially across borders and a precedent established in one region does not ensure an automatic transfer of legal weight. Each territory may impose independent compliance requirements or require its own test case. Furthermore, the project's success

depends on navigating various international postures toward blockchain governance. However, successful execution of this will position Singapore as a pioneer in formalising blockchain-anchored admissibility. This reinforces the state's standing at the forefront of legal technology governance while providing the global judiciary with a cryptographically reliable alternative to traditional forensic intermediaries.

6. Limitations, Mitigation Strategies, and Future Trajectory

This section identifies the limitations within Bedrock's current POC, presenting a detailed risk analysis of technical exposures while exploring the potential for the system's future trajectories.

6.1 Known Limitations

The following limitations are inherent to the current POC implementation (as of 15 April 2026) , grouped by feature. The risks that arise from these limitations, together with their assessment and any compensating controls, are addressed in the subsequent Section 6.2.

Access Control and Identity Management	
Limitation(s)	**Assessment**
No User Authentication System	***Description***: The current implementation has no dedicated user authentication layer. Any party with access to Bedrock's platforms can initiate a recording session without identifying themselves. Recordings are not linked to authenticated accounts, and no identity-linked audit trail exists for session initiation or record downloads. ***Rationale***: User authentication was excluded from the POC scope to prioritise the core evidentiary mechanisms. This version demonstrates the feasibility of blockchain -anchored chain of custody rather than serving as a production deployment.
Liveness Verification	
Limitation(s)	**Assessment**
Single-gated Liveness Check	***Description***: The architecture performs a single liveness check before recording begins. Once the system enters the LIVE_HASHING phase, no further liveness verification occurs for the remainder of capture. ***Rationale***: Periodic or continuous liveness reverification was considered but rejected. Mid-recording prompts would disrupt the capture of legal proceedings or other contexts requiring uninterrupted recording. The single-gate design was retained for the current iteration, with liveness captured as the opening segment of the primary video stream. The transition from liveness check to main recording is therefore captured as a single, immutable event within the same rolling hash chain.

Hashing Chain	
Limitation(s)	**Assessment**
Cross-Platform Segment Size Uniformity	***Description:*** All Bedrock prototypes (web app, Android, or Raspberry Pi) must utilise identical segment sizes as it predetermines the success of the verification module. To appropriately recompute the hashing chain to complete the verification process, Bedrock needs to segment the uploaded video in the same manner as the way the video was captured on Bedrock. ***Rationale:*** Any deviation to the segment size across platforms would cause the verification module to fail, even if the video was not altered. While the Raspberry Pi prototype might perform better with larger segments to save memory, optimising the segment size specifically for it would disable the verification feature. This uniformity is a permanent rule, not a setting to be tweaked, and all future devices must follow it.
Blockchain Anchoring	
Limitation(s)	**Assessment**
Hash-Only On-Chain Footprint	***Description:*** Bedrock anchors only the cumulative SHA-256 hashes of each video segment to the subcontract on GTN. The underlying video content is never transmitted to or stored on the blockchain. A verifier with only the on-chain record cannot reconstruct or view the recording, and can only confirm whether a file in their possession matches the anchored hash chain. ***Rationale:*** Audio-visual recordings routinely contain personal data within the meaning of Section 2 of the PDPA, and committing such content to an immutable public ledger (STABILITY) would breach the Retention Limitation Obligation under Section 25, the right to withdraw consent under Section 16, and the Protection Obligation under Section 24, all of which presuppose that the controller retains the technical capacity to act on data-subject requests. Bedrock's Privacy Policy (as documented in Appendix S) covers such details. Anchoring only the hash resolves these tensions because a SHA-256 output is a one-way cryptographic fingerprint from which the original content cannot be reconstructed and which carries no personally identifiable information.
Tampering Detected by Segments, Not Frame-by-Frame Detection	***Description:*** Cumulative hashes are anchored at segment boundaries rather than individual frames. When verification identifies a divergence, the system can localise the point of alteration to the affected segment but cannot identify the precise frames modified within it. ***Rationale:*** Frame-by-frame anchoring would exceed GTN's practical throughput and introduce prohibitive latency, requiring an on-chain transaction per frame. The

	current segment size balances verification granularity against blockchain throughput and client-side load. Segment-level localisation remains sufficient for evidentiary purposes to establish that the file is not bitwise identical to the sealed original.
Recording Pipeline	
Limitation(s)	**Assessment**
Single Continuous Recording Sessions	***Description:*** The system supports only one continuous recording session per subcontract. It does not permit pausing and resuming, stitching multiple segments into a unified record, or capturing from multiple cameras in parallel. ***Rationale:*** A continuous session produces a single, unambiguous hash chain from the opening liveness segment to the final frame. Pause-and-resume functionality would create temporal discontinuities in the hash chain that the system could not distinguish from mid-session substitution. The continuous session constraint is therefore an integrity control, not a feature gap.
Evidence Packaging	
Limitation(s)	**Assessment**
POIC Delivered as Separate Artefact	***Description:*** The POIC is delivered as a standalone PDF alongside the master video rather than embedded within the video file's metadata or container. The user receives two separate artefacts that must be retained and submitted together. ***Rationale:*** Embedding would change even one byte of the video, which would cause the hash to no longer match the blockchain record - and Bedrock would then flag the video as tampered.
Verification Module	
Limitation(s)	**Assessment**
No Backward Compatibility Across Segment Size Changes	***Description:*** Once a recording has been captured and anchored under a specific segment size, that segment size becomes a permanent property of the verification requirement for that recording. The verification module must apply the same segment size during recomputation to reconstruct the cumulative hash chain correctly. Any future adjustment of the capture-side segment size would require the verification module to retain backward-compatible logic for every historical segment size used across the system's operational history. ***Rationale:*** The blockchain record is immutable and does not store the segment size used during recording. To keep verification simple and reliable, we have fixed the segment size at ~1.3 MB for the entire system. While this makes the system less

	flexible for future changes, it ensures that every video is processed identically and can always be verified without error.
Cryptographic Sensitivity to Transcoding and Re-encoded Media	***Description:*** Bedrock's verification requires an exact byte-for-byte match, any post-capture processing renders the record unverifiable. Standard actions such as compression, re-encoding, or watermarking alter the file's underlying byte-stream. This causes the system to flag the record as "tampered" or unrecognised even if the visual and audible content remains identical to the original capture and was recorded on Bedrock. ***Rationale:*** Byte-level integrity was a deliberate choice to ensure a zero-trust security posture. This prioritises the detection of any and all modifications over flexibility of media handling. In a judicial context, a perfect copy is the safest baseline.

Table 22: Known Limitations of Bedrock's Features against Bad Actors

6.2 Risk Analysis and Mitigations

6.2.1 Risk Rating Classification Legend

Risk ratings were grouped into four severity bands, each defined by the level of attention it warrants.

Score Range	Severity	Definition
1 - 4	Low	Minor exposure with no material impact on evidentiary integrity or system operation. Standard monitoring and no active remediation is required.
5 - 9	Moderate	Tangible exposure that warrants active management and documented compensating controls. Does not threaten the core evidentiary guarantee but should be reviewed periodically.
10 - 15	High	Significant exposure that could materially compromise user trust, operational continuity, or the admissibility posture of individual recordings. Remediation should be prioritised on the production roadmap.
16 - 25	Critical	Severe exposure that could invalidate the evidentiary guarantee at the architectural level, undermine legal admissibility, or constitute a statutory compliance breach. Immediate remediation required.

Table 23: Risk Rating Classification Legend

6.2.2 Risk Register

Likelihood is scored on a scale from 1 (rare) to 5 (almost certain) and **Impact** on a scale from 1 (negligible) to 5 (severe). The two axes are multiplied to produce a **Risk Rating** between 1 and 25 that is then mapped to one of the four severity bands defined above. Each risk is presented with inherent and residual scores so the effectiveness of compensating controls can be evaluated independently of the underlying exposure. The five risks prioritised here (not in order) represent the most material exposures to the twin objectives of evidentiary integrity and legal admissibility.

Risk ID #1: Operational Dependency on STABILITY's GTN	
Description	Bedrock's recording flow depends on a successful connection to the GTN at every session start. If STABILITY's GTN is unreachable, the subcontract cannot be deployed and the session cannot proceed. Hashes will not be stored on-chain. The exposure is a hard block on the system during network unavailability.
Inherent Risk	Likelihood (4) × Impact (5) = 20 **[Critical]**
Compensating Controls	The system maintains active configurations for both GTN and testnet environments, allowing the team to redirect the blockchain anchoring to an alternate network when one becomes unavailable. Hence, being able to preserve recording capability when the primary network is experiencing downtime. Furthermore, STABILITY is actively attempting to fix this in response to growing aggregate demand across its product suite.
Residual Risk	Likelihood (2) × Impact (5) = 10 **[High]**
Justification	Residual likelihood (2): Likelihood reduced as STABILITY's capacity upgrades and Bedrock's manual testnet fallback significantly lower the frequency of connection failures. Residual impact (5): No change to impact score as a failure to deploy the subcontract prevents recording entirely. This exposure was observed during the roadshow when high demand for STABILITY's Monolith caused downtime. While the primary remediation is external to Bedrock, these infrastructure improvements decrease the probability of recurrence, even though the high consequence of a hard system block remains an architectural constant.

Risk ID #2: Mid-Session Capture Substitution	
Description	A genuine user passes the initial liveness challenge and subsequently redirects the camera input to a pre-recorded or synthetic feed. The manipulated recording therefore concludes with a complete and valid hash chain on-chain, producing a file that has been certified authentic by Bedrock but whose underlying content was not demonstrably live for the full duration of the session. Though the electronic record has become cryptographically verifiable, the captured content has been compromised.
Inherent Risk	Likelihood (4) × Impact (5) = 20 **[Critical]**

Compensating Controls	The liveness challenge is captured as the opening segment of the primary video stream rather than as a separate pre-recording step, whereby the liveness footage and the subsequent recording form a single continuous video that is hashed and anchored as one unbroken sequence. Any discontinuity introduced by a mid-session feed substitution, whether a temporal jump, a change in lighting, or an inconsistency in the physical environment, is preserved within the master video file itself and is therefore available for visual review by auditors or the court at the point of verification.
Residual Risk	Likelihood (2) × Impact (4) = 8 **[Moderate]**
Justification	Residual likelihood (2): Advanced deepfakes can bypass liveness checks without leaving detectable artefacts. This score reflects the future trajectory of AI, where synthetic media eventually becomes indistinguishable from reality which renders the existing visual controls ineffective. Residual impact (4): A deepfake "certified" as authentic is the system's most severe failure mode. While human review may catch some errors, it cannot mitigate the damage to the system's credibility when a cryptographic guarantee fails to detect synthetic content.

Risk ID 3: Absence of Liveness Verification on Mobile and Hardware Prototypes	
Description	Recordings produced through the Android APK or Raspberry Pi CCTV prototypes do not carry the full “legitimate stream” guarantee, as neither form factor implements facial liveness verification. Evidence captured through these prototypes may therefore be admissible on integrity grounds but not on provenance grounds.
Inherent Risk	Likelihood (3) × Impact (4) = 12 **[High]**
Compensating Controls	For the Raspberry Pi CCTV prototype, the physical enclosure and embedded software function as a trusted execution environment, whereby the integrity of the stream is anchored to the anti-tampering nature of the hardware unit itself. For the Android APK, the application is restricted to non-jailbroken and non-rooted devices as a short-term control against virtual camera injection, with GPS and telemetry correlation proposed as a future provenance substitute tying recordings to a verified location and timestamp. The formalisation of these context-aware policies is submitted to STABILITY for post-project development.
Residual Risk	Likelihood (2) × Impact (3) = 6 **[Moderate]**

Justification	Residual likelihood (2): Both form factors retain narrow but real attack surfaces. The Raspberry Pi remains vulnerable to physical USB compromise or remote SSH access if the network address is known, while the Android APK's restriction to non-rooted devices does not defend against purpose-built hardware interception. Residual impact (3): A successful exploit would fully bypass the provenance guarantee on the affected device, and no court has yet ruled on the admissibility of hardware-attested recordings.

Risk ID 4: Client-Side Computational Bottleneck During Real-Time Hashing	
Description	On lower-specification platforms or under sustained recording load, the hashing pipeline may fall behind the rate at which video data is generated. The consequences range from increased latency in the anchoring queue, to dropped or delayed segments, to session failures where the queue overflows or the browser tab becomes unresponsive. Hence, resulting in a failure in verification.
Inherent Risk	Likelihood (3) × Impact (4) = 12 **[High]**
Compensating Controls	The segment size has been deliberately tuned to approximately ~1.3 MB to balance verification granularity against client computational load and blockchain throughput, prioritising sustained operation on mainstream consumer hardware over finer-grained tamper localisation. Hash computation utilises SHA-256 and leverages the Web Crypto API within Chromium-based browsers, where performance characteristics are well understood and consistent.
Residual Risk	Likelihood (2) × Impact (3) = 6 **[Moderate]**
Justification	Residual likelihood (2): The combination of architectural choices and the standardisation on Chromium-based environments, which collectively ensure that the system performs reliably on the hardware and browser configurations it was designed for. The risk persists as the system cannot control the user's hardware or background processes. External factors like weak device specs or extended recordings remain unavoidable edge cases. Residual impact (3): A bottleneck event affects only the recording in progress at the time of failure. Previously completed recordings retain their full integrity and the on-chain record is unaffected.

Risk ID 5: Judicial Non-Receptivity to Blockchain-Anchored Authentication Mechanism	
Description	The Singapore courts may decline to utilise blockchain-anchored hashes despite Bedrock's flawless functionality as a self-contained proof of authority without prior legislative clarification or judicial guidance on the evidentiary status of distributed ledger records. The admissibility pathway set out in Section 5.4 have dependencies that lie outside of the team's direct control. A failure to secure judicial recognition would materially limit the practical value of the work by leaving Bedrock-captured evidence subject to the same forensic re-verification burden that the architecture was designed to eliminate.
Inherent Risk	Likelihood (4) × Impact (4) = 16 **[Critical]**
Compensating Controls	Though lying outside of the team's direct control, the three-phase engagement pathway as documented in Section 5.4 are structured and well-prepared. Bedrock's architectural design directly addresses the concerns raised by Justice Aidan Xu in his July 2025 IT Law Series address on AI-fabricated evidence which establishes a substantive alignment between the system and the judiciary's stated priorities. The system's compliance posture under IMDA's Model AI Governance Framework, as documented in Section 5.2, provides a recognised governance reference point that strengthens the case for institutional acceptance. Additionally, STABILITY's standing as a sponsor with existing legal-sector relationships through its Chief Strategy Officer and General Counsel provides a credible channel for judicial engagement that an unsponsored academic project would lack.
Residual Risk	Likelihood (3) × Impact (3) = 9 **[Moderate]**
Justification	Residual likelihood (3): Reduced as the structured engagement strategy and the alignment between Bedrock's value proposition and the Judiciary's articulated concerns regarding synthetic media. Residual impact (3): The evidentiary value of Bedrock-captured recordings does not depend exclusively on a single statutory pathway. Section 116A(5) empowers the Minister to make regulations governing the processes by which documents may be recorded or stored, including the appointment of certifying bodies to approve such processes. While Bedrock could in principle seek certification as an "approved process" under any regulations made pursuant to this provision, the absence of such certification does not preclude admissibility. Courts retain broad discretion to accord weight to the POIC and the on-chain verification record when assessing the authenticity of electronic evidence submitted as primary evidence under Section 64.

6.3 Future Trajectory

The web application is merely one expression of the underlying engine rather than the engine itself, where our prototypes serve as working demonstrations that the Bedrock core-engine is inherently platform-agnostic, where we are able to provide integrity guarantees on video evidence across multiple platforms.

This portability opens a substantial expansion surface.

(1) In the legal and corporate domain, the pipeline can be embedded directly into video-conferencing platforms to seal depositions, hearings, mediation sessions, and board meetings at the point of capture, consequently producing tamper-evident corporate records that are admissible by construction rather than by retrospective forensic certification.

(2) In the public safety domain, the embedded hardware deployment model already validated by the Raspberry Pi prototype extends naturally to law enforcement body cameras, drone footage, and security surveillance systems, where the integrity of the capture stream is presently dependent on chain-of-custody documentation rather than cryptographic proof.

(3) In the media and journalism domain, the verification module can be deployed as a service that newsrooms integrate into their editorial pipelines to authenticate source footage against the synthetic media threats documented in Section 5.2, providing a scalable alternative to the manual provenance checks on which the industry currently relies (Kishor, 2026).

Each of these expansions is achievable without re-engineering the core verification logic, which means the path from POC to broad deployment is one of integration work rather than architectural redesign.

The international dimension follows the same logic. The European Union's AI Act, the United Kingdom's Online Safety Act, and the converging Asia-Pacific frameworks on synthetic media governance are aligning around a common requirement for verifiable proof that content has not been altered since creation. Bedrock's model of cryptographic anchoring at the point of capture satisfies this requirement by design.

Once domestic legal precedent is established through the pathway in Section 5.4, the same technical architecture can support adoption across jurisdictions. The platform expansion described above can therefore proceed in parallel across multiple regulatory environments rather than sequentially within a single one, and the commercial and institutional reach of the underlying engine is bounded only by the range of capture devices on which it can run.

7. Conclusion

As synthetic media tools improve in fidelity and accessibility, detection-based countermeasures face the inherent disadvantage of having to perpetually adapt to increasingly sophisticated forgeries. This asymmetry means that reactive forensic analysis alone cannot sustain the evidentiary standards required by modern legal systems. A proactive approach, one that establishes authenticity at the point of creation rather than reconstructing it after dispute, is both technically necessary and legally overdue.

This project demonstrated the feasibility of such an approach. Bedrock delivers a working POC that fuses liveness verification, continuous real-time hashing, and blockchain anchoring into a single unbroken pipeline across three deployment platforms. User Acceptance Testing with eight external participants validated the system's usability (average SUS score of 87.5) and functional correctness across all four main tasks. All in all, the technical architecture provides a deterministic, reproducible, and independently verifiable mechanism for establishing file integrity.

Several limitations warrant acknowledgement. Liveness is verified once at session initiation, leaving mid-session substitution as a partially unmitigated vector. Liveness attestation data remains off-chain and the CCTV blockchain writer requires further production hardening. As adversarial techniques evolve, so must the verification layers. New controls such as continuous liveness monitoring, device-level attestation, and multi-source corroboration represent natural extensions of the architecture presented here. These constraints and recommendations, as documented in Section 6.1, are priorities for future development rather than fundamental architectural weaknesses. They are undertakings the sponsor and the team hope to explore should this POC generate enough commercial/policy buy-in to warrant the investment.

In conclusion, the absence of native court infrastructure for authenticating electronic evidence at the point of submission represents both a vulnerability in the current legal system and an opportunity. With continued investment, judicial engagement through the phased admissibility pathway outlined in Section 5.4, and the platform-agnostic extensibility demonstrated by the mobile and hardware prototypes, this project establishes a credible foundation for a new evidentiary standard in the age of synthetic media.

Try Bedrock Today!

Scan the QR code or visit the live application at to record and verify tamper-proof video evidence.

https://bedrockapp.space/

Get in Touch

Bedrock.

Bedrock Team | bedrockapp.space@gmail.com

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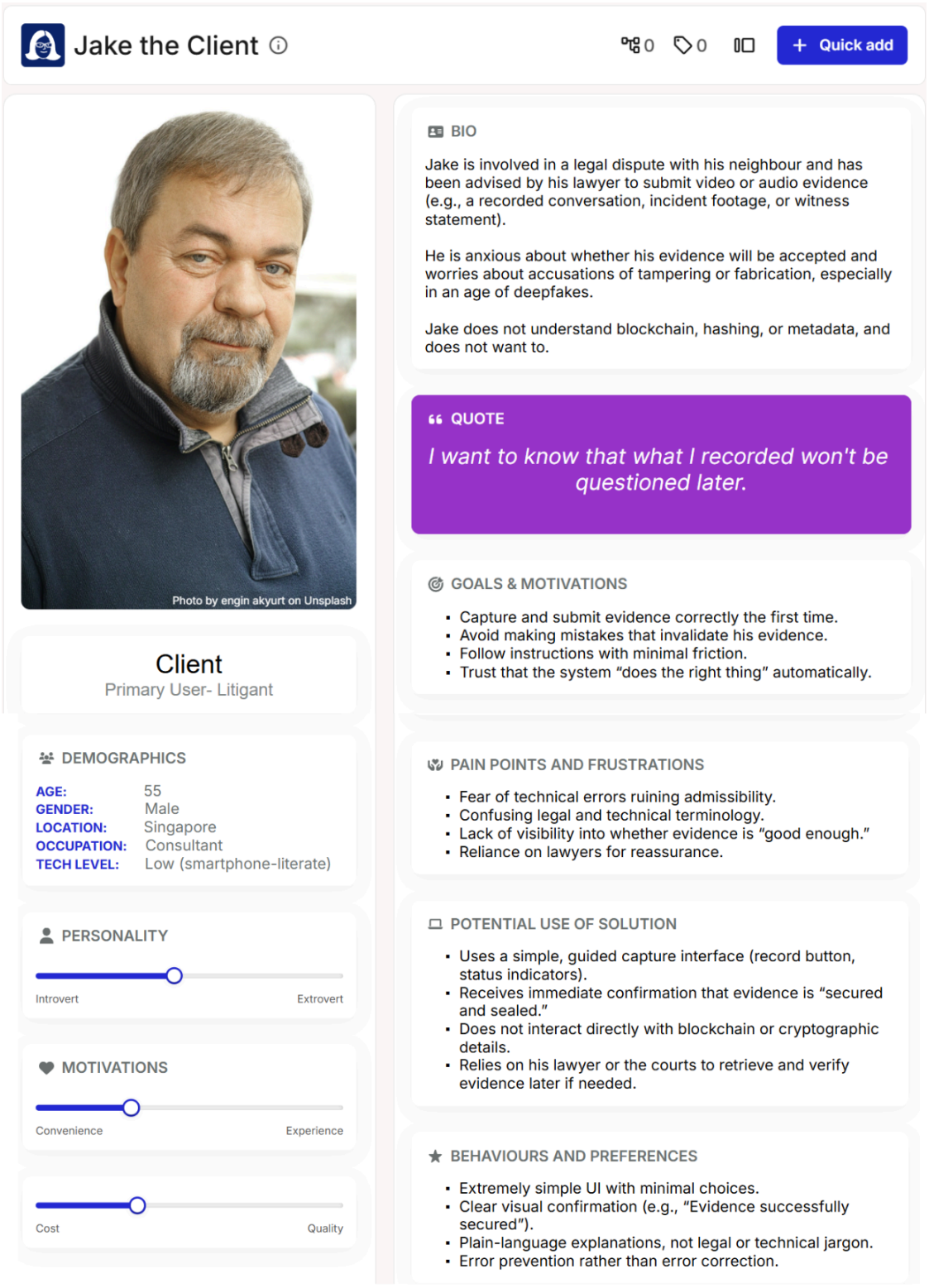
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9. Appendices

Appendix A: User Personas

A.1 Jake the Client



A.2 Maya the Lawyer

Maya the Lawyer

0 0 + Quick add



Senior Legal Counsel
Primary User- Litigant

DEMOGRAPHICS

AGE:	40
GENDER:	Female
LOCATION:	Singapore
OCCUPATION:	Lawyer
TECH LEVEL:	Intermediate (comfortable with SaaS tools)

GOALS & MOTIVATIONS

- Pre-empt challenges to electronic evidence authenticity.
- Reduce reliance on slow, expensive forensic experts.
- Present clear, court-explainable proof of evidence integrity.
- Maintain defensible chain-of-custody documentation.

PAIN POINTS AND FRUSTRATIONS

- Post-hoc verification is reactive and inefficient.
- Evidence provenance is often reconstructed under time pressure.
- Technical proofs are difficult to explain to judges and opposing counsel.
- Fragmented evidence tools (DMS, email, forensic vendors).

POTENTIAL USE OF SOLUTION

- Uses the in-app capture interface to create "file-of-first-instance" evidence.
- Confidently relies on GTN-anchored hashes to demonstrate integrity.
- Generates Proof-of-Integrity Certificates (POICs) for Affidavits of Evidence-in-Chief (AEICs) and submissions.
- Uses the verification dashboard to rebut authenticity challenges quickly.

BIO

Maya regularly manages litigation and arbitration matters involving digital-first evidence: CCTV footage, video witness statements, call recordings, emails, and system logs. She has previously worked with forensic experts and understands both their value and their cost.

Maya is highly aware of Section 116A EA and anticipates challenges to electronic evidence, especially in cases involving AI-generated or manipulated media.

She is not a blockchain expert, but she understands hashing, timestamps, audit trails, and why they matter legally.

QUOTE

I don't need the judge to understand blockchain. I need them to trust the evidence without calling three experts.

BEHAVIOURS AND PREFERENCES

- Transparent audit logs with timestamps and identities.
- Human-readable verification reports suitable for court submission.
- Ability to tag evidence by matter ID and export bundles.
- Clear explanation of "what this proves" vs. "what it does not."

PERSONALITY

Introvert — Extrovert

MOTIVATIONS

Convenience — Experience

Cost — Quality

A.3 Ron the Data Verification Specialist

Ron the Data Verification Specialist

0 0 + Quick add



Data Specialist
Secondary Users - Courts/Enforcement Agencies

DEMOGRAPHICS

AGE: 30
GENDER: Male
LOCATION: Singapore
OCCUPATION: Data Verification Specialist at the Singapore Courts
TECH LEVEL: High (hashing, metadata analysis, system logs, audit trails)

PERSONALITY

Introvert — Extrovert

MOTIVATIONS

Convenience — Experience

Cost — Quality

BIO

Ron's role is to independently assess disputed electronic evidence when authenticity or integrity is challenged. He operates at the intersection of law and technology, supporting the courts during litigation by validating whether evidence has been altered, substituted, or improperly handled.

Traditionally, Ron and his team rely on manual chain-of-custody documents, file comparisons, metadata inspection, and forensic toolkits, often under tight deadlines. He is highly aware of how generative AI and deepfake technologies complicate evidentiary disputes and increase scrutiny of digital files.

Ron is not decision-making counsel, but his technical conclusions often underpin affidavits, expert opinions, and litigation strategy.

QUOTE

I don't want to infer integrity. I want to verify it against an immutable source of truth. As quick and as reliably as possible.

GOALS & MOTIVATIONS

- Determine quickly and conclusively whether evidence has been tampered with.
- Provide defensible, repeatable verification outcomes.
- Reduce ambiguity and subjective interpretation in evidence disputes.
- Support legal teams with technically sound, court-defensible findings.

PAIN POINTS AND FRUSTRATIONS

- Incomplete or inconsistent chain-of-custody records.
- Reliance on manual, time-consuming forensic workflows.
- Pressure to explain complex technical findings to non-technical stakeholders.
- Late-stage discovery that evidence provenance is unclear.

POTENTIAL USE OF SOLUTION

- Uses the verification dashboard to run disputed evidence files through verification hashing algorithms.
- Compares uploaded files against GTN-anchored first-instance hashes.
- Reviews timestamped audit trails covering capture, storage, and access events.
- Generates Proof-of-Integrity Certificates (POICs) as objective verification artifacts.

BEHAVIOURS AND PREFERENCES

- Fast, reliable hash comparison with clear pass/fail results.
- Access to detailed metadata and event logs.
- Ability to verify single files or batch evidence bundles.
- Exportable technical reports that integrate into legal submissions.

Appendix B: User Stories

B.1 Easy Flow (Persona: Jake the Client)

Jake Lim is involved in an ongoing noise dispute with his neighbour, who frequently plays the drums late at night. After several informal complaints, the neighbour agrees to meet Jake in person to discuss and negotiate acceptable quiet hours. Jake is concerned that the agreement may later be disputed and wants to ensure that whatever is agreed upon can be relied on if needed.

Before the meeting begins, Jake loads the web application and is presented with 2 options: a create option (to create a new video) or a verify option (to verify whether a video has been tampered with). The user navigates to the record function, and the interface presents a pop-up with clear instructions and a disclaimer form, to which the user agrees. Following this, a simple camera view appears and a “verify live” checker scans Jake’s face (to prevent artificial pre-recorded video) before Jake can record. He places his laptop on the table, capturing the discussion.

During the meeting, the neighbour agrees to limit drumming to between 10:00 a.m. and 8:00 p.m. Jake does not interact with the application while recording; the system automatically handles the recording process in the background. The video is hashed one way, and the subsequently produced hash (not the video itself) is uploaded *LIVE* to the smart contract made on blockchain. All this is happening in the background.

Once the discussion ends, Jake stops the recording. The system prompts Jake to wait for the verifiable video and the Proof-of-Integrity Certificate (POIC) to be downloaded. The interface concludes that the evidence has been successfully secured and is tamper-evident. Both files are downloaded to Jake’s computer.

Jake leaves the meeting reassured that, should the agreement be broken, he has a trustworthy record of what was agreed upon.

Features

- Create a new video recording via webcam
- Background recording with no user interaction
- Pre-recording consent, disclaimer, and risk disclosure
- Face liveness check
- Downloadable verifiable master video
- Downloadable POIC
- Tooltips / short instructions during capture

B.2 Hard Flow (Persona: Maya the Lawyer)

Several weeks after the initial agreement, the neighbour repeatedly violates the agreed quiet hours. Jake schedules a follow-up meeting and brings his lawyer, Maya, to lead the discussion and formally restate the agreed terms. Anticipating potential denial, Maya decides that the meeting should be recorded using the evidence capture system to ensure legal defensibility. At the follow- up meeting, Maya supervises the evidence capture process. Using the web application, she starts the recording through the controlled capture interface. The meeting ends with the neighbour reaffirming the noise restrictions on record. The video is published and sealed immediately while it is being recorded, creating verifiable proof of origin.

Despite the recording and second agreement, the neighbour violates the terms again a week later, denies the agreement, and escalates the matter. Jake initiates legal proceedings for harassment. During litigation, opposing counsel challenges the reliability and integrity of the recorded video, claiming possible tampering and AI manipulation of the agreed- upon terms.

As such, Maya needs to find the relevant recording to respond to the dispute. She verifies the recording she took to the portal, and it goes through with no issue. Jake sends her the verifiable video to Maya over WhatsApp, and she proceeds to the Verification Portal to confirm its authenticity for the courts. However, WhatsApp compressed the variable file, stripping the smart contract addresses in the metadata. Fortunately, the portal immediately detects the mismatch against the sealed reference file on the blockchain, displaying a clear verification status message and guidance on restarting the process, along with common errors. Maya then has the option for Jake to send over the original recording or input the smart contract address from the metadata of the variable file. Jake sends over the original verifiable file to Maya, and she runs it through the verification portal to attempt the verification again.

Within seconds, the application displays a successful verification result, confirming that the video has not been altered since its creation. Maya then downloads and generates another Proof-of-Integrity Certificate (POIC), which clearly documents the hash, timestamp, and verification result. This POIC is submitted to the court as part of Jake's evidence bundle, enabling the judge to assess the authenticity of the video without requiring extensive forensic testimony.

Features

- Create new video recording via webcam
- Background recording with no user interaction
- Upload video for verification OR manual smart contract address input
- Detection of material changes → binary Verified / Not Verified
- Downloadable POIC
- POIC includes hash, timestamp, contract address, verification result, and audit data
- Court-ready without forensic expert reliance
- Guided recovery when verification fails
- Explanation of common errors
- Regenerate POIC from original recording

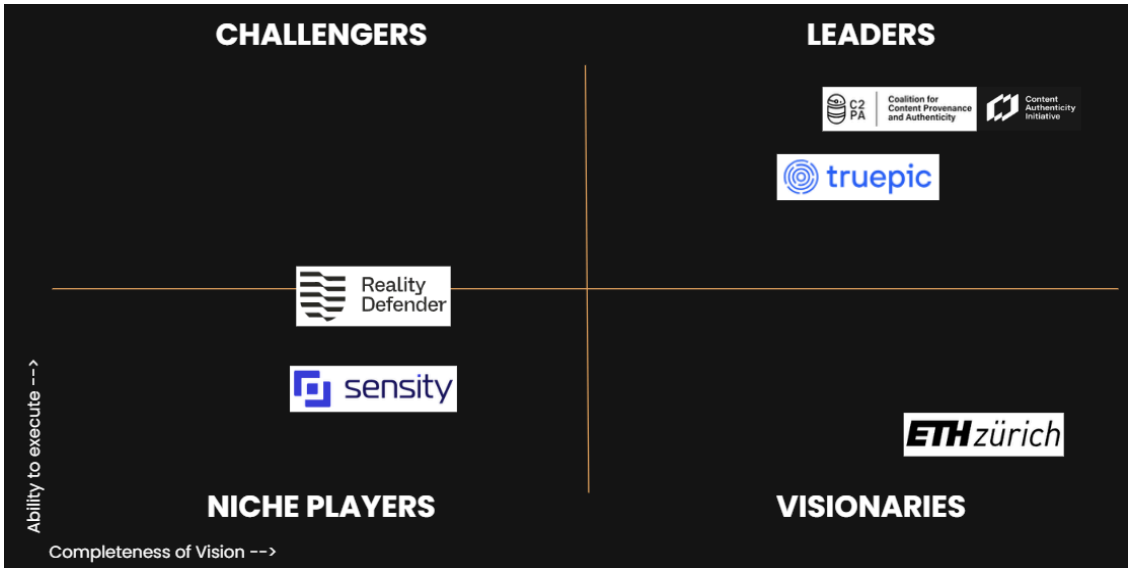
Appendix C: Competitive Analysis Matrix and Gartner Magic Quadrant

The competitor solutions are graded based on a matrix which accounts for:

Company	Verification Approach	Evidence Strength	Real-Time Capability	Ecosystem & Adoption	Price	Total
C2PA and Content Authenticity Initiative	4 Provenance metadata	3 Provenance chain (weak), Deterministic	3 At signing	5 Open standard	5 Very Low/Free	**20**
Truepic	5 Proactive (PKI)	4 Cryptographic seal, Deterministic	4 At capture (one-time)	3 Established in the industry	2 High	**18**
ETH Zurich Chip	5 Proactive (hardware)	4 Cryptographic signature, Deterministic	5 At capture (on-chip)	1 Research stage	1 N/A	**16**
Reality Defender	3 Reactive detection	3 Confidence score, Probabilistic, Forensic report	1 N/A	4 SG/Asean presence, Established	4 Low to Medium	**15**
Sensity AI	3 Reactive detection	3 Confidence score, Probabilistic, Forensic report	1 N/A	2 Limited presence	2 High	**11**

Competitive Analysis Matrix Based on Selected Criteria

Based on the rankings a Gartner Magic Quadrant depicting the four competitors can be drawn up:



Gartner Magic Quadrant

Appendix D: Consent and Data Disclosure Form

Bedrock: User Terms & Consent

Last Updated: 20 Feb 2026

1. Data Collection

Bedrock enables you to create evidential recordings by capturing video and audio from your device and generating technical integrity data, including cryptographic hashes, timestamps, and verification references. This integrity data is anchored to a blockchain network to establish a tamper-evident record of the file's existence and condition at a specific point in time. Blockchain references are persistent and may be publicly accessible.

2. Third-Party Liveness Check

A mandatory facial liveness check is performed during recording to confirm that a live individual is present. To conduct this check, limited imagery or video frames may be transmitted securely to an independent third-party provider, which returns only a liveness result to the App.

3. Legal Implications

You represent and warrant that you have the legal right and authority to make the recording and that you will obtain any required consents or provide any required notices to individuals being recorded, in accordance with

4. Disclaimer

Verification results generated by Bedrock are automated technical assessments of file integrity. They do not constitute legal advice, forensic certification, or a guarantee of admissibility in any legal proceeding. In addition, Bedrock shall not be liable for legal outcomes, disputes, or loss or misuse of recordings stored on your device. Responsibility for how recordings are used and relied upon remains with the user.

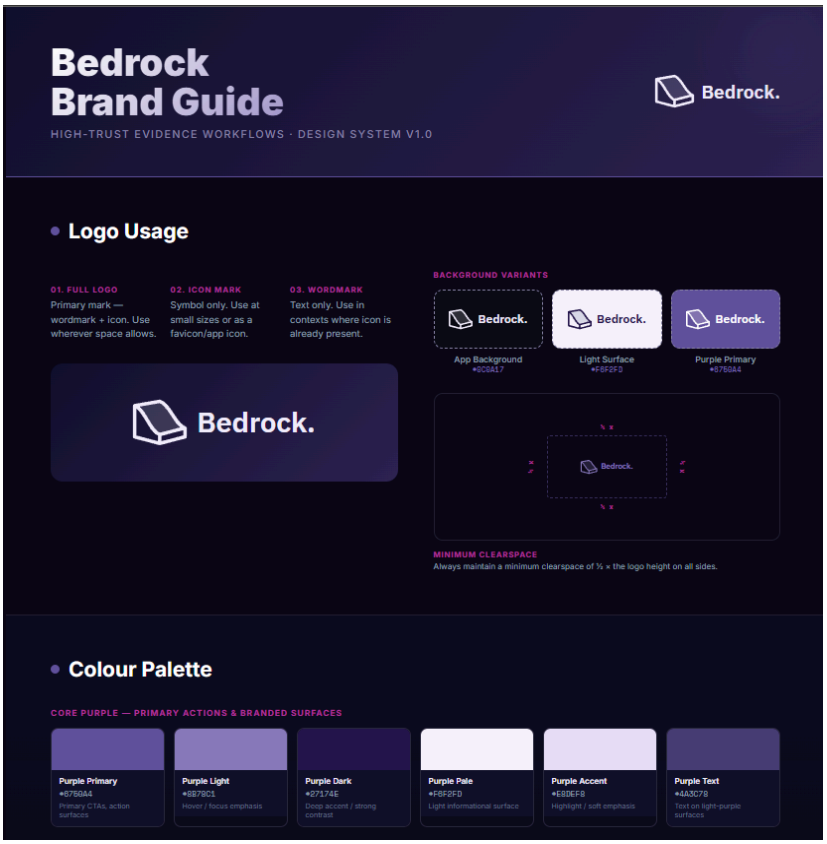
5. Consent Acknowledgment

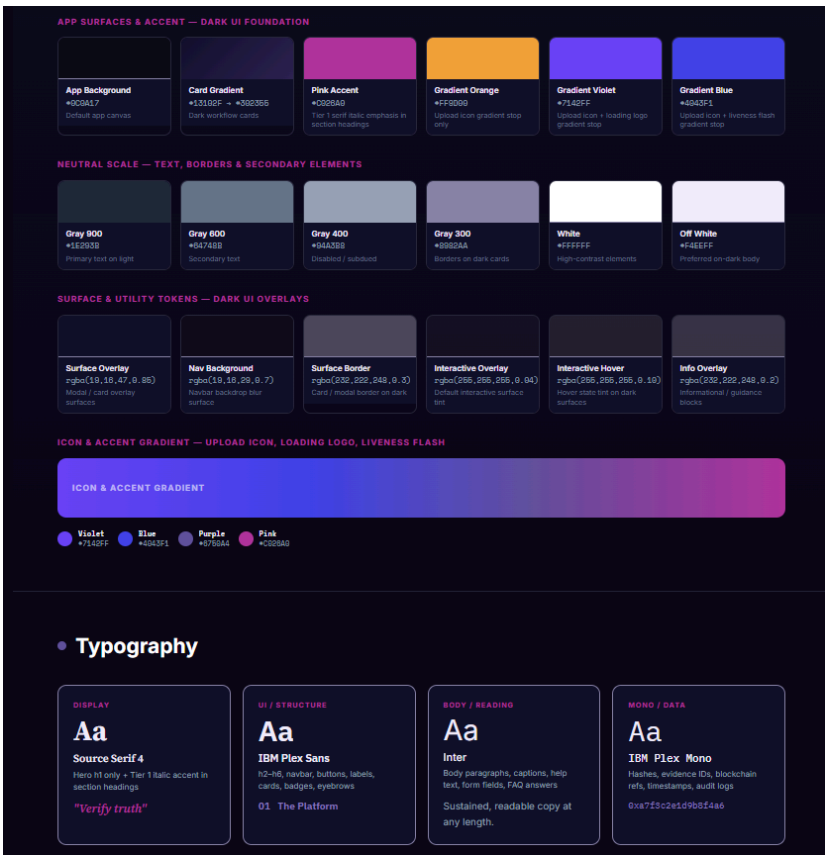
By selecting "I Agree" or continuing to use the App, you confirm your consent to the recording, integrity processing, liveness verification, and blockchain anchoring described above, and you accept responsibility for lawful use of the platform.

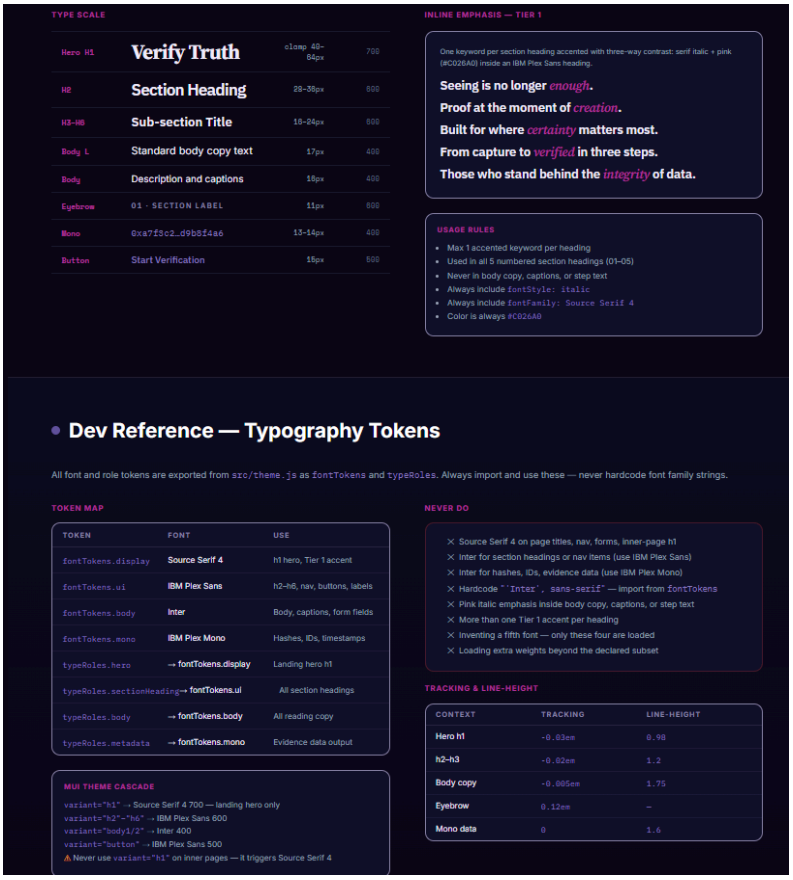
For enquiries, please contact: verify@bedrock.com.sg

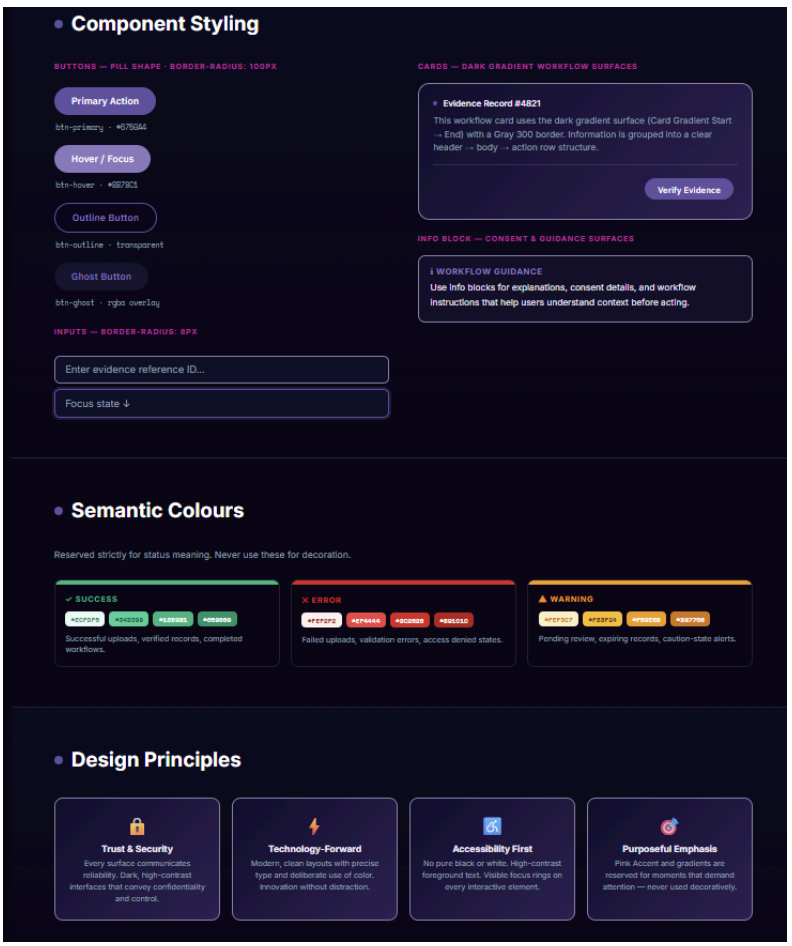
Deny | I Agree

Appendix E: Brand Guide

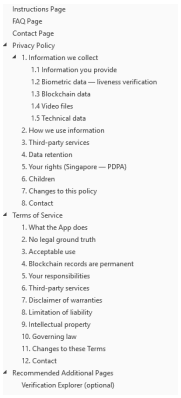





TYPE SCALE
Hero H1
Verify Truth
clamp 40–64px
700
H2
Section Heading
28–36px
600
H3–H6
Sub-section Title
18–24px
600
Body L
Standard body copy text
17px
400
Body
Description and captions
16px
400
Eyebrow
01 · SECTION LABEL
11px
600
Mono
0xa7f5c2…d9b5f4a6
13–14px
400
Button
Start Verification
15px
500
INLINE EMPHASIS — TIER 1
One keyword per section heading accented with three-way contrast: serif italic + pink (#C026A0) inside an IBM Plex Sans heading.
Seeing is no longer enough.
Proof at the moment of creation.
Built for where certainty matters most.
From capture to verified in three steps.
Those who stand behind the integrity of data.
USAGE RULES
Max 1 accented keyword per heading
Used in all 5 numbered section headings (01–05)
Never in body copy, captions, or step text
Always include fontStyle: italic
Always include fontFamily: Source Serif 4
Color is always #C026A0
Dev Reference — Typography Tokens
All font and role tokens are exported from src/theme.js as fontTokens and typeRoles. Always import and use these — never hardcode font family strings.
TOKEN MAP
TOKEN
FONT
USE
fontTokens.display
Source Serif 4
h1 hero, Tier 1 accent
fontTokens.ui
IBM Plex Sans
h2–h6, nav, buttons, labels
fontTokens.body
Inter
Body, captions, form fields
fontTokens.mono
IBM Plex Mono
Hashes, IDs, timestamps
typeRoles.hero
→ fontTokens.display
Landing hero h1
typeRoles.sectionHeading
→ fontTokens.ui
All section headings
typeRoles.body
→ fontTokens.body
All reading copy
typeRoles.metadata
→ fontTokens.mono
Evidence data output
MUI THEME CASCADE
variant="h1" → Source Serif 4 700 — landing hero only
variant="h2"–"h6" → IBM Plex Sans 600
variant="body1/2" → Inter 400
variant="button" → IBM Plex Sans 500
⚠ Never use variant="h1" on inner pages — it triggers Source Serif 4
NEVER DO
× Source Serif 4 on page titles, nav, forms, inner-page h1
× Inter for section headings or nav items (use IBM Plex Sans)
× Inter for hashes, IDs, evidence data (use IBM Plex Mono)
× Hardcode `'Inter', sans-serif` — import from fontTokens
× Pink italic emphasis inside body copy, captions, or step text
× More than one Tier 1 accent per heading
× Inventing a fifth font — only these four are loaded
× Loading extra weights beyond the declared subset
TRACKING & LINE-HEIGHT
CONTEXT
TRACKING
LINE-HEIGHT
Hero h1
-0.03em
0.98
h2–h3
-0.02em
1.2
Body copy
-0.005em
1.75
Eyebrow
0.12em
–
Mono data
0
1.6


Component Styling
BUTTONS — PILL SHAPE · BORDER-RADIUS: 100PX
Primary Action
btn-primary · #6759A4
Hover / Focus
btn-hover · #6B7BC1
Outline Button
btn-outline · transparent
Ghost Button
btn-ghost · rgba overlay
INPUTS — BORDER-RADIUS: 8PX
Enter evidence reference ID...
Focus state ↓
CARDS — DARK GRADIENT WORKFLOW SURFACES
Evidence Record #4821
This workflow card uses the dark gradient surface (Card Gradient Start → End) with a Gray 300 border. Information is grouped into a clear header → body → action row structure.
Verify Evidence
INFO BLOCK — CONSENT & GUIDANCE SURFACES
i WORKFLOW GUIDANCE
Use info blocks for explanations, consent details, and workflow instructions that help users understand context before acting.
Semantic Colours
Reserved strictly for status meaning. Never use these for decoration.
✓ SUCCESS
#ECFDF5
#A4EBD9
#2EB981
#059669
Successful uploads, verified records, completed workflows.
✕ ERROR
#FEF2F2
#EF4444
#DC2626
#991B1B
Failed uploads, validation errors, access denied states.
▲ WARNING
#FEF3C7
#F3BF24
#F59E0B
#D77706
Pending review, expiring records, caution-state alerts.
Design Principles
Trust & Security
Every surface communicates reliability. Dark, high-contrast interfaces that convey confidentiality and control.
Technology-Forward
Modern, clean layouts with precise type and deliberate use of color. Innovation without distraction.
Accessibility First
No pure black or white. High-contrast foreground text. Visible focus rings on every interactive element.
Purposeful Emphasis
Pink Accent and gradients are reserved for moments that demand attention — never used decoratively.

Appendix F: Website Copy Brief



Blockchain Video Verification

Website Copy Brief — v2

This document contains proposed copy for five supporting pages of the app: Instructions, FAQ, Contact, Privacy Policy, and Terms of Service. It is intended for the copywriting team to review, refine, and adapt for final publication.

Audience: Legal professionals, law enforcement, institutions, and general users seeking tamper-proof video evidence.

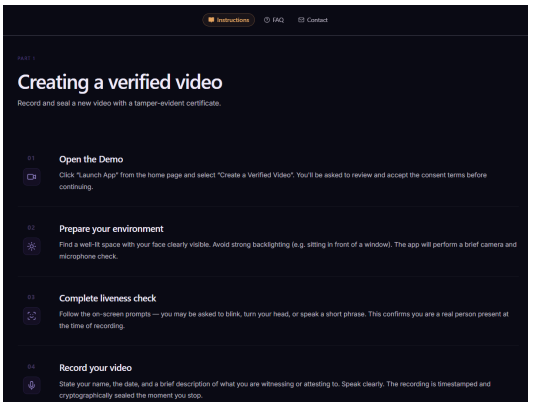
Tone: Clear, trustworthy, technically precise without being alienating. Avoid marketing fluff — this audience values credibility.

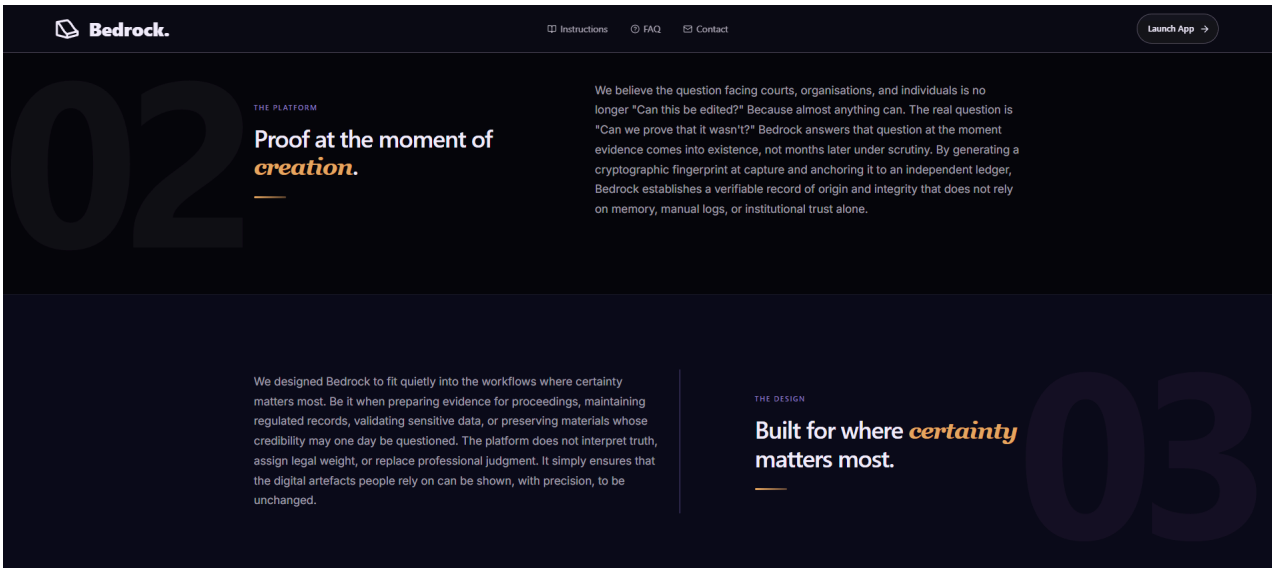
Key principle: The app creates cryptographic proof of video integrity — not legal ground truth. Copy must never overstate what blockchain anchoring provides.

Placeholders: Text marked in **RED BOLD** requires confirmation before publishing (legal entity name, dates, contact details).

Contents

- Instructions Page
- FAQ Page
- Contact Page
- Privacy Policy
- Terms of Service
- Recommended Additional Pages

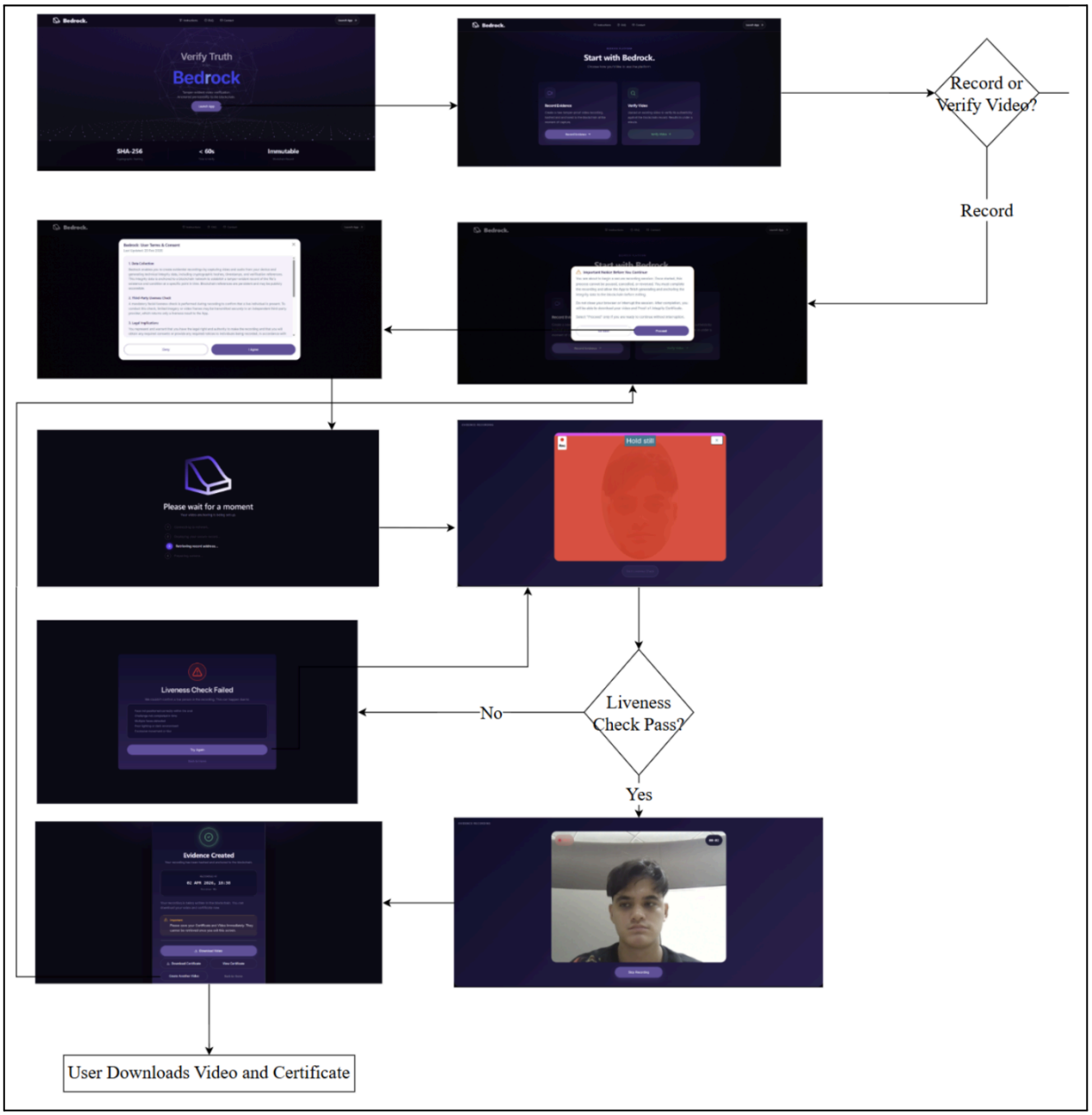




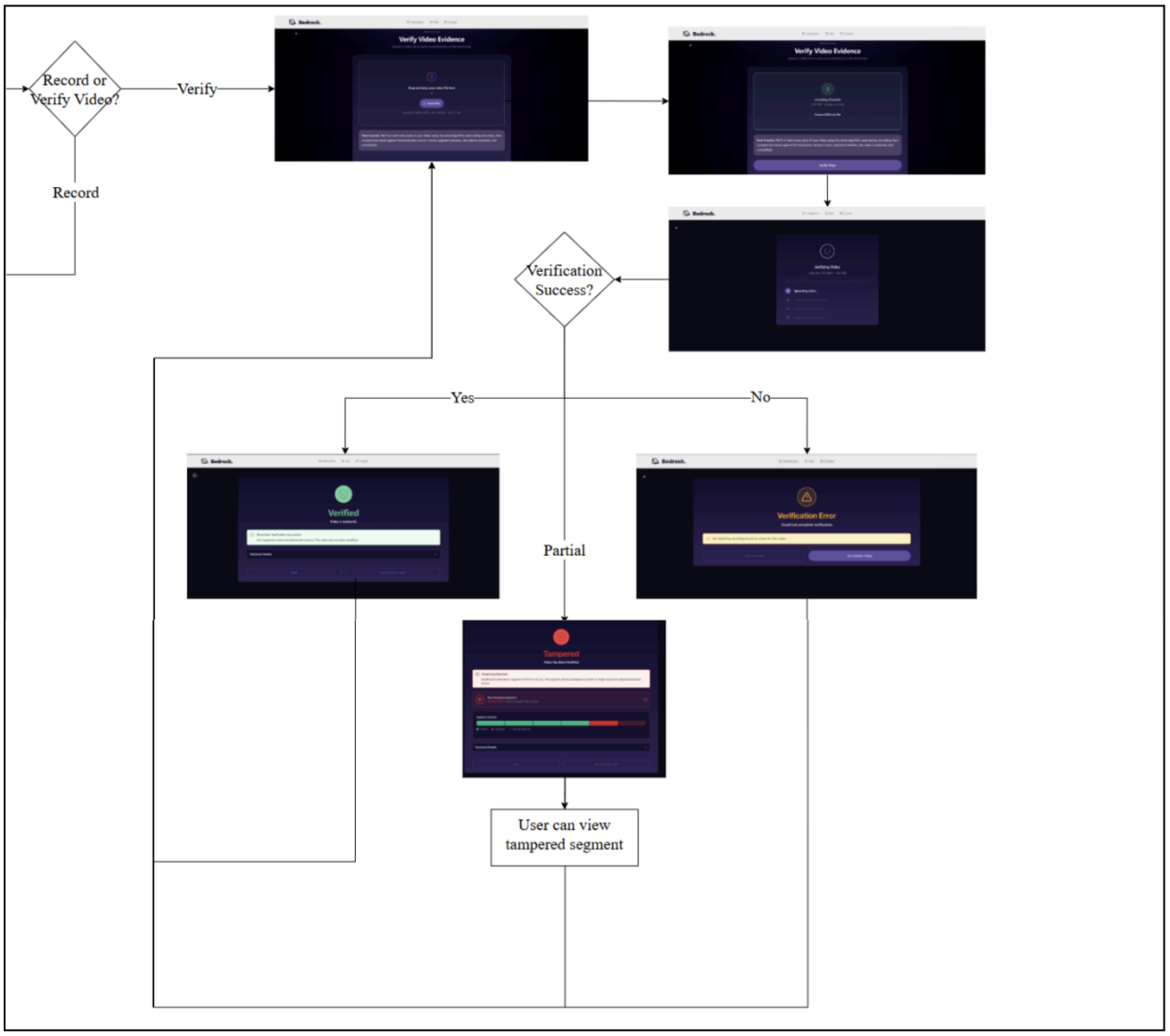
Snapshots of Website Copy Brief in Action

Appendix G: User Flow Diagrams

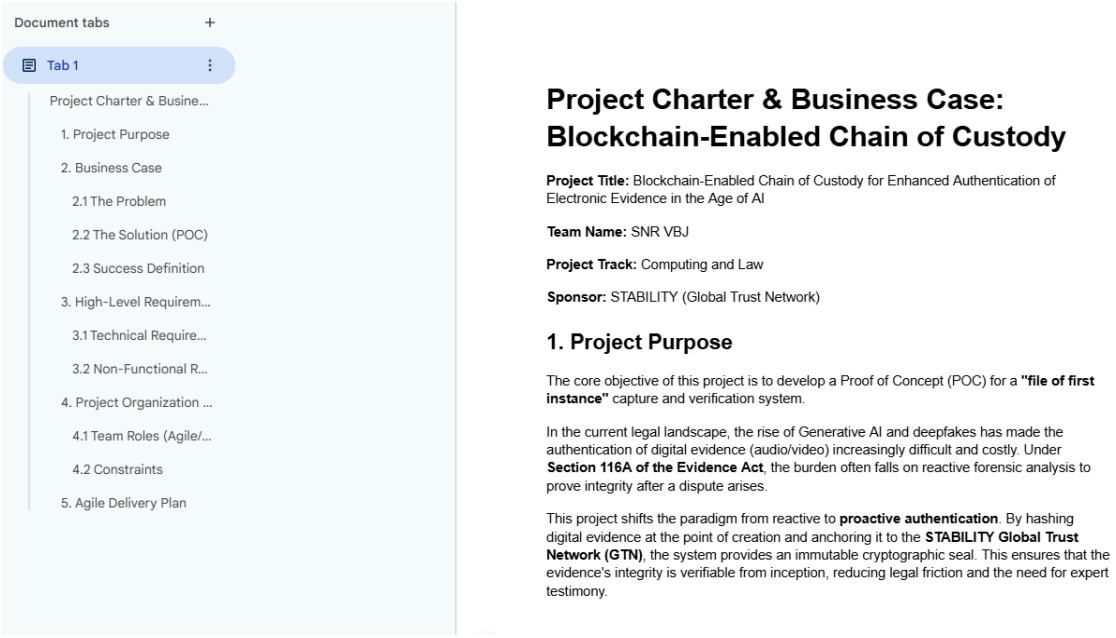
G.1 Record Module User Flow Diagram



G2: Verify Module User Flow Diagram



Appendix H: Project Charter



Project Charter & Business Case: Blockchain-Enabled Chain of Custody

Project Title: Blockchain-Enabled Chain of Custody for Enhanced Authentication of Electronic Evidence in the Age of AI

Team Name: SNR VBJ

Project Track: Computing and Law

Sponsor: STABILITY (Global Trust Network)

1. Project Purpose

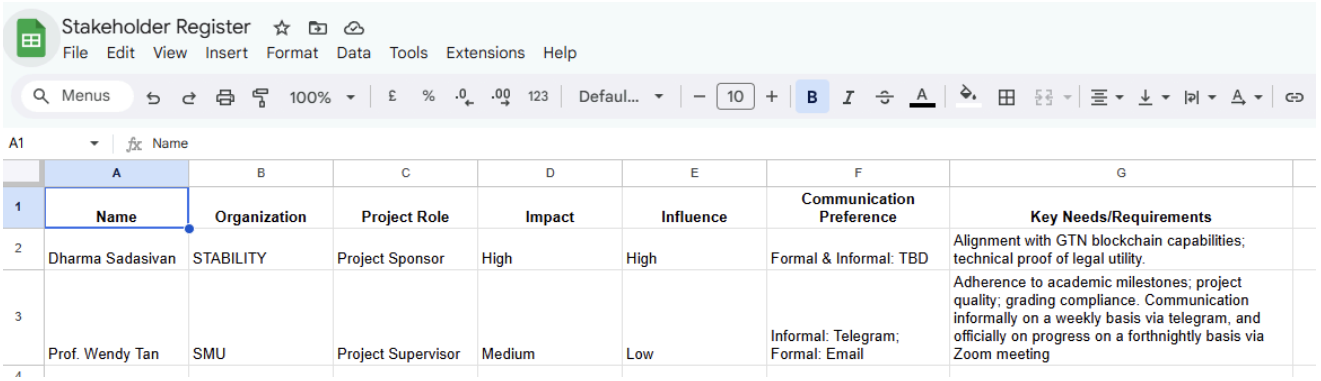
The core objective of this project is to develop a Proof of Concept (POC) for a **"file of first instance"** capture and verification system.

In the current legal landscape, the rise of Generative AI and deepfakes has made the authentication of digital evidence (audio/video) increasingly difficult and costly. Under **Section 116A of the Evidence Act**, the burden often falls on reactive forensic analysis to prove integrity after a dispute arises.

This project shifts the paradigm from reactive to **proactive authentication**. By hashing digital evidence at the point of creation and anchoring it to the **STABILITY Global Trust Network (GTN)**, the system provides an immutable cryptographic seal. This ensures that the evidence's integrity is verifiable from inception, reducing legal friction and the need for expert testimony.

Snapshot of Project Charter

Appendix I: Stakeholder Register

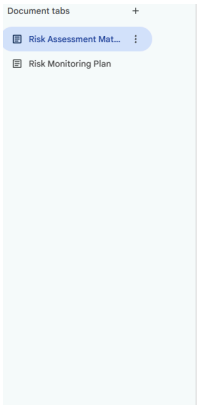


	A	B	C	D	E	F	G
1	Name	Organization	Project Role	Impact	Influence	Communication Preference	Key Needs/Requirements
2	Dharma Sadasivan	STABILITY	Project Sponsor	High	High	Formal & Informal: TBD	Alignment with GTN blockchain capabilities; technical proof of legal utility.
3	Prof. Wendy Tan	SMU	Project Supervisor	Medium	Low	Informal: Telegram; Formal: Email	Adherence to academic milestones; project quality; grading compliance. Communication informally on a weekly basis via telegram, and officially on progress on a forthnightly basis via Zoom meeting

Name	Role(s)	Impact	Influence	Communication Preference	Primary Responsibility	Secondary Responsibility
Ranveer Bains	PM & Smart Contract Dev	High	High	daily Telegram/ Weekly Zoom	Project Management: Project artifacts, documentation, Sprint reviews, and backlog management.	Blockchain Dev: Writing smart contracts on Stability GTN; REST API integration with Backend.
Vaibhav Mittal	Backend Lead	High	High	daily Telegram/ Weekly Zoom	Backend/Infrastructure: Database architecture (Supabase), system integration, and authentication.	PM Support: Work product validation and technical feasibility assessments.
Nicole Daduya	Frontend & UIUX Lead	High	High	daily Telegram/ Weekly Zoom	FE/UIUX: Delegation of FE deliverables, User Personas, Storylines, and Figma prototyping.	PM Support: Coordination with PM on feature prioritisation.
John Shutler	Lead Report Writer & FE Dev	Medium	High	daily Telegram/ Weekly Zoom	Content & Reporting: Compiling final submission report and all presentation materials.	Frontend Dev: Assisting with Front End deliverables.
S Bharath Raj	Backend Dev & Tech Doc	Medium	High	daily Telegram/ Weekly Zoom	Backend Dev: Assisting with BE deliverables.	Tech Reporting: Drafting system architecture and API documentation.
Sarah Do	Compliance Res. & Dev	Medium	High	daily Telegram/ Weekly Zoom	Compliance: Legal research for implementation of tech dealing with User Data.	Encryption: Implementing encryption style for the application given the legal use case.

Snapshot of Stakeholder Register

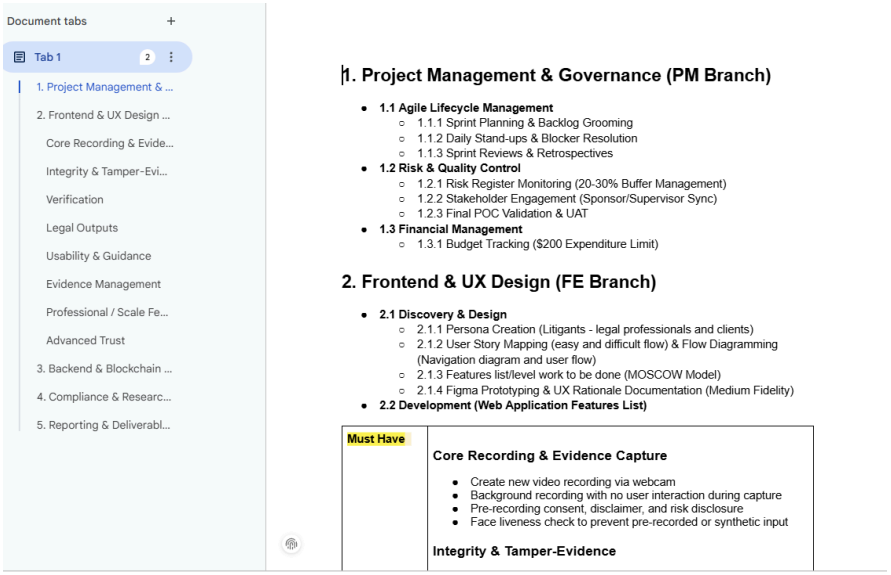
Appendix J: Risk Register



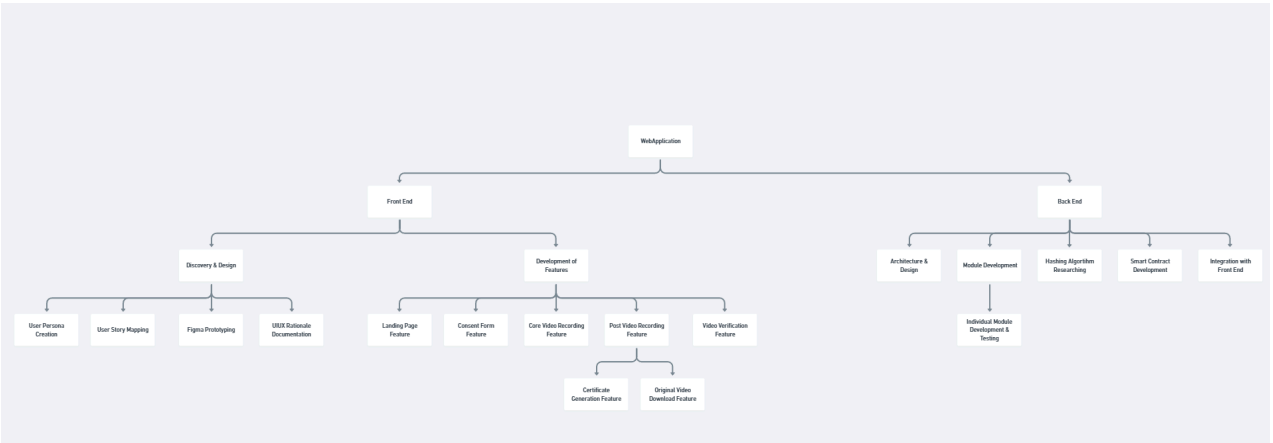
ID	Risk Description	Category	Impact	Likelihood	Mitigation Strategy	Owner	Resolved?
R01	**Universal Tech Learning Curve:** No team members have prior experience with Stability GTN or smart contract dev.	Project	High	High	Sprint 1 dedicated to team-wide "Stability Bootcamp"; PM to facilitate knowledge sharing sessions.	Ranveer Bains	
R02	**Budget Overrun:** Unexpected costs (e.g., domain, specific storage) exceeding the $200 limit.	Project	High	Low	Use free-tier services (Supabase free tier); avoid any paid API credits without PM approval.	Ranveer Bains	
R03	**Computation Bottleneck:** Real-time video hashing/encryption failing on client-side machines due to the use of SHA-256	Technical	High	Medium	Conduct feasibility testing and optimisation research for the proposed rolling cumulative hashing design (H_i = SHA-256(H_{i-1} ‖ chunk)), focusing on chunk sizing, on-device throughput, and	Sarah Do	

Snapshot of Risk Register

Appendix K: Work Breakdown Structure



Snapshot of WBS



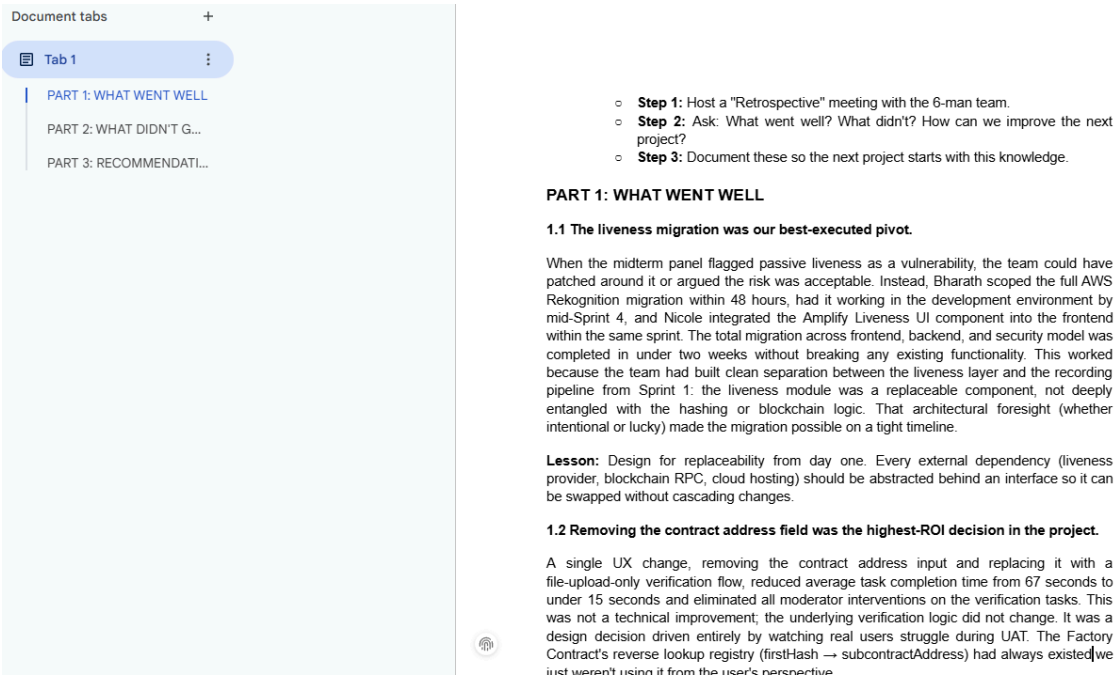
Snapshot of Whimsical Breakdown Tree

Appendix L: RACI Matrix

Project Management & Delivery								
Task / Deliverable	**Ranveer (PM)**	**Nicole (FE Lead)**	**Vaibhav (BE Lead)**	**John**	**Bharath**	**Sarah**	**Sponsor (Dharma)**	**Supervisor (Wendy)**
Project Charter & Roadmap	**A/R**	C	C	C	I	I	C	I
Sprint Backlog & Planning	**A/R**	R	R	I	I	I	I	I
Risk Register Management	**A/R**	C	C	I	I	I	C	I
20-30% Time Buffer Oversight	**A/R**	C	C	I	I	I	I	I
Stakeholder Communications	**A/R**	I	I	I	I	I	R	R
Development of POC								
Task / Deliverable	**Ranveer**	**Nicole**	**Vaibhav**	**John**	**Bharath**	**Sarah**	**Sponsor**	**Supervisor**
GTN Smart Contract Dev	**A/R**	I	C	I	I	I	C	I
UI/UX Design & Prototyping	I	**A/R**	I	C	I	I	I	I
Backend & DB (Supabase)	C	I	**A/R**	I	R	I	I	I
API & System Integration	C	R	**A/R**	I	R	I	C	I
Encryption Research & Dev	I	C	C	I	I	**A/R**	C	I
Real-time Hashing Logic	C	R	R	I	I	**A/R**	C	I
Legal Compliance & Reporting								
Task / Deliverable	**Ranveer**	**Nicole**	**Vaibhav**	**John**	**Bharath**	**Sarah**	**Sponsor**	**Supervisor**
Compliance/PDPA Research	I	I	I	C	I	**A/R**	C	I
Technical Documentation	I	C	C	I	**A/R**	I	I	I
Final Submissions & Report	C	I	I	**A/R**	R	I	I	I
Proof-of-Integrity (POIC)	C	R	R	I	I	**A/R**	C	I

Snapshot of RACI Matrix

Appendix M: Lessons Learned Register



Snapshot of Lessons Learned Register

Appendix N: Usability Testing Raw Data

Task Raw Results

Participant	Role	Task 1: UVP	Task 2: Capture	Task 3: Download Files
P1	Non-technical	Accurate	Completed	9s
P2	Lawyer	Accurate	Completed	7s
P3	Non-technical	Inaccurate	Completed	21s
P4	Non-technical	Partial	Completed	11s
P5	Non-technical	Inaccurate	Completed	31s (failed)
P6	Lawyer	Accurate	Completed	8s
P7	Lawyer	Partial	Completed	10s

Task Pass Rates

Task	Pass Rate	Status
Task 1: Navigate & Describe UVP	3/7 (43%)	Needs Attention
Task 2: Capture Video	7/7 (100%)	Passed
Task 3: Download POIC & Find Master File	6/7 (86%)	Needs Attention

SUS Results

Benchmark: A SUS score of 68 is considered average. Scores above 70 are rated "Good"; above 80 are "Good/Excellent"; above 85 are "Excellent."

SUS Question	P1	P2	P3	P4	P5	P6	P7
Q1: I think I would like to use this system frequently	4	5	3	4	3	5	4
Q2: I found the system unnecessarily complex	2	1	3	2	3	1	2

Q3: I thought the system was easy to use	4	5	4	4	3	5	4
Q4: I think I would need technical support to use this	2	1	4	2	4	1	2
Q5: I found the various functions well integrated	4	4	4	4	3	5	4
Q6: I thought there was too much inconsistency	2	1	3	2	3	1	2
Q7: Most people would learn this quickly	4	5	3	4	3	5	4
Q8: I found the system cumbersome to use	2	1	3	2	3	1	2
Q9: I felt confident using this system	4	5	4	4	3	5	4
Q10: I needed to learn a lot before using this	2	1	4	2	4	1	2
SUS Score	80	92.5	65	80	60	97.5	80

Average SUS Score: 79.3 (Good)

Appendix O: User Acceptance Testing Raw Data

Task Raw Results

ID	Role	T1: Describe UVP	T2: Record + Download	T3: Verify Authentic	T4: Verify Tampered
P1	Law Student	Accurate	Completed (12s)	Authentic (14s)	Tampered (11s)
P2	Lawyer	Accurate	Completed (9s)	Authentic (12s)	Tampered (13s)
P3	Non-technical	Accurate	Completed (15s)	Authentic (18s)	Tampered (16s)
P4	Law Student	Accurate	Completed (11s)	Authentic (13s)	Tampered (14s)
P5	Non-technical	Partial	Completed (18s)	Authentic (14s)	Tampered (15s)
P6	Lawyer	Accurate	Completed (8s)	Authentic (11s)	Tampered (12s)
P7	Lawyer	Accurate	Completed (14s)	Authentic (15s)	Tampered (14s)
P8	Non-technical	Inaccurate	Completed (20s)	Needed help (27s)	Confused by result (21s)

Task Pass Rates

Task	Pass Rate	vs. Usability Testing	Status
Task 1: Describe Bedrock's UVP	**6/8 (75%)**	↑ from 43%	Target Met
Task 2: Capture Video + Download Files	**8/8 (100%)**	↑ from 71%	Target Met
Task 3: Verify Authentic File	**7/8 (88%)**	New task	Target Met
Task 4: Verify Modified File	**7/8 (88%)**	New task	Target Met

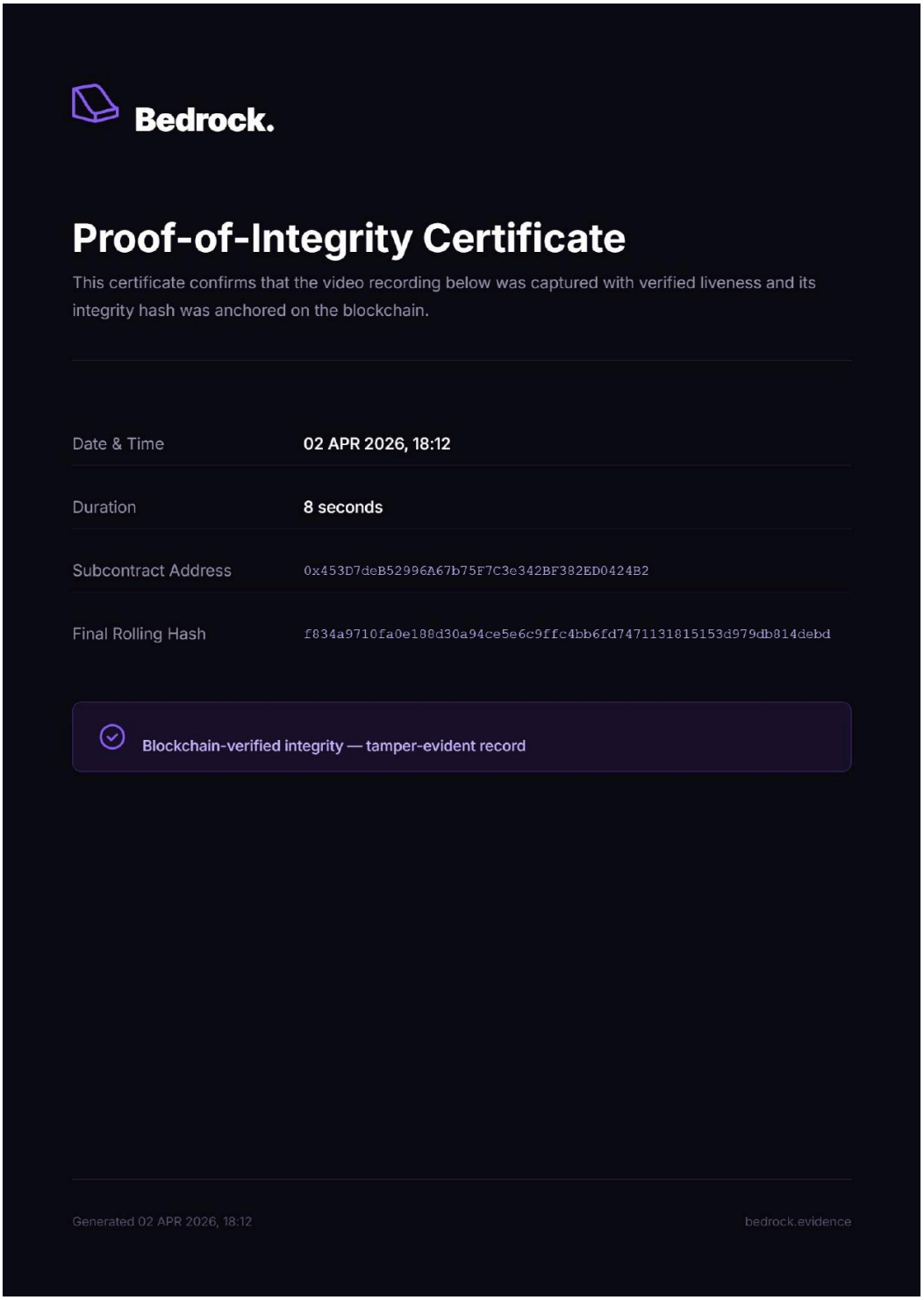
SUS Results

Benchmark: A SUS score of 68 is considered average. Scores above 70 are rated "Good"; above 80 are "Good/Excellent"; above 85 are "Excellent."

SUS Question	P1	P2	P3	P4	P5	P6	P7	P8
Q1: I would like to use this system frequently	5	5	4	5	4	5	5	5
Q2: I found the system unnecessarily complex	1	1	1	1	1	1	1	2
Q3: I thought the system was easy to use	5	5	4	5	4	5	5	4
Q4: I would need technical support to use this	2	1	2	1	2	1	2	4
Q5: I found the functions well integrated	4	5	4	4	4	5	4	4
Q6: There was too much inconsistency	2	1	2	1	1	1	2	2
Q7: Most people would learn this quickly	5	5	4	5	4	5	5	4
Q8: I found the system cumbersome to use	2	1	2	1	2	1	2	2
Q9: I felt confident using this system	5	5	4	5	5	5	5	4
Q10: I needed to learn a lot before using this	2	2	3	1	2	2	1	4
SUS Score	87.5	97.5	77.5	97.5	85.0	97.5	90.0	67.5

Average SUS Score: 87.5 (Excellent)

Appendix P: Proof of Integrity Certificate



Appendix Q: Meeting Notes Summary

Attached herewith is a summary of relevant meeting notes that have been recorded with STABILITY and our Supervising Professor Wendy Tan.

Meeting Date	STABILITY (Sponsor) Meeting Summary
21/1	Introduced the team to Dharma (sponsor, CSO). Team structure confirmed: Nicole & John on frontend (mid-fi Figma prototypes, personas, user stories); Vaibhav, Bharath, Ranveer & Sarah on backend (Stability blockchain architecture across capture, hashing, anchoring, verification); Ranveer leading smart contracts; Sarah on hashing research and PDPC compliance. Adopted a Scrum-ban approach in Jira with 8 tickets for the sprint. Dharma flagged UCFR.I/O as a reference and offered R&D support from Stability's Singapore team. Bi-weekly sponsor meetings agreed.
30/1	Backend agreed on SHA256 with segmented hashing (10–20 frames per segment) over MD5, with cumulative hashes written to chain every 2–4 seconds. Ranveer proposed a "grandfather contract" indexing user addresses to subcontracts and final video hashes to solve the contract address retrieval problem. Dharma offered to upgrade the API plan for higher throughput given the 3–5 second blockchain write latency. Bharath raised search efficiency concerns for large datasets. Frontend presented mid-fi prototype covering record and verify flows with consent form and webcam liveness check. Dharma suggested engaging Singapore courts post-project given the shift to online hearings.
13/2	No notes recorded
27/2	On track. Sponsor pushed the team to aim for delighters in the final presentation.
27/3	Web app and phone app largely complete, pending a byte-mismatch verification bug; Nicole progressing on Raspberry Pi CCTV implementation. Target release before 13 April. Planned logistics for the final presentation at 1:45 PM, including security clearance for dignitary attendees (team had invited the dean). Dharma agreed to arrive early. Discussed presentation structure, continuation plans (Bharath and Nicole willing to continue; Sarah stepping away), and equipment expense claims.

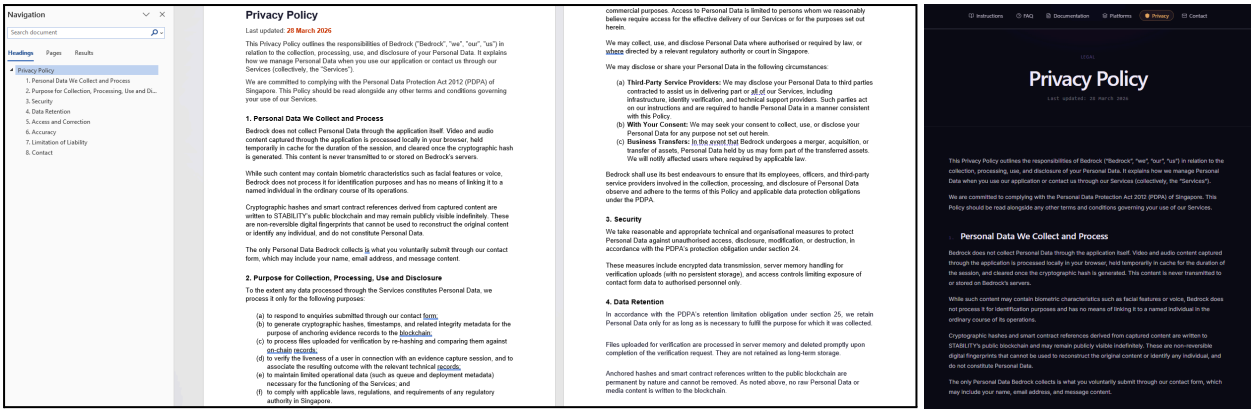
Meeting Date	Supervisor Meeting Summary
16/1	Wendy outlined academic requirements: midterm, final, and Poster Day (with 1-minute pitch video), weekly Friday updates, and fortnightly sprint reviews. Roles confirmed (Ranveer: lead, smart contracts, sponsor liaison; Bharath & Vaibhav: backend + reporting; Nicole: UI/UX; Sarah: encryption). Telegram for updates, email for official comms. Wendy offered escalation to Prof. Randall for complex blockchain issues.

	Action items: break project into weekly deliverables, factor in Vaibhav's travel (20–29 Jan) with a 30% buffer for finals, review Stability documentation, and send the first weekly update by Friday.
30/1	Team demoed the Bedrock mid-fi prototype covering record and verify modules across three personas (Jake the client, Maya the lawyer, a data verification specialist). Sponsor responded positively, prioritising feature versatility over UX polish. Technical decisions confirmed: SHA256 segmented hashing, smart contract mapping workaround for contract addresses, hashing on frontend with blockchain interactions on backend. Vaibhav to update architecture to identify tampering points. Ranveer to produce Gantt chart targeting first user flow by the midterm (week 7). Next meeting moved to 13/2 at 2:30 PM.
13/2	90% of sprint work on track. Frontend delayed by John's illness but design improvements made. Backend liveness check working via an imported online package to resolve false negatives. API usage within the 500-check monthly limit; iBeta-certified liveness meets ISO 30107. Sarah confirmed SHA256 as final choice despite longer hash times. Wendy clarified $200 expense allowance covers API keys, cloud subscriptions, and AI debugging tools. Ranveer to verify against Prof. Kiru's claimable list. Mock presentation scheduled for Thursday 8 PM.
26/2	Verbal approval to use budget for AI coding assistance subscription. Agreed to illustrate UAT in an agile context during the presentation
13/3	Final presentation confirmed for 13 April at 1:45 PM; team moved to the last slot (1:40–2:10) for extra Q&A time. Attendee list includes Patrick Lim (CEO, Ace Singapore), IMDA directors, and a judge so larger room needed, SR room pending release two weeks out. Team to produce a separate technical document on backend architecture for Wendy and Prof. Antoine a few days ahead of the presentation. Pitch video to open the final. Improvements agreed on energy, transitions, and demo lag explanation. Second-phase scope confirmed: video verification, tamper detection, Android APK, AWS liveness exploration, and Raspberry Pi + CCTV hardware attestation. Team raised concerns about evaluator domain expertise and observer roles; Ranveer and Nicole to meet Wendy 1:1 on Tuesday.
27/3	Web and phone apps complete pending bug fixes; Nicole soldering and installing Raspberry Pi CCTV; John on slides and tech deck for Wendy and Prof. Antoine. Run-through and tech deck handover scheduled for next meeting. Booth presentation to include a physical poster or brochure (Wendy awaiting Kiru's clarification on requirements). Final presentation logistics: book Room 3.3 opposite the presentation room to host attendees beforehand, pre-register all attendees online with QR codes, and inform security in advance to streamline entry for industry guests.

Appendix R: Sponsor Handover Pack

Component	Description
Source code (full repository)	Complete GitHub repository with frontend, backend, smart contracts, Android APK, and CCTV pipeline
Full report	Problem statement, scope, business requirements and value, technical deep dive, testing and validation documentation
Technical documentation	Technical brief (simple and in-depth versions), architecture diagrams
Environment configuration guide	Instructions for replicating the development environment, including AWS credentials, GTN RPC endpoint, Redis configuration
Brand guide (see Appendix E)	Bedrock UI/UX brand guide with colour codes, typography, component library references
Website copy brief (see Appendix F)	Complete copywriting for Instructions, FAQ, Privacy Policy, Terms of Service
POIC template	PDF certificate template
CCTV hardware guide	Bill of materials, 3D print files, assembly instructions, software setup guide

Appendix S: Bedrock Privacy Policy



Snapshot of Privacy Policy